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ABSTRACT

Two Tales in Scientific Computing

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This thesis narrates two stories that emerge from the exploration of two distinct problems in the domain of scientific computing.

The first story begins with a question: What happens when we leave the real line for numerical time integration? Most numerical methods for time integration use real time steps. Complex time steps provide an additional degree of freedom, as we can select the magnitude of the step in both the real and imaginary directions. By time stepping along specific paths in the complex plane, integrators can gain higher orders of accuracy and achieve expanded stability regions. We show how to derive these paths for explicit and implicit methods, discuss computational costs, and demonstrate clear advantages in solving complex-valued systems. We highlight the particular case of the nonlinear Schrodinger equation for which complex time integrators can be uniquely optimal. We also explore how complex time stepping allows us to break the Runge-Kutta order barrier for real-valued differential equations, enabling fifth order accuracy using only five function evaluations.

The second story explores how to effectively model and simulate the reaction environment in an electrochemical reactor. The goal is to gain insights into the optimal functionality of a catalyst. Catalysis is an enigmatic process that involves a multitude of hypothesized scenarios. Furthermore, electrochemical phenomena are difficult to model and simulate because of the various physical processes occurring at vastly differently spatial and time scales. We derive a mathematical model and construct an associated numerical method to effectively resolve the multiple spatial and temporal scales in this problem. Finally, we compare our mathematical simulations to experimental data to understand the dominant physical processes at play.

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CHAPTER 1

Introduction

This thesis details two of the main problems I encountered during my PhD. Both of these problems can be classified as scientific computing problems. Scientific computing and numerical analysis are occasionally dismissed by some as studies of rounding error. The definition of scientific computing has been a source of great philosophical debate [3] and like all good philosophical debates, there are often multiple valid points of view. To me, scientific computing is the art and science of using a computer to explore and understand the world we experience. While it is not the only perspective to explore the world with, it has provided its student, me, with a license to work in virtually any field with a background in mathematics. Despite being scientific computing problems, the two stories narrated in this thesis could not be any more different from each other.

The second chapter revolves around the numerical time integration of differential equations. This is arguably one of the most extensively studied topics in numerical analysis and is a very well-established field at this point. In the first chapter, I explore a simplistic question that originated from a more complicated problem I was trying to understand: When we take numerical time steps to simulate our systems of interest, we typically take real time steps. Are there any benefits in leaving the real axis and taking time steps in the complex plane?

Naturally, it turned out that I was not the first person to ask such a question in this well-studied field. Complex steps and complex Runge-Kutta coefficients were originally used for

circumventing singularities along the path of time integration (pole-vaulting) [4]. Nowadays they are predominantly used in splitting methods for linear evolution equations [5, 6, 7, 8, 9].

In the second chapter, the basic methodology and potential benefits of complex time stepping are introduced using accessible examples, derivations, and proofs. I show that along certain paths in the complex plane, integrators like Euler and Runge-Kutta gain higher orders of accuracy and expanded stability regions. I discuss how to derive these paths and talk about trade-offs like computational cost. Unlike most work on complex time integrators that focus on linear and integrable problems, I discuss complex Runge-Kutta methods for general nonlinear differential equations and show how we can gain more flexibility for real-valued problems by cleverly restricting the error to be purely imaginary.

Restricting error to be purely imaginary allows us to create a fifth-order accurate method for real-valued differential equations using only five function evaluations, thereby breaking through the order barrier, which limits traditional Runge-Kutta methods. The second chapter concludes by highlighting another domain where the use of complex time steps offers a distinct advantage. Complex time integrators allow complex-valued differential equations to have uniquely optimal stability regions. I highlight the example of the nonlinear Schrodinger equation and show a factor of 2 reduction in computational time for the same accuracy. Please note that our paper [10] is reproduced in this thesis.

The third and fourth chapters revolve around the mathematical modeling and efficient numerical simulation of the reaction environment during electrocatalysis which is a catalytic process involving oxidation or reduction through the direct transfer of electrons, usually at an electrode under some potential that makes the reaction chemically possible. Catalysis is quite

an enigmatic process because the key reactions happen at a nano-scale interface where observation and measurement by dynamic probing over time are difficult. There are a medley of scenarios that are hypothesized to occur in that interface. This is a mathematically understudied field where all the efforts (including the ones in this thesis) can arguably be described as pioneering, because of the vast unknown mathematical landscapes still waiting to be explored and understood.

Mathematical exploration in this field is a path towards understanding what would make certain crucial chemical reactions more efficient. This is highly relevant for global sustainability goals [11] such as climate action, clean and affordable energy and clean water. I spent most of my childhood in Tanzania where I learnt how important these goals were to sustainable and stable human lifestyles. Growing up in a different part of the world is a unique blessing because of the cultures, lifestyles and values you get exposed to at young age. Despite being in a middle class family, we'd encounter scheduled and unscheduled power cuts most weeks and in times of low rainfall, have to ration the water we had to use. The same issues were mirrored in my home country in India and are no doubt present in most developing countries across the world. This makes them very important global problems to work on.

Electrochemical reactions have captured interest worldwide because of their potential to be a clean alternative for energy storage and utilization, and for the creation of valuable products. They offer pathways for capturing carbon dioxide from the atmosphere and for the desalination of salt water which could have a huge impact on our response to climate change and our ability to enable wider access to clean drinking water. The vast majority of the research in this field has been experimental in nature. There has been relatively less of a focus on advancing

the mathematical description of electrocatalytic phenomena. While there has been recent commercial interest in using machine learning methods to identify low-cost catalysts, a complex black-box model which is purely predictive is not the point of the work in this thesis. I build and simulate a mathematical model that we know to be far from complete in its description of electrochemical phenomena. The overarching goal is to create a simplistic yet tunable mathematical and numerical framework - a virtual laboratory of sorts, to identify and mathematically describe the key dominant physical processes occurring under various experimental conditions. I describe in this thesis how the simultaneous evolution of both the mathematical model and the numerical method is necessary to create a robust and reliable framework for computational experimentation.

The work described in the third and fourth chapters is a collaborative effort with the experimental researchers in the Seitz lab at Northwestern. While this work is primarily on the oxygen reduction reaction, the mathematical modeling and numerical methods developed in this thesis can be naturally extended to other electrocatalytic reactions like the carbon dioxide reduction reaction.

In the pages of this thesis, I aim to recount my journey exploring these two challenges. Each story is distinctive, encompassing its unique frustrations, revelations, and insights. Together, they depict the broad spectrum of experiences one gets to encounter in a career in scientific computing. Hopefully, I can leave the reader with a feeling that scientific computing is so much more than the study of rounding error.

CHAPTER 2

Walking into the complex plane to ‘order’ better time integrators.

2.1. Introduction

The real line is a tiny sliver in the whole of the complex plane. Wandering into the complex plane has often been employed to solve real-world problems. Two direct examples are the solution of complicated real-valued integrals via contour integrals [12] and the usage of the intuition gained from poles in the complex plane in control-theory applications [13]. The complex plane provides wonderful insights to real problems, allowing mathematicians to impact practical applications. One of the motivations for this idea was a simple numerical application highlighted in a 2018 SIAM News article [14] where Nick Higham describes how taking a numerical complex derivative neatly gets rid of round-off error. However, most methods for numerically solving a differential equation from time a to time b , construct a path utilizing only the real line. We show how to use the complex plane to our advantage during numerical time integration by taking time steps in the complex plane.

This thesis is not the first to talk about complex time steps. Early work on time stepping in the complex plane focused on systematically avoiding singularities during numerical integration of singular differential equations [4]. A relatively common use of complex time steps is in the field of operator splitting methods. One of the earliest works in this field is [15], where Chambers explores operator splitting in symplectic integrators using complex substeps. Taking real substeps often results in negative substeps and substeps larger than the original step causing

the symplectic integrator go back and forth along the direction of integration. By using complex substeps, Chambers showed how to avoid negative and unstable substeps, significantly reducing the truncation error. Chambers's work inspired others to consider complex time steps and complex coefficients as part of operator splitting methods for various classes of parabolic and nonlinear evolution problems [5, 6, 7, 8, 9, 16]. Complex coefficients have found uses in the construction of numerical schemes for stiff differential algebraic systems [17] and for Runge-Kutta-Nystrom methods [18]. More recently, [19] used Cauchy's integral formula as inspiration to introduce a general framework for deriving integrators using interpolating polynomials in the complex plane. This results in parallel Backward Differentiation Formula (BDF) methods with improved stability regions. While powerful, these methods have been developed for specific classes of problems and we explore a more general and accessible approach.

In this chapter, we ask fundamental questions about the utility of numerical time-stepping in the complex plane for general differential equations, including: 'What paths lead to the least error?' and 'Are there general strategies and trade offs for complex time-stepping?'. Complex time steps have been previously studied in a general manner in [20] where Filatov derives complex paths for linear differential equations where the Euler method gains higher orders of accuracy. Relatedly, [21] uses linear analysis to show that complex time steps based on specific roots of unity allows Runge-Kutta methods to gain superconvergence, increasing the order of the method by one as the number of steps go to infinity. In our work, we go beyond the simple linear problem and demonstrate how to systematically derive higher-order paths with Runge-Kutta integrators for general nonlinear vector-valued differential equations.

We show that stepping into the complex plane provides an extra dimension that can be exploited to improve accuracy for general differential equations. To begin, we introduce the

concept of complex time-stepping using Euler-stepping and linear examples which should be accessible to a broad audience (Section 2.2). Next, we demonstrate the construction of higher-order time integrators for general nonlinear non-autonomous equations using Euler steps and include an explanatory proof that 2nd-order accuracy could not be achieved by real-valued Euler time steps (Section 2.3). We then extend our results to implicit methods (Section 2.4) and make connections to Butcher Series and rooted trees for deriving Runge-Kutta methods (Section 2.5). We discuss the tradeoffs of computational cost for complex operations (Section 2.7) and show that complex time integrators can circumvent the Runge-Kutta order barrier [22, 23] for real-valued systems enabling fifth order methods in five stages (Section 2.8). Finally, we use the additional degree of freedom to design expanded stability regions, beyond those attainable by state-of-the-art Runge-Kutta methods (Section 2.9). We demonstrate that complex time integrators can be uniquely optimal for complex-valued systems like the nonlinear Schrodinger equation, and show a factor of 2 reduction in computational time for the same accuracy. The code to reproduce the numerical experiments in this chapter is available on GitHub¹.

2.2. Building intuition: Taking complex time steps with the Euler method

The idea of a complex time step can be unusual and unintuitive. To make the unusual idea of complex time-stepping accessible, we first illustrate simple paths with complex time steps and demonstrate that applying the forward Euler method along the complex paths improves accuracy compared to forward Euler with real steps. The paths we choose in the complex plane must intersect the real line at the time points where we desire a solution. Our primary question is ‘what is the optimal n -step path in the complex plane over which to integrate between two real time points?’

¹https://github.com/Dirivian/complex_time_integrators

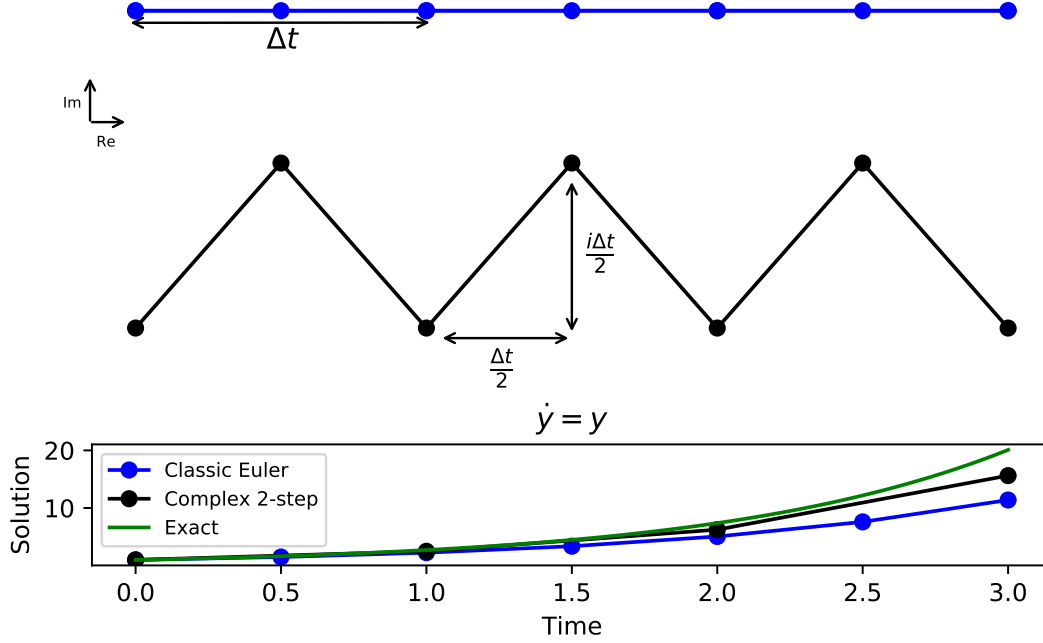


Figure 2.1. Taking complex time steps that return to the real line results in increased accuracy.

To illustrate the simplest possible case, suppose we want to solve $\dot{y} = \lambda y$ with the initial condition y_0 at time $t = 0$. To find the solution $y(\Delta t)$, we can either take two real time steps of length $\frac{\Delta t}{2}$ or take a complex time step of length $\frac{\Delta t}{2} + i\frac{\Delta t}{2}$ followed by a complex time step of length $\frac{\Delta t}{2} - i\frac{\Delta t}{2}$ returning back to the real line (paths in Fig. 2.1 (top)). We numerically integrate the differential equation $\dot{y} = y$ along these paths (Fig. 2.1 (bottom)). By eye, the complex time path is closer to the exact solution than the real-time path in this linear case. The improvement holds for nonlinear and non-autonomous ODEs (Fig. 2.2) and nonlinear PDEs, such as the viscous Burgers equation (Fig. 2.3). In section 2.3, we describe the generalizations needed to solve nonlinear equations, but first, we provide theoretical insight into the simple linear case.

Why does time stepping on the complex path improve accuracy of the numerical solution so significantly? To answer that, we analyze the linear problem $\dot{y} = y$ with initial condition $y(t_0) = y_0$. Consider time stepping from the initial condition y_0 at time t_0 to time $t + \Delta t$ using two real time steps. The discrete values of $y(t)$ are given by the following at $t = t_0$, $t_0 + \frac{\Delta t}{2}$, and $t_0 + \Delta t$:

$$\begin{aligned}
 y(t_0) &= y_0 \\
 y\left(t_0 + \frac{\Delta t}{2}\right) &= y_0 \left(1 + \frac{\Delta t}{2}\right) \\
 (2.1) \quad y(t_0 + \Delta t) &= y_0 \left(1 + \frac{\Delta t}{2}\right) \left(1 + \frac{\Delta t}{2}\right) = y_0 \left(1 + \Delta t + \frac{\Delta t^2}{4}\right).
 \end{aligned}$$

Now consider timestepping from y_0 at time t_0 to time $t + \Delta t$ using the two complex time steps.

$$\begin{aligned}
 y(t_0) &= y_0 \\
 y\left(t_0 + \frac{\Delta t}{2} + i\frac{\Delta t}{2}\right) &= y_0 \left(1 + \frac{\Delta t}{2} + i\frac{\Delta t}{2}\right) \\
 (2.2) \quad y(t_0 + \Delta t) &= y_0 \left(1 + \frac{\Delta t}{2} + i\frac{\Delta t}{2}\right) \left(1 + \frac{\Delta t}{2} - i\frac{\Delta t}{2}\right) = y_0 \left(1 + \Delta t + \frac{\Delta t^2}{2}\right)
 \end{aligned}$$

The exact solution to the problem is given by the Taylor expansion

$$(2.3) \quad y(t_0 + \Delta t) = y_0 e^{\Delta t} = y_0 \left(1 + \Delta t + \frac{\Delta t^2}{2} + \dots\right).$$

The secret behind the higher accuracy of the complex 2-step method is that the polynomial obtained through the numerical approximation matches the Taylor series of the exact solution to the second-order term, making the forward Euler method second-order accurate. The only other unique path with 2nd order accuracy is the permutation of the complex time steps in Eq. 2.2: $(\frac{\Delta t}{2} - i\frac{\Delta t}{2}, \frac{\Delta t}{2} + i\frac{\Delta t}{2})$, which leads to the same polynomial. This 2-step 2nd order complex Euler path originally appeared in [20] where the author derives it as a higher-order path for linear differential equations. In Subsection 2.3.2, we show this path satisfies 2nd-order accuracy for general nonlinear differential equations.

By stepping into the complex plane, we gain an extra degree of freedom in the coefficients that allows us to design time-steps with improved accuracy. This extra dimension can be used to construct higher order paths for linear and nonlinear differential equations (Section 2.3), circumvent the order barrier for Runge-Kutta methods (Section 2.8) and even design improved stability regions (Section 2.9). In the next section, we demonstrate how to construct paths with higher-order accuracy for linear and nonlinear systems.

2.3. Constructing a higher order time integrator

Is it possible then to construct 3-step path on which forward Euler would be 3rd-order accurate? Would it be nth order accurate on an n-step complex path? To build intuition for the simplest higher-order construction, we continue analyzing the linear problem in Subsection 2.3.1 and prove that achieving higher order requires complex time-stepping. Later in Subsections 2.3.2, 2.4.2 and Section 2.5, we extend the construction of complex time steppers to the general nonlinear problem. We also describe implicit methods in Section 2.4.

The complex 2-step method vs the forward Euler method

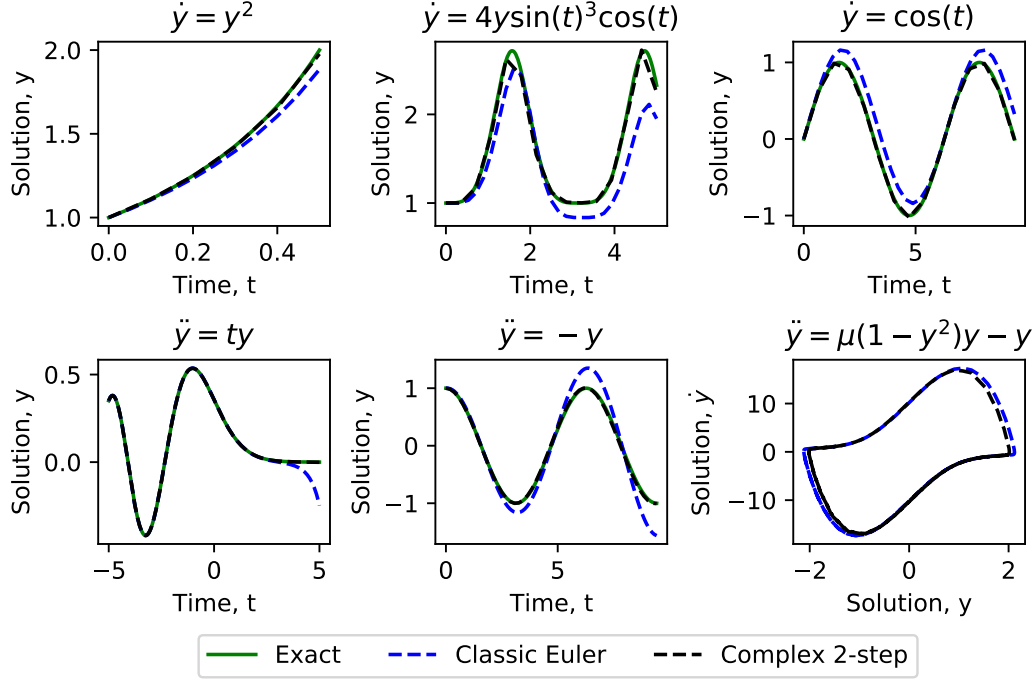


Figure 2.2. Comparison of real-valued classic Euler method and 2-step complex-valued Euler method. The number of real timepoints for the classic scheme are roughly double, $2n - 1$, the number for the 2-step complex scheme so that the net number of timesteps and function evaluations are the same for both schemes.

2.3.1. Higher order integrators for linear differential equations using complex Euler steps

Here, we construct a third-order time integrator for the differential equation $\dot{y} = \lambda y$ by taking three forward Euler time steps of variable sizes in the complex plane that return to the real line at every desired time point.

For three complex steps $w_1 \Delta t$, $w_2 \Delta t$ and $w_3 \Delta t$ going from $y_0(t_0)$ to $y(t_0 + \Delta t)$, the approximate numerical solution $\bar{y}$ at time $t + \Delta t$ is given by

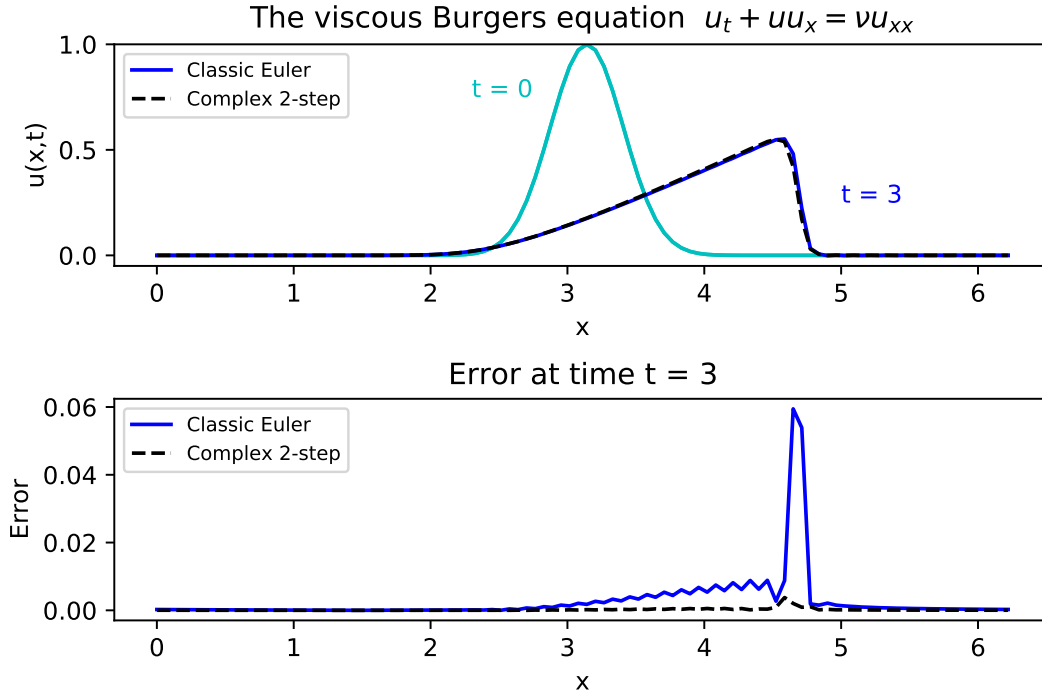


Figure 2.3. The complex time steps reduce the error for the nonlinear viscous Burgers equation.

$$(2.4) \quad \bar{y}(t_0 + \Delta t) = y_0(1 + w_1 \lambda \Delta t)(1 + w_2 \lambda \Delta t)(1 + w_3 \lambda \Delta t)$$

where the w_i 's are complex variables. Collecting $\lambda \Delta t$ terms results in

$$(2.5) \quad \bar{y}(t_0 + \Delta t) = y_0(1 + (w_1 + w_2 + w_3)\lambda \Delta t + (w_1 w_2 + w_2 w_3 + w_1 w_3)\lambda^2 \Delta t^2 + w_1 w_2 w_3 \lambda^3 \Delta t^3).$$

The exact solution of the linear problem is given by

$$(2.6) \quad y(t_0 + \Delta t) = y_0 e^{\lambda \Delta t} = y_0(1 + \lambda \Delta t + \frac{1}{2} \lambda^2 \Delta t^2 + \frac{1}{6} \lambda^3 \Delta t^3 + \dots)$$

For the numerical approximation to be third-order accurate, the coefficients of $y_0(\lambda\Delta t)^n$ for $n = 1, 2, 3$ in equations (2.5) and (2.6) need to match. This results in the following order conditions:

$$\begin{aligned}
 w_1 + w_2 + w_3 &= 1 \\
 w_1 w_2 + w_2 w_3 + w_1 w_3 &= \frac{1}{2} \\
 w_1 w_2 w_3 &= \frac{1}{6}.
 \end{aligned}
 \tag{2.7}$$

One of the solutions to the above nonlinear system is

$$(2.8) \quad (w_1, w_2, w_3) = (0.186731 + 0.480774i, 0.626538, 0.186731 - 0.480774i).$$

However, as the nonlinear system 2.7 is symmetric about w_1, w_2 and w_3 , all 6 permutations of $(0.186731 + 0.480774i, 0.626538, 0.186731 - 0.480774i)$ result in unique 3-step 3rd order complex time stepping paths. In Fig 2.4, all six 3-step 3rd order paths are shown along with the two 2-step 2nd order paths and the single step forward Euler path.

For a general n -step n th order path, the order conditions are

$$\begin{aligned}
 \sum_i w_i &= 1 \\
 \sum_{i \neq j} w_i w_j &= \frac{1}{2} \\
 \sum_{i \neq j \neq k} w_i w_j w_k &= \frac{1}{6} \\
 &\vdots \\
 \prod_{i=1}^n w_i &= \frac{1}{n!}.
 \end{aligned}
 \tag{2.9}$$

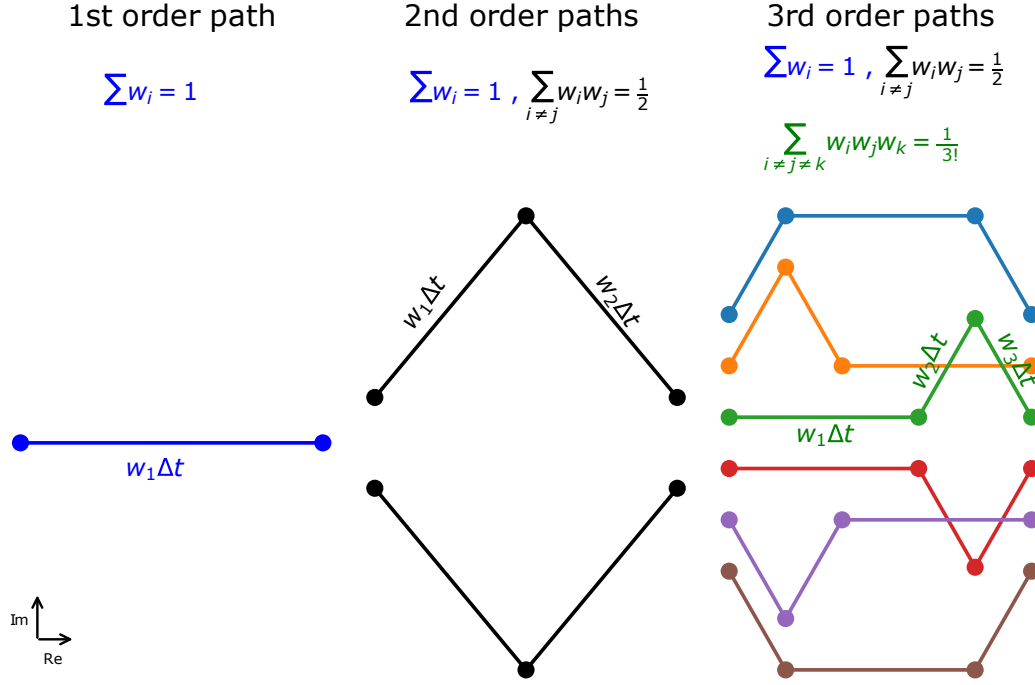


Figure 2.4. All possible first, second, and third-order complex Euler methods for linear differential equations. Each order requires an additional step and adds an additional constraint on the coefficients.

Equivalent formulae for higher-order Euler paths for linear problems appeared in [20], but in a less-compact fractional-step form. For higher orders, solving the nonlinear system of constraints on the coefficients becomes an increasingly difficult symbolic task. However, solutions can be found numerically up to a certain precision and stored in a library for general use.

These results beg the theoretical question: is there any way to achieve similar results with real-valued time steps? Can one take steps of varying size on the real line and achieve a higher-order time integrator? We prove orders 2 or greater cannot be achieved with real-valued Euler steps.

Theorem 1. *It is impossible to construct a time integrator with order greater than or equal to 2 using real-valued Euler time steps of variable sizes.*

PROOF. We prove the above theorem by showing that it is impossible to construct a second-order time integrator using forward Euler steps of variable size on the real line. Suppose we use n time steps with sizes $w_1, w_2, \dots, w_n$. The approximate solution $\bar{y}$ for $\dot{y} = \lambda y$ is given by

$$(2.10) \quad \bar{y}(\Delta t) = (1 + w_1 \lambda \Delta t)(1 + w_2 \lambda \Delta t) \dots (1 + w_n \lambda \Delta t),$$

where the first-order condition is satisfied: $w_1 + w_2 + \dots + w_n = 1$. Simplifying, we find

$$(2.11) \quad \bar{y}(\Delta t) = 1 + \lambda \Delta t + (w_1 w_2 + w_1 w_3 + \dots + w_{n-1} w_n) \lambda^2 \Delta t^2 + \dots$$

For second-order accuracy, the coefficient on $\lambda^2 \Delta t^2$ must be $\frac{1}{2}$. Expressing w_n as $1 - (w_1 + w_2 + \dots + w_{n-1})$ using the first-order condition, we arrive at

$$(2.12) \quad w_1 + w_2 + \dots + w_{n-1} - w_1^2 - w_2^2 \dots - w_{n-1}^2 - w_1 w_2 - w_1 w_3 \dots - w_{n-1} w_{n-2} = \frac{1}{2}.$$

Expressing $w_j = k_j w_1$ where k_j is real, we obtain a quadratic equation in terms of w_1 :

$$(2.13) \quad w_1^2(k_1^2 + k_2^2 \dots + k_{n-1}^2 + k_1 k_2 + k_1 k_3 \dots + k_{n-1} k_{n-2}) - (k_1 + k_2 + \dots + k_{n-1}) w_1 + \frac{1}{2} = 0.$$

To ensure real solutions for w_1 , a positive discriminant is required. However, the discriminant Δ is always negative:

$$(2.14) \quad \Delta = -(k_1^2 + k_2^2 + \dots + k_{n-1}^2).$$

Thus, w_1 cannot be a purely real time step and must to be complex to satisfy our second-order condition. Therefore, not only *can* we achieve higher order integrators through complex time stepping, but we *must* step into the complex plane to achieve higher order. $\square$

2.3.2. Higher order explicit integrators for general nonlinear non-autonomous differential equations

We have shown how to construct higher-order complex paths on which to time integrate linear ordinary differential equations and partial differential equations. However, paths derived for linear problems are not guaranteed to have the same order of accuracy for nonlinear problems. Indeed, for nonlinear and non-autonomous equations, the 3-step forward Euler paths result in only second-order accuracy. To achieve 3rd-order accuracy for nonlinear, non-autonomous equations, we generalize the previously described procedure.

A general nonlinear differential equation is given by $\dot{y} = f(t, y)$, $y(t_0) = y_0$. Not every real-valued, nonlinear differential equation can be extended into the complex plane. For the time-integral of a differential equation to be the same along real and complex paths, $f(t, y)$ needs to be analytic over the domain enclosed by the two paths. Furthermore, some differential equations (e.g. $\dot{y} = \max(1, y)$) are only defined over the real line, because $f(t, y)$ is only defined for real-valued arguments (e.g. $\max(x)$). For this reason, we consider only differential equations

with natural extensions in the complex plane. Let us also assume the function f is sufficiently analytic near the region of integration i.e it does not have any singularities close to the path of integration. At time $t = t_0 + \Delta t$, we have

$$(2.15) \quad y(t_0 + \Delta t) = y_0 + \Delta t f(t_0, y_0) + \frac{1}{2} \Delta t^2 \dot{f}(t_0, y_0) + \frac{1}{3!} \Delta t^3 \ddot{f}(t_0, y_0) + \dots$$

where the total derivative is

$$(2.16) \quad \dot{f}(t, y) = \frac{d}{dt} f(t, y) \Big|_{t=t_0, y=y_0} = f(t_0, y_0) \frac{\partial}{\partial y} f(t, y) \Big|_{t=t_0, y=y_0} + \frac{\partial}{\partial t} f(t, y) \Big|_{t=t_0, y=y_0}.$$

Expanding equation 2.15, we get

$$(2.17) \quad \begin{aligned} y(t_0 + \Delta t) = & y_0 + \Delta t f(t_0, y_0) + \frac{1}{2} \Delta t^2 \left(f(t, y) \frac{\partial}{\partial y} f(t, y) + \frac{\partial}{\partial t} f(t, y) \right) \Big|_{t=t_0, y=y_0} \\ & + \frac{1}{3!} \Delta t^3 \left(f^2(t, y) \frac{\partial^2}{\partial y^2} f(t, y) + \left(\frac{\partial}{\partial y} f(t, y) \right)^2 f(t, y) \right. \\ & \left. + 2f(t, y) \frac{\partial^2}{\partial y \partial t} f(t, y) + \frac{\partial}{\partial t} f(t, y) \frac{\partial}{\partial y} f(t, y) + \frac{\partial^2}{\partial t^2} f(t, y) \right) \Big|_{t=t_0, y=y_0} + O(\Delta t^4). \end{aligned}$$

Now, consider a complex time stepper that takes 3 complex time steps $w_1 \Delta t, w_2 \Delta t, w_3 \Delta t$ starting from y_0 to go to y_1, y_2, y_3 . These steps are given by:

$$(2.18) \quad y_0 = y(t_0)$$

$$(2.19) \quad y_1 = y_0 + w_1 \Delta t f(t_0, y_0)$$

$$(2.20) \quad y_2 = y_1 + w_2 \Delta t f(t_0 + w_1 \Delta t, y_1)$$

$$(2.21) \quad y_3 = y_2 + w_3 \Delta t f(t_0 + w_1 \Delta t + w_2 \Delta t, y_2).$$

One can expand $f(t_n, y_n)$ around $f(t_0, y_0)$ using the multivariate Taylor series. Doing so,

$$\begin{aligned}
 y_3 = & y_0 + \Delta t (w_1 + w_2 + w_3) f(t_0, y_0) \\
 & + \Delta t^2 (w_1 w_2 + w_1 w_3 + w_2 w_3) \left(f(t, y) \frac{\partial}{\partial y} f(t, y) + \frac{\partial}{\partial t} f(t, y) \right) \Big|_{t=t_0, y=y_0} \\
 & + \Delta t^3 \left(w_1 w_2 w_3 \left(f(t, y) \left(\frac{\partial}{\partial y} f(t, y) \right)^2 + \frac{\partial}{\partial t} f(t, y) \frac{\partial}{\partial y} f(t, y) + \frac{\partial}{\partial t} f(t, y) \frac{\partial}{\partial y} f(t, y) \right) \right. \\
 & + (w_1^2 w_2 + w_1^2 w_3 + 2w_1 w_2 w_3 + w_2^2 w_3) \left(\frac{1}{2} f^2(t, y) \frac{\partial^2}{\partial y^2} f(t, y) \right. \\
 (2.22) \quad & \left. \left. + \frac{1}{2} \frac{\partial^2}{\partial t^2} f(t, y) + f(t, y) \frac{\partial^2}{\partial y \partial t} f(t, y) \right) \right) \Big|_{t=t_0, y=y_0} + O(\Delta t^4).
 \end{aligned}$$

For this three step complex integrator to have 3rd order accuracy, the coefficients of the partial derivatives in 2.22 should match that of equation 2.17. Equating coefficients, we get the system

of nonlinear equations

$$\begin{aligned}
 w_1 + w_2 + w_3 &= 1 \\
 w_1 w_2 + w_1 w_3 + w_2 w_3 &= \frac{1}{2} \\
 w_1 w_2 w_3 &= \frac{1}{6} \\
 (2.23) \quad w_1^2 w_2 + w_1^2 w_3 + 2w_1 w_2 w_3 + w_2^2 w_3 &= \frac{1}{3}.
 \end{aligned}$$

The first three equations are the same as those derived for the linear system equation 2.7, and the 4th equation arises from the nonlinear terms in the multivariate expansion. Unfortunately, this system is overconstrained and does not have a solution, implying that we need more than three steps for 3rd order accuracy. However, for systems with real solutions, we can take further advantage of the degree of freedom in the imaginary part of the step and find many 3-step paths with 3rd order accuracy in the real-valued part. If we allow imaginary truncation error beyond $O(\Delta t)$, we can remove this imaginary error by taking only the real part of the numerical solution at the end of the 3-step sequence. Let y be the true solution and $\bar{y}$ be the approximate numerical solution. Then, allowing imaginary $O(\Delta t^3)$ truncation error, we have $y(t + \Delta t) = \bar{y}(t + \Delta t) + ik\Delta t^3 + O(\Delta t^4)$, where k is real. Taking the real part on both sides, $\mathbf{Re}(y(t + \Delta t)) = \mathbf{Re}(\bar{y}(t + \Delta t)) + O(\Delta t^4)$. If the true solution to our differential equation and its initial condition are real-valued at real time points, then $y(t + \Delta t) = \mathbf{Re}(y(t + \Delta t))$. This trick of enforcing purely imaginary error at $O(\Delta t^3)$ produces a system with fewer constraints on the coefficients.

The coefficients must satisfy the following relaxed conditions:

$$\begin{aligned}
 w_1 + w_2 + w_3 &= 1 \\
 w_1 w_2 + w_1 w_3 + w_2 w_3 &= \frac{1}{2} \\
 \mathbf{Re}(w_1 w_2 w_3) &= \frac{1}{6} \\
 (2.24) \quad \mathbf{Re}(w_1^2 w_2 + w_1^2 w_3 + 2w_1 w_2 w_3 + w_2^2 w_3) &= \frac{1}{3}.
 \end{aligned}$$

There are many 3rd-order paths satisfying these equations and more could be found by allowing $O(\Delta t^2)$ error to be purely imaginary. Of the 3rd order paths found to satisfy the linear order conditions (Fig. 2.4), the top and bottom (cyan and brown) paths also satisfy the 3rd order nonlinear order conditions given in 2.24. We have tested our complex time-stepping schemes on a variety of differential equations to show that they satisfy the order conditions (Fig.2.5 with details of equations in Table 2.1)

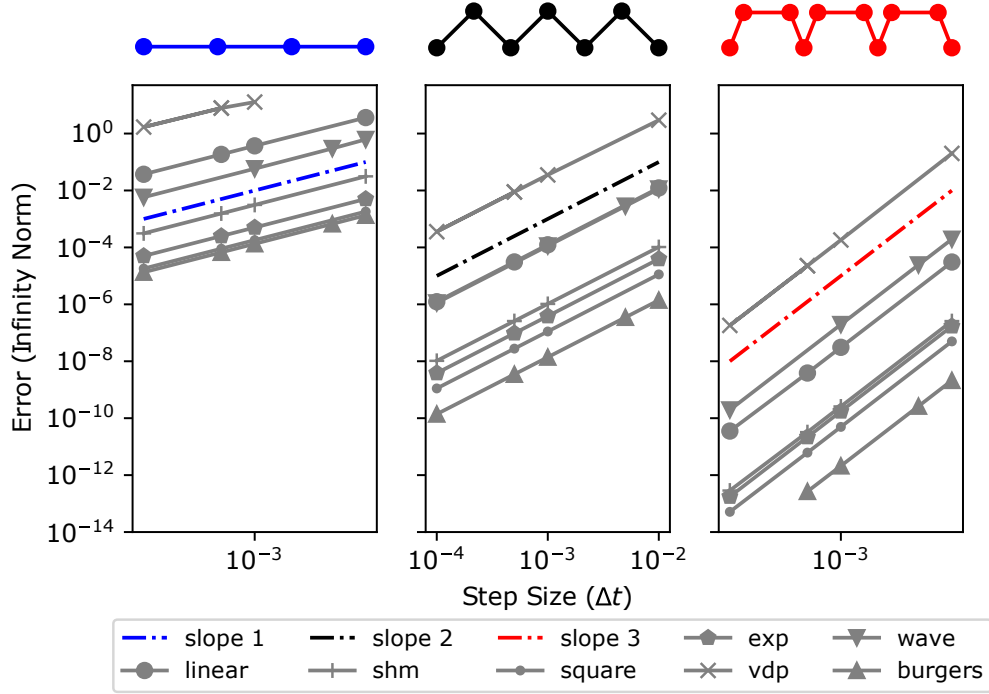


Figure 2.5. Convergence of various ODEs and PDEs (Sec. 2.3.2) for 1-step (forward Euler), 2-step, and 3-step complex time steps. The path/method used is displayed above the corresponding convergence plot. The orders of accuracy are as expected.

The error for all ODEs and PDEs apart from the Van Der Pol oscillator is calculated by comparing the numerical solution to the exact solution. For the Van Der Pol oscillator, the numerical solution using complex paths is compared to the numerical solution obtained by using the ‘RK44’ 4th order method from Numpy [24] with $\Delta t = 10^{-6}$.

2.4. Dealing with stiff systems: Implicit methods

Stiff systems require integrators that allow for large stable steps. Explicit methods usually have narrow stability regions. Hence, implicit methods are often preferred for stiff problems

Table 2.1. List of differential equations

1. Linear ODEs • The Dahlquist test problem ('linear') $\dot{y} = \lambda y, \quad y(0) = 1$ • Simple harmonic oscillator ('shm') $\ddot{y} = -y, \quad y(0) = 1, \quad y'(0) = 0$
2. Nonlinear ODEs • Nonlinear autonomous ode 1('square') $\dot{y} = -y^2, \quad y(0) = 1$ • Nonlinear autonomous ode 2 ('exp') $\dot{y} = -e^y, \quad y(0) = 1$ • Nonlinear sine('nlsin') $\dot{y} = 4y \sin(t)^3 \cos(t), \quad y(0) = 1$ • The Van Der Pol Oscillator (vdp'') $y_1 = y_2, \dot{y}_2 = \mu(1 - y_1^2)y_2 - y_1, \quad (y_1, y_2)(0) = (2, 0)$
3. Linear PDE : The advection equation ('wave') This is a linear PDE given by $(2.25) \quad u_t = cu_x, \quad x \in (0, 2\pi), \quad t \in (0, 1)$ This PDE is solved with periodic boundary conditions and the initial condition $u(x, t=0) = e^{7(x-\pi)^2}$. The exact solution is $u(x, t) = e^{7(x-t-\pi)^2}$. The parameter c is set to 1. The Fourier spectral method is used for the spatial differencing with 70 modes.
3. Nonlinear PDE : Viscous Burgers equation ('burgers') This is a nonlinear PDE given by $(2.26) \quad u_t + uu_x = \nu u_{xx}, \quad x \in (0, 2\pi), \quad t \in (0, 2)$ This PDE is solved with periodic boundary conditions and the initial condition $u(x, t=0) = 2\mu \sin x / (1.5 + \cos x)$. The exact solution is $u(x, t) = 2\mu e^{-\mu t} \sin x / (1.5 + e^{-\mu t} \cos x)$. The parameter ν is set to 0.1. The Fourier spectral method is used for the spatial differencing with 70 modes.

because they are often A-stable (their stability regions contain the entire left half plane) and L-stable (their solution goes to zero in a single step as the step size goes to infinity). In this section, we demonstrate how to extend the analysis we did for higher-order explicit complex Euler methods to implicit methods by constructing a 2-step complex path for the implicit midpoint method that is 4th-order accurate. First, we study linear ODEs (Sec. 2.4.1), and then we extend to general, nonlinear, non-autonomous differential equations (Sec. 2.4.2).

2.4.1. Higher order implicit integrators for linear differential equations

Consider the linear equation $\dot{y} = \lambda y$. Applying the implicit midpoint method to this equation produces the following difference equation,

$$(2.27) \quad y_{n+1} = y_n + \frac{\lambda \Delta t}{2} (y_n + y_{n+1})$$

which can also be written as

$$(2.28) \quad y_{n+1} = y_n \frac{1 + \lambda \Delta t / 2}{1 - \lambda \Delta t / 2}.$$

Now, suppose we take two complex steps $w_1 \Delta t$ and $w_2 \Delta t$ to go from y_n to y_{n+1} . Our numerical approximation at y_{n+1} is

$$(2.29) \quad y_{n+1} = y_n \frac{(1 + \lambda w_1 \Delta t / 2)(1 + \lambda w_2 \Delta t / 2)}{(1 - \lambda w_1 \Delta t / 2)(1 - \lambda w_2 \Delta t / 2)}.$$

If $y(t) = y_n$, the true solution at time $t + \Delta t$ would be $y_n e^{\lambda \Delta t}$. This solution can be represented to 4th order accuracy by the following rational function expansion of the exponential function, known as the (2,2) Pade approximant [25],

$$(2.30) \quad y_{t+\Delta t} = y_n \frac{1 + \lambda \Delta t/2 + \lambda^2 \Delta t^2/12}{1 - \lambda \Delta t/2 + \lambda^2 \Delta t^2/12} + O(\Delta t^5).$$

Setting the right hand sides of (2.29) and (2.30) equal to each other, gives us the 2-step path in the complex plane on which the implicit midpoint method has 4th order accuracy.

By solving

$$(2.31) \quad w_1 + w_2 = 1 \quad , \quad w_1 w_2 = \frac{1}{3}$$

we find that the two 2-step 4th order paths are $(\frac{1}{2} + i\frac{1}{2\sqrt{3}}, \frac{1}{2} - i\frac{1}{2\sqrt{3}})$ and $(\frac{1}{2} - i\frac{1}{2\sqrt{3}}, \frac{1}{2} + i\frac{1}{2\sqrt{3}})$.

Equations (2.29) and (2.30) highlight a clear connection between the optimal k-step complex path for the implicit midpoint method and the (k,k) Pade approximant.

In a similar way, we can derive the optimal k-step complex path for the Backward Euler method by setting it equal to the (0,k) Pade approximate. Taking 3 complex step with the Backward Euler method before returning to the real line, the numerical solution is given by

$$(2.32) \quad y_{n+1} = \frac{y_n}{(1 - w_1 \lambda \Delta t)(1 - w_2 \lambda \Delta t)(1 - w_3 \lambda \Delta t)}.$$

The exact solution can be approximated to 3rd order accuracy by the (0,3) Pade approximant.

$$(2.33) \quad y_{t+\Delta t} = \frac{y_n}{1 - \lambda \Delta t + \lambda^2 \Delta t^2/2 + \lambda^3 \Delta t^3/6} + O(\Delta t^4).$$

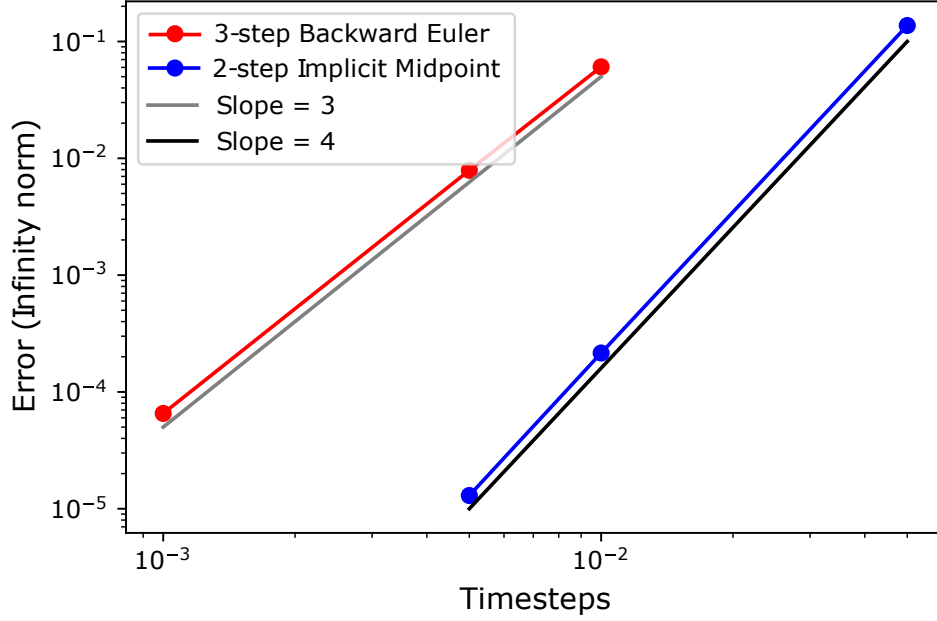


Figure 2.6. The 2-step implicit midpoint method has 4th order accuracy while the 3-step Backward Euler has 3rd order accuracy for the heat equation.

This results in the same equations and paths as derived in Subsection 2.3.1 for the forward Euler method. We test out our newfound implicit paths on the heat equation with Dirichlet boundary conditions.

$$(2.34) \quad u_t = u_{xx}, \quad x \in (0, 1)$$

We choose the initial condition $u(x, 0) = \sin(\pi x)$, which has the exact solution $u(x, t) = e^{-\pi^2 t} \sin(\pi x)$. The spatial discretization is performed using 4th-order finite differencing, periodic boundary conditions and a grid size of 10000 cells. As expected, the complex 3-step backward Euler method has 3rd-order accuracy and the 2-step complex implicit midpoint method has 4th-order accuracy (Fig. 2.6).

2.4.2. Higher order implicit integrators for general nonlinear non-autonomous differential equations

In the previous subsection, we found that the path $(\frac{1}{2} + i\frac{1}{2\sqrt{3}}, \frac{1}{2} - i\frac{1}{2\sqrt{3}})$ allows the implicit midpoint method to be 4th order accurate when solving linear differential equations. Does the expected 4th order accuracy of this 2-step implicit method hold for general nonlinear differential equations? Numerically, we confirm the method is accurate for real-valued systems, provided we allow for imaginary error as we did in Section 2.3.2 from the main text (Fig. 2.7). To demonstrate 4th order accuracy analytically, we analyze the midpoint method following similar reasoning to the analysis for explicit methods (Equations (2.18) - (2.21)) from the main text. Taking two implicit steps $w_1\Delta t$ and $w_2\Delta t$ from y_n at t_n to y_{n+1} at $t + \Delta t$ with intermediate solution y_m at $t + w_1\Delta t$,

$$(2.35) \quad y_n = y(t_n)$$

$$(2.36) \quad y_m = y_n + w_1\Delta t f(t_n + w_1\Delta t/2, (y_n + y_m)/2)$$

$$(2.37) \quad y_{n+1} = y_m + w_2\Delta t f(t_n + w_1\Delta t + w_2\Delta t/2, (y_m + y_{n+1})/2).$$

Unlike the previous analysis, equations (2.36) and (2.37) are implicit equations and cannot be directly Taylor expanded around $f(t, y_n)$. To calculate the desired w_1 and w_2 step sizes, we approximate the implicit equations with explicit versions that are at least 4th order accurate. The explicit approximations are found by recursively substituting the implicit equation into itself in an iterative process similar to Picard Iteration. For example, recursively applying the implicit equation $y = y_0 + \Delta t g(y)$ twice results in the following explicit equation, $y = y_0 + \Delta t g(y_0 + \Delta t g(y_0 + \Delta t g(y_0)))$ which is accurate to the implicit equation to the 3rd order.

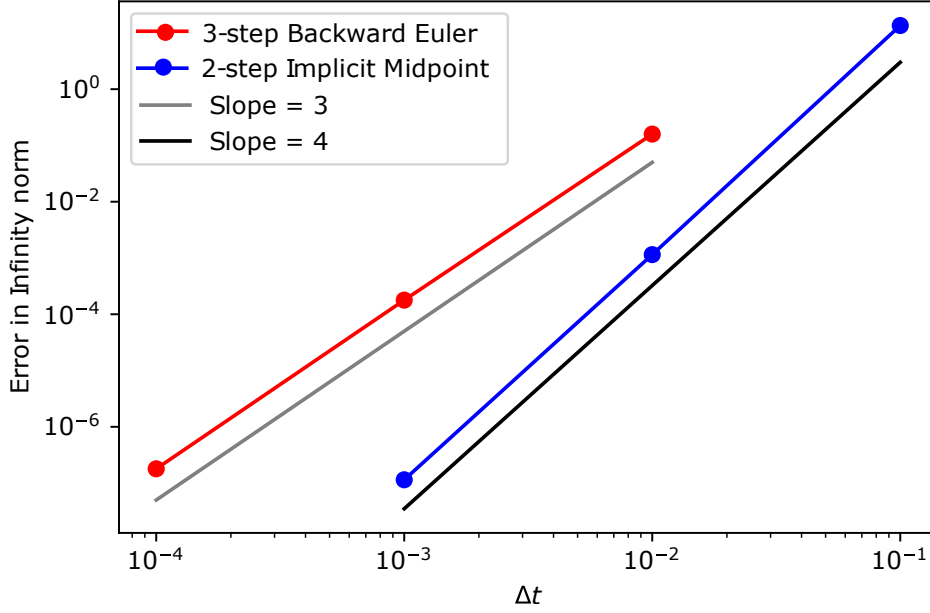


Figure 2.7. The 2-step implicit midpoint method has 4th order accuracy while the 3-step Backward Euler has 3rd order accuracy for the Van Der Pol oscillator.

In order to obtain the 4th order accurate explicit approximation of equations (2.36) and (2.37), they need to be recursively applied at least three times. Note that the explicit equations are 4th order accurate approximations of the original implicit difference equations and not the original differential equation. To show that the 2-step path is 4th order accurate for real-valued systems, we substitute $(w_1, w_2) = (\frac{1}{2} + i\frac{1}{2\sqrt{3}}, \frac{1}{2} - i\frac{1}{2\sqrt{3}})$ into the explicit approximation of equation (2.37) and obtain

$$\begin{aligned}
y_{n+1} = & y_0 + \Delta t f(t_n, y_n) + \frac{1}{2} \Delta t^2 \left(f(t, y) \frac{\partial}{\partial y} f(t, y) + \frac{\partial}{\partial t} f(t, y) \right) \Big|_{t=t_n, y=y_n} \\
& + \frac{1}{3!} \Delta t^3 \left(f^2(t, y) \frac{\partial^2}{\partial y^2} f(t, y) + \left(\frac{\partial}{\partial y} f(t, y) \right)^2 f(t, y) \right. \\
& \left. + 2f(t, y) \frac{\partial^2}{\partial y \partial t} f(t, y) + \frac{\partial}{\partial t} f(t, y) \frac{\partial}{\partial y} f(t, y) + \frac{\partial^2}{\partial t^2} f(t, y) \right) \Big|_{t=t_n, y=y_n} \\
& + \frac{1}{4!} \Delta t^4 \left((1 + 0.096225i) f^3(t, y) \frac{\partial^3}{\partial y^3} f(t, y) \right. \\
& + (4 - 0.288675i) f^2(t, y) \frac{\partial}{\partial y} f(t, y) \frac{\partial^2}{\partial y^2} f(t, y) + (3 + 0.288675i) f^2(t, y) \frac{\partial^3}{\partial y^2 \partial t} f(t, y) \\
& + 3f(t, y) \frac{\partial}{\partial t} f(t, y) \frac{\partial^2}{\partial y^2} f(t, y) + f(t, y) \left(\frac{\partial}{\partial y} f(t, y) \right)^3 \\
& + (5 - 0.57735i) f(t, y) \frac{\partial}{\partial y} f(t, y) \frac{\partial^2}{\partial y \partial t} f(t, y) + (3 + 0.288675i) f(t, y) \frac{\partial^3}{\partial y \partial t^2} f(t, y) \\
& + \frac{\partial}{\partial t} f(t, y) \left(\frac{\partial}{\partial y} f(t, y) \right)^2 + 3 \frac{\partial}{\partial t} f(t, y) \frac{\partial^2}{\partial y \partial t} f(t, y) + (1 - 0.096225i) \frac{\partial^3}{\partial t^3} f(t, y) \\
(2.38) \quad & \left. + (1 - 0.288675i) \frac{\partial^2}{\partial t^2} f(t, y) \frac{\partial}{\partial y} f(t, y) \right) \Big|_{t=t_n, y=y_n} + O(\Delta t^5).
\end{aligned}$$

Comparing Equation (2.38) to the first 5 terms of Taylor expansion in equation (2.17) from the main text, we see that the error at $O(\Delta t^4)$ is purely imaginary, which demonstrates we can get 4th order accuracy for real-valued systems by extracting the real component of the solution.

Now, that we have proved our two-step implicit midpoint method to be 4th order accurate for real-valued nonlinear systems, we test it on the Van Der Pol oscillator,

$$(2.39) \quad \dot{y} = x - \frac{x^3}{3} - \frac{\dot{x}}{\mu}$$

which can be rewritten as the following system

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= \mu(1 - x^2)y - x. \end{aligned} \tag{2.40}$$

We set $\mu = 10$ and the initial condition as $(x, y) = (2, 0)$. We achieve the expected 4rd order accuracy for the 2-step complex implicit method (Fig. 2.7).

2.5. Complex paths and coefficients for general Runge-Kutta methods

So far, we constructed paths in the complex plane that satisfy the order conditions for scalar non-autonomous differential equations. We also applied these paths to vector-valued non-autonomous equations including systems and partial differential equations, and demonstrated up to 3rd-order accuracy with the numerical results in subsection 2.3.2 from the main text. However, Butcher showed in [23] that the order conditions for scalar non-autonomous differential equations are the same as those for vector-valued differential equations only up to order 4. Thus, the approach used so far is insufficient to derive the correct order conditions for a fifth-order or higher complex path for vector-valued differential equations. In order to obtain the order conditions for general time-steppers for vector-valued equations, we need to utilize the theory of Butcher series (B-Series) and rooted trees [26, 27].

B-Series allow us to represent both the exact solutions to general differential equations and their numerical solutions created from a broad class of integrators, as a series expansion in Δt similar to the Taylor series analysis we have been doing so far. Runge-Kutta methods and methods created by taking complex time steps with Runge-Kutta methods fall into the class of integrators whose numerical solutions can be represented by B-Series. In this section, we

describe the process of deriving order conditions for complex time-steppers using B-Series and show how the software BSeries.jl [28] simplifies that process tremendously. Finally, as an example, we re-derive the order conditions necessary for the 3-step 3rd order complex Euler method using and show that it is the same as that obtained by our Taylor series analysis (Equations 2.23 from the main text).

Consider the first order system of differential equations given by

$$(2.41) \quad \dot{y} = f(y), \quad y(t_0) = y_0.$$

The exact solution at time $t = t_0 + \Delta t$ is given by

$$(2.42) \quad y(t_0 + \Delta t) = y_0 + \Delta t \dot{y}(t_0) + \frac{1}{2} \Delta t^2 \ddot{y}(t_0) + \frac{1}{3!} \Delta t^3 \dddot{y}(t_0) + \dots$$

Each term in the above expansion can be written as the combination of products of partial derivatives of f , known as elementary differentials. Using f' as short-hand for $\frac{\partial f}{\partial t}$, we can rewrite each term in the expansion in terms of their elementary differentials.

$$(2.43) \quad \begin{aligned} \dot{y} &= f \\ \ddot{y} &= f'(f) \\ \dddot{y} &= f''(f, f) + f'(f'(f)) \end{aligned}$$

The set of elementary differentials is isomorphic to the set of rooted trees and thus, each elementary differential has an associated rooted tree. We can rewrite equations 2.43 in terms of the associated trees.

$$\begin{aligned}
 \dot{y} &= F_f(\bullet) \\
 \ddot{y} &= F_f(\mathfrak{z}) \\
 (2.44) \quad \ddot{y} &= F_f(\mathfrak{y}) + F_f\left(\begin{smallmatrix} \mathfrak{z} \\ \mathfrak{z} \end{smallmatrix}\right)
 \end{aligned}$$

where $F_f(\tau)$ is the elementary differential associated with the rooted tree τ .

We can also write the exact solution at time $t = t_0 + \Delta t$ in terms of the associated rooted trees, creating a B-Series for the exact solution.

$$(2.45) \quad y(t_0 + \Delta t) = y_0 + \sum_{\tau \in T} \frac{\Delta t^{|\tau|}}{\sigma(\tau)\gamma(\tau)} F_f(\tau)(y)(t_0)$$

where $F_f(\tau)$ is the elementary differential associated with the rooted tree τ , T is the set of all rooted trees, $|\tau|$ is the number of nodes in the rooted tree τ , $\sigma(\tau)$ is the symmetry of the rooted tree τ and $\gamma(\tau)$ is the density of the rooted tree τ . The rooted trees, elementary differentials, symmetries and densities can all be defined recursively and a more elaborate discussion of them can be found in [29, 30, 28, 26, 27].

Numerical time-stepping methods that have an associated B-Series, such as Runge-Kutta methods, are called B-Series methods [26, 27]. For a Runge-Kutta method with elementary weights Φ , its solution at time $t = t_0 + \Delta t$ can be written in terms of its B-Series.

$$(2.46) \quad y(t_0 + \Delta t) = y_0 + \sum_{\tau \in T} \frac{\Delta t^{|\tau|}}{\sigma(\tau)} \Phi(\tau) F_f(\tau)(y)(t_0)$$

One can now derive the order conditions needed to be satisfied by the elementary weights by matching the elementary differentials in equation 2.46 with the elementary differentials in equation 2.45 up to the desired order.

The complex time steppers that we have so far mentioned also have associated B-Series. We can derive their B-Series using the composition of the B-Series associated with each individual Euler steps. Once we construct the B-Series, we can derive the order conditions needed to be satisfied by the complex steps by matching the B-Series to the B-Series of the exact solution up to the desired order. We can do this same process for any complex time integrator obtained by taking complex time steps with any Runge-Kutta method. We can also let all the coefficients of the Runge-Kutta method be complex valued and further increase our degrees of freedom.

Ketcheson and Ranocha recently released a package in Julia named BSeries.jl [28] that allows for the symbolic construction, composition and manipulation of B-Series. We now demonstrate how BSeries.jl can be used to construct B-Series for a general complex time stepper and obtain order conditions.

Runge-Kutta methods can be completely defined by their coefficients A, b and c . The following code constructs the B-Series for a Forward Euler method taking a time step of $w_1 \Delta t$ from its coefficients A, b and c . The coefficient b which is 1 for a time step of Δt , is modified to w_1 for the time step of $w_1 \Delta t$. An alternate way of obtaining the B-Series for the numerical solution after a time step of $w_1 \Delta t$ is to replace every instance of Δt with $w_1 \Delta t$ in the B-Series corresponding to a time step of Δt .

```

w_1 = symbols("w_1", real=true)
A = [0;]
A = A[:, :]
b = [w_1]
c = [0]

coefficients_w1 = bseries(A, b, c, 5)

latexify(coefficients_w1, cdot=false)

```

This gives us the following B-Series.

$$(2.47) \quad y(t + w_1 \Delta t) = y_0 + \Delta t w_1 F_f(\bullet)$$

Similarly, we can find B-Series for the numerical solution from the Forward Euler method after a time step of $w_2 \Delta t$ and a time step of $w_3 \Delta t$. We can then compose the B-Series together to get the B-Series for the numerical solution after taking three complex Euler time steps.

We can run the following code to find the B-Series for the Forward Euler method after two time steps of size $w_1 \Delta t$ and $w_2 \Delta t$

```

coeff_w1w2 = compose(coefficients_w1, coefficients_w2);
latexify(coeff_w1w2, cdot=false)

```

$$(2.48) \quad y(t + w_1 \Delta t + w_2 \Delta t) = y_0 + \Delta t (w_1 + w_2) F_f(\bullet) + \Delta t^2 w_1 w_2 F_f(\bullet) + \dots$$

Similarly, we can find the B-Series for the Forward Euler method after the time steps $w_1 \Delta t$, $w_2 \Delta t$ and $w_3 \Delta t$.

```
coeff_w1w2w3 = compose(coeff_w1w2,coefficients_w3);
latexify(coeff_w1w2w3, cdot=false)
```

$$\begin{aligned}
 y(t + (w_1 + w_2 + w_3)\Delta t) = & y_0 + \Delta t (w_1 + w_2 + w_3) F_f(\bullet) + \Delta t^2 (w_1 w_2 + w_3 (w_1 + w_2)) F_f(\mathfrak{z}) \\
 & + \Delta t^3 w_1 w_2 w_3 F_f(\mathfrak{z}) + \Delta t^3 \left(\frac{w_1^2 w_2}{2} + \frac{w_3 (w_1 + w_2)^2}{2} \right) F_f(\mathfrak{z}) \\
 & + \dots
 \end{aligned}
 \tag{2.49}$$

You can also find the B-Series of the exact solution with a single line.

```
latexify(ExactSolution(coeff_w1w2w3), cdot = false)
```

$$y(t + \Delta t) = y_0 + \Delta t F_f(\bullet) + \frac{\Delta t^2}{2} F_f(\mathfrak{z}) + \frac{\Delta t^3}{6} F_f(\mathfrak{z}) + \frac{\Delta t^3}{6} F_f(\mathfrak{z}) + \dots
 \tag{2.50}$$

Matching the coefficients in equations 2.50 and 2.48 gives us the conditions necessary to obtain 2nd order accuracy. Matching the coefficients in equations 2.50 and 2.49 gives us the conditions necessary to obtain 3rd order accuracy and we can see that the order conditions are exactly those we previously derived in equation 2.23 from the main text.

The theory of B-Series and the symbolic package BSeries.jl thus provides us with a systematic way to find the higher-order complex paths for general Runge-Kutta methods without much difficulty.

2.6. Higher order integrators tailored to specific problems

For specific problems, it is possible to derive complex time steppers that are uniquely accurate with fewer steps. For example, we can derive a 2-step complex Euler method with 3rd-order accuracy for the following nonlinear problem,

$$(2.51) \quad y' = -y^2 \quad y(t_0) = y_0.$$

The true solution at time $t_0 + \Delta t$ can be approximated to third-order accuracy using Eq (2.17) to obtain

$$(2.52) \quad y(t_0 + \Delta t) = \frac{y_0}{1 + y_0 \Delta t} = y_0 - y_0^2 \Delta t + y_0^3 \Delta t^2 - y_0^4 \Delta t^3 + O(\Delta t^4).$$

Note that one does not need the true solution to obtain Eq (2.52) as this expansion can be obtained from the right-hand side of the differential equation.

Now, we apply the following two-step complex method to integrate from y_0 at time $t = t_0$ to y_2 at time $t = t_0 + \Delta t$ using forward Euler steps.

$$(2.53) \quad \begin{aligned} y_0 &= y_0 \\ y_1 &= y_0 + w_1 \Delta t y_0^2 \\ y_2 &= y_1 + w_2 \Delta t y_1^2 \end{aligned}$$

where $w_1\Delta t, w_2\Delta t$ are complex time steps. Writing y_2 in terms of the initial condition, we get

$$(2.54) \quad y_2 = y_0 + (w_1 + w_2)y_0^2\Delta t + 2w_1w_2y_0^3\Delta t^2 + w_1^2w_2y_0^4\Delta t^3.$$

This two-step complex Euler method would have 3rd order accuracy if the coefficients in Eq. 2.52 and 2.54 match. However, there are no values w_1, w_2 that satisfy these conditions. So, we assume the case in which the true solution to (2.51) is real at real time points (valid if the initial condition is real). For this case, we can let the error be purely imaginary beyond order 1 and use the trick of taking the real part of the numerically calculated values every time we complete a complex path and return to the real line. This relaxed constraint produces the following a nonlinear system of equations that can be solved,

$$(2.55) \quad \begin{aligned} w_1 + w_2 &= 1 \\ \mathbf{Re}(w_1w_2) &= \frac{1}{2} \\ \mathbf{Re}(w_1^2w_2) &= 1. \end{aligned}$$

One of the many solutions to this system is

$$(2.56) \quad w_1 = 1 - \frac{1}{\sqrt{2}}i \quad w_2 = \frac{1}{\sqrt{2}}i.$$

Applying this new 2-step time stepper to the nonlinear problem, it indeed has 3rd-order accuracy (Fig 2.8). Similar 2-step paths are derived for $\dot{y} = f(t)$ and $\dot{y} = -e^y$ and they also show expected 3rd order accuracy in Fig. 2.8.

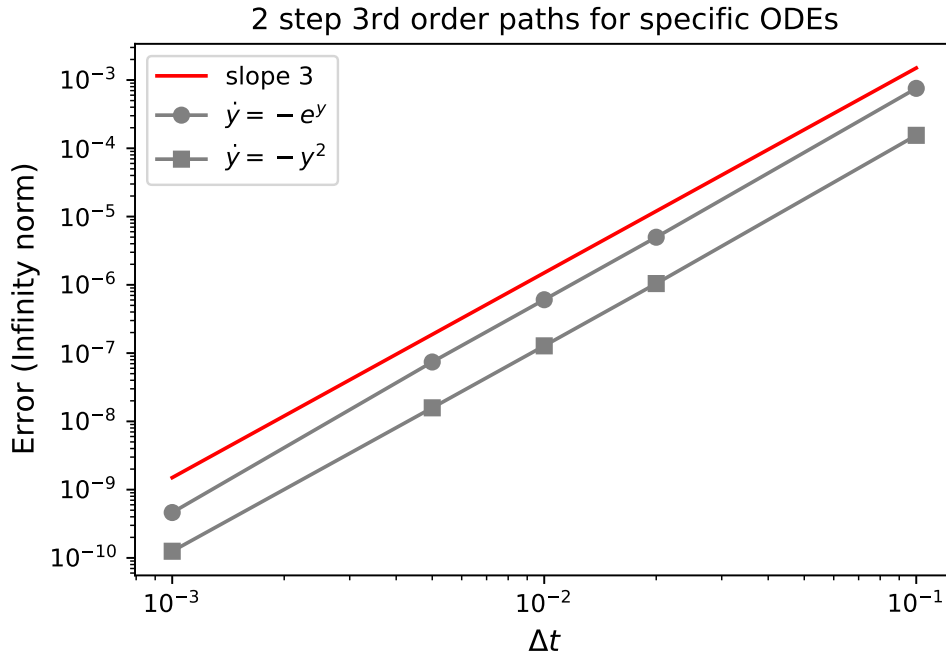


Figure 2.8. Two-step complex method Euler methods that have third-order accuracy for two specific problems.

In the process of constructing this integrator, we needed to plug in the right-hand side of the differential equation into Eq. 2.17 to derive the expansion. This may be an intensive symbolic task for more complicated problems. However, the right complex steps could be ‘learned’ for specific and difficult problems through approaches similar to [31] where a machine learning approach is used to obtain Runge-Kutta methods tailored to particular ODE families.

2.7. Computational cost of complex timestepping schemes

In this section, we discuss a major drawback of the complex time stepping schemes: the increased computational cost for performing complex operations.

Consider the forward Euler method applied to the differential equation $\dot{y} = f(t, y)$, yielding the iteration scheme $y_{i+1} = y_i + \Delta t f(t, y)$. If the differential equation is linear, the most expensive part of the scheme is the multiplication $\Delta t f(t, y) = \Delta t \lambda y$. Naively, multiplying two complex numbers costs roughly four times as much as multiplying two real numbers. If we split the complex difference equation into its real and imaginary parts, we get the following systems made up of purely real variables and we can clearly see that the number of multiplications needed at each step has increased by a factor of 4.

$$(2.57) \quad \begin{bmatrix} y_r \\ y_i \end{bmatrix}^{k+1} = \begin{bmatrix} y_r \\ y_i \end{bmatrix}^k + \begin{bmatrix} \lambda_r \Delta t_r & -\lambda_i \Delta t_i \\ \lambda_i \Delta t_i & \lambda_r \Delta t_r \end{bmatrix} \begin{bmatrix} y_r \\ y_i \end{bmatrix}^k$$

So, we expect the cost of solving linear differential equations using complex time integration could be as high as 4 times that of their real equivalents. Indeed, when solving the real-valued equation $\dot{y} = y$, a 2-step complex Euler integrator has running time about twice as long compared to the real-valued mid-point method, which has comparable order (Fig 2.9). However, when we solve the complex valued differential equation $\dot{y} = \left(\frac{1}{2} - \frac{1}{2}i\right)y$, the running times are closer for the real and complex integrators.

For nonlinear systems, the dominant cost comes from function evaluations. Functions can be designed to minimize the computational cost, but the real equivalent of any complex-valued function can always be calculated using fewer operations. So, for real valued differential equations, the complex time stepper will always be more expensive than an explicit Runge-Kutta

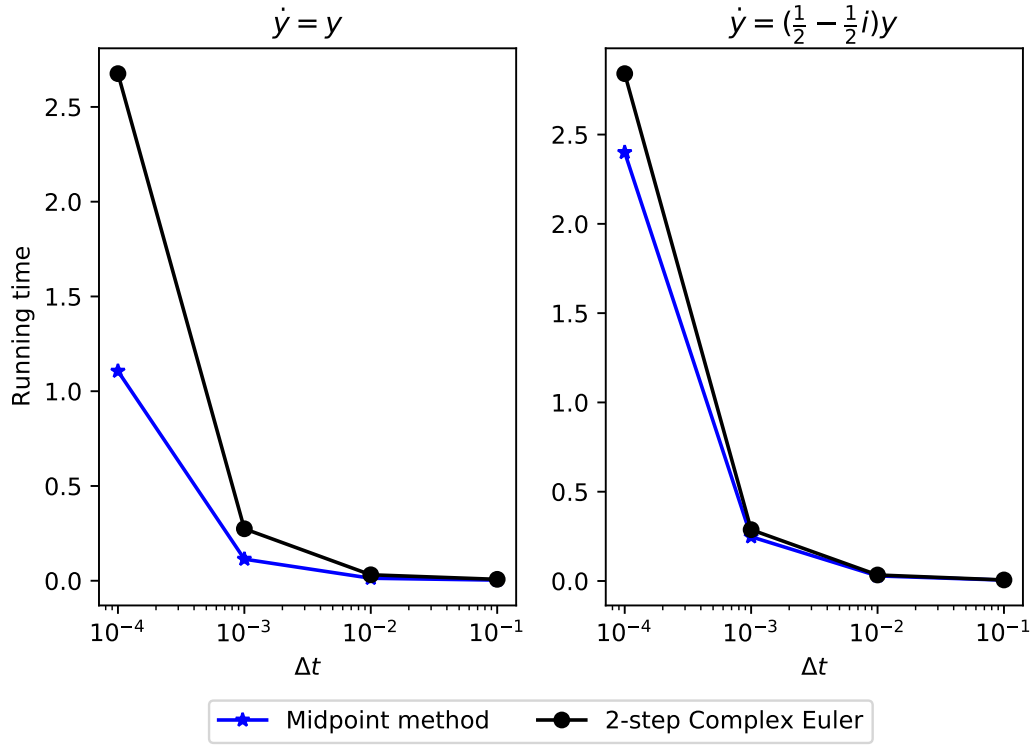


Figure 2.9. Running times for the 2-step complex Euler method and the real explicit midpoint method for the differential equation $\dot{y} = y$ and $\dot{y} = (\frac{1}{2} - \frac{1}{2}i)y$ solved from time 0 to 2 with various time steps (Δt). We see the additional computational cost of complex time integrators for solving real differential equations. The additional cost is significantly less when solving complex valued differential equations.

method with the same number of function evaluations. Therefore, complex time steppers are more likely to be advantageous for the numerical solution of complex differential equations. In the next two sections, we discuss unique advantages obtained by using complex integrators.

In Section 2.8, we demonstrate that complex time integrators require fewer function evaluations than the classical Runge-Kutta methods of the same order by circumventing the Runge-Kutta order barrier [23]. However, integrators which break the order barrier are only exactly accurate for real-valued systems. If the cost of additional function evaluations is more than

the cost of complex operations, the complex integrators become the less expensive option. In Section 2.9, we show how complex time integrators can expand stability regions for complex differential equations.

2.8. Breaking the order barrier for explicit Runge-Kutta methods for real-valued systems

2.8.1. What is the Runge-Kutta Order barrier?

An s -stage Runge-Kutta method of order p uses the free parameters in its s stages to satisfy the order conditions necessary to achieve p -th order where $s \geq p$. Table 2.2 lists the number of conditions needed to be satisfied at each order and the number of free parameters associated with the stages for a feasible Runge-Kutta method. Up to fourth order, one can construct explicit Runge-Kutta methods where $s = p$. For fifth order and beyond, the number of stages needs to be more than the order desired, i.e. $s > p$ [23]. The inability to construct Runge-Kutta methods with $s = p$ for order 5 or higher is known as the order barrier for Runge-Kutta methods [22]. In the following subsections, we show that it is possible to construct a 5-stage, 5th-order complex time integrator for real-valued differential equations.

Table 2.2. Conditions for various orders of accuracy and free parameters associated with Runge-Kutta methods with various stages

Order	1	2	3	4		5	6
Conditions	1	2	4	8		17	37
Stages	1	2	3	4	5	6	7
Parameters	1	3	6	10	15	21	28

2.8.2. Circumventing the Runge-Kutta order barrier for real-valued differential equations

A five stage Runge-Kutta method has 15 free parameters which are insufficient to satisfy the 17 order conditions for fifth order accuracy. Therefore, achieving fifth order requires six stages and hence six function evaluations. What if we allowed some of the Runge-Kutta coefficients in a 5-stage Runge Kutta step to be complex variables and constrained the error to be purely imaginary beyond order 1? For real-valued differential equations, we can take the real part of the numerical solution at the end of each time step, remove the imaginary error, and achieve fifth order accuracy. This increases the number of free parameters from 15 to between 17 and 29 in order to solve the 17 order conditions! For higher order, it also nearly doubles the free parameters when using Runge-Kutta methods for real-valued problems (Table 2.3).

Table 2.3. Increased free parameters when using Runge-Kutta methods with complex coefficients for real valued differential equations

Stages	1	2	3	4	5	6	7
Parameters with real coefficients	1	3	6	10	15	21	28
Parameters with complex coefficients	1	5	11	19	29	41	55

The following equations describe the complex Runge-Kutta-5 method for real-valued systems,

$$(2.58) \quad y_n = y_n$$

$$(2.59) \quad k_1 = f(t_n, y_n)$$

$$(2.60) \quad k_2 = f(t_n + c_2\Delta t, y_n + \Delta t a_{21}k_1)$$

$$(2.61) \quad k_3 = f(t_n + c_3\Delta t, y_n + \Delta t(a_{31}k_1 + a_{32}k_2))$$

$$(2.62) \quad k_4 = f(t_n + c_4\Delta t, y_n + \Delta t(a_{41}k_1 + a_{42}k_2 + a_{43}k_3))$$

$$(2.63) \quad k_5 = f(t_n + c_5\Delta t, y_n + \Delta t(a_{51}k_1 + a_{52}k_2 + a_{53}k_3 + a_{54}k_4))$$

$$(2.64) \quad y_{n+1} = \mathbf{Re}(y_n + b_1\Delta t k_1 + b_2\Delta t k_2 + b_3\Delta t k_3 + b_4\Delta t k_4 + b_5\Delta t k_5).$$

With the constraints $c_i = \sum_j a_{ij}$ and $\sum_i b_i = 1$, these equations describe 5th-order 5-stage Runge-Kutta methods for real-valued differential equations with complex-valued coefficients. The coefficients for one such method are listed in Table 2.4 and fifth order accuracy is satisfied for the differential equations in Fig 2.10.

Complex coefficients thus allow us to circumvent the Runge-Kutta order barrier and create a 5-stage fifth order method for real-valued differential equations.

2.8.3. Five stage, approximately fifth order complex Runge-Kutta methods

In the previous subsection, we saw that complex coefficients can be used to break the order barrier for real-valued differential equations. Even larger benefits may occur at higher orders, where s is much larger than p for existing Runge-Kutta methods with real coefficients. However,

Table 2.4. 5-stage Runge-Kutta method for real differential equations.

Parameters	Coefficients
a_{21}	0.4359927813681785+0.18820134969500546i
a_{31}	0.5984581874875472-0.6801332593573275i
a_{32}	0.09443736474929139+0.9536785997657906i
a_{41}	-0.5318588311678385+0.06199640671232824i
a_{42}	0.7090327838155295+0.17964710178664897i
a_{43}	0.7502336256211084+0.014717632306291894i
a_{51}	0.11597306658216743+0.19224587759603343i
a_{52}	-1.211955728302135+0.6697664876487938i
a_{53}	1.2481894547610273-1.0517638511367862i
a_{54}	1.1414853262483962+0.48897430346527126i
b_1	0.14051930946802596+0.047034144968353016i
b_2	0.5387707041084535+0.40236901283300025i
b_3	0.28423712936738976-0.23543136671378956i
b_4	0.06199686687229152-0.21051296375579337i
b_5	-0.02552400981616073-0.003458827331770331i

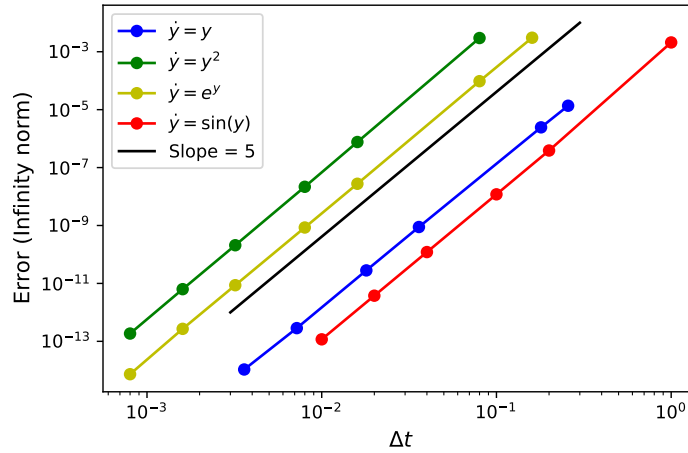


Figure 2.10. The 5-stage 5th order Runge-Kutta method with complex coefficients has 5th order accuracy for real-valued differential equations.

since we take the real part of the solution at the end of every complex path, this approach cannot be used for complex-valued differential equations. As the additional computational cost of complex time integration is negligible for complex-valued systems, it is desirable to create a

5-stage Runge Kutta method where both the real and imaginary parts of the error are as close to zero as possible, approaching order 5.

Complex solutions have the potential to satisfy the order conditions when there are no real solutions, as real solutions are a subset of complex solutions. Consequently, our goal was to find approximate complex solutions to the order conditions for fifth-order accuracy using five stages. We numerically searched for possible complex solutions using a combination of root-finding and minimization algorithms from Python's Scipy package [32] and Matlab's optimization toolbox [33]. We found complex coefficients that satisfy the order conditions to a total error of 2.1×10^{-11} . This means that for the differential equation $\dot{y} = f(t, y)$, the error behaves like $2.1 \times 10^{-11} \Delta t^4 \ddot{f} + O(\Delta t^5)$. Finding these approximate solutions is a highly non-convex optimization problem, and it is unclear if this is the global minima or if more accurate solutions could be found. We also found real solutions that satisfy the order conditions to a total error of 1.8×10^{-9} . The coefficients for both methods are listed in Table 2.5. We tested the complex five stage Runge-Kutta method on a few differential equations to obtain the convergence plot shown in Fig 2.11.

Here, we have shown the existence of five-stage fifth-order methods up to certain precision for both real and complex-valued differential equations. To our knowledge, this has not been shown before. We tested our approximately fifth order complex Runge-Kutta method on the linear Schrodinger equation $u_t = iu_{xx}, x \in [0, 2\pi], t \in [0, 10]$ with periodic boundary conditions. The spatial differentiation is done using the Fast Fourier Transform (FFT) with 100 modes. The exact solution is given by $u(x, t) = e^{i(x-t)} + e^{2i(x-2t)}$. In Table 2.6, we compare the computational time (averaged over 100 simulations) and error (L^1 grid function norm) from ten simulations of the Schrodinger equation by our approximately fifth-order complex five stage Runge-Kutta

Table 2.5. 5-stage Runge-Kutta method with complex and real coefficients that approximately satisfy the order conditions for fifth order accuracy.

Parameters	Complex Coefficients	Real Coefficients
a_{21}	$1.856587156265275e-07+1.5309457192095022e-07i$	$5.254899676102671e-07$
a_{31}	$355378.2918682022+744398.7276677284i$	-282414.4914234111
a_{32}	$-355377.7953985455-744399.1156280392i$	282415.0362283838
a_{41}	$10087.244864198223+2889.0099565661917i$	2300.659307961569
a_{42}	$-10086.873754015176-2889.502710365815i$	-2300.39437640888
a_{43}	$0.6299769187106239+0.4890885486059816i$	0.355521993237099
a_{51}	$16933.145111205715+9895.134727417835i$	-47221.11292217593
a_{52}	$-16932.764260866286-9895.630239734079i$	47221.41809024295
a_{53}	$0.6179505431419234+0.49914380654207474i$	-0.5826235568166092
a_{54}	$0.001199117424035724-0.003631490298717103i$	1.277455493703932
b_1	$-46564.847414291915+214551.5532581192i$	-51977.8184877715
b_2	$46565.24321098434-214551.70058574365i$	51978.11194824268
b_3	$0.20881428641527866+0.0021225559323642816i$	0.1667650923273279
b_4	$5.083449173489563-12.796017531317302i$	0.4161357937120537

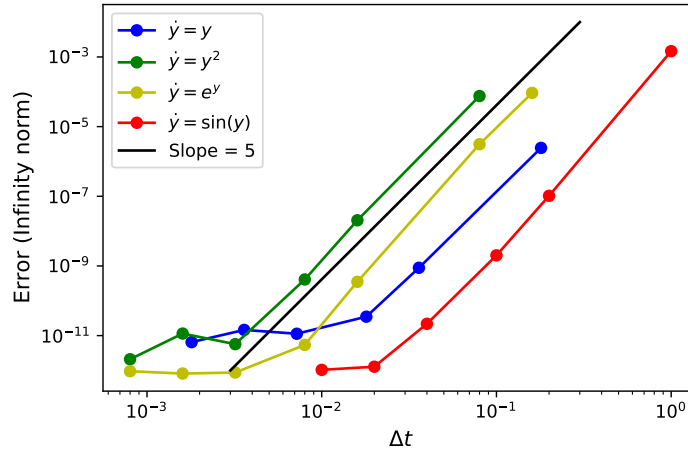


Figure 2.11. The 5-stage Runge-Kutta method that satisfied the order conditions for fifth-order accuracy to an error of 2.1×10^{-11}

method (CRK5) to a traditional 5th order six stage Runge-Kutta method, Fehlberg's method with Formula 2 (RKF) [34].

Although the complex 5-stage method is limited by its precision (Table 2.6), it is less computationally expensive when compared to a traditional fifth-order Runge-Kutta method. Methods

Table 2.6. Computational time (seconds) and error in the simulation of the Schrodinger equation by the complex Runge-Kutta method (CRK5) and the Fehlberg method (RKF)

Step size	Time (CRK5)	Time (RKF5)	Error (CRK5)	Error (RKF)
0.0002	6.41	7.52	2.44e-08	1.01e-13
0.0001	13.01	15.53	1.28e-08	1.24e-13
0.00005	25.91	30.29	9.99e-09	1.64e-13

like these may become favorable options when high order of accuracy is desired for convergence but limited precision is satisfactory. We hope this example motivates searches for complex solutions to order conditions, especially in the case of complex-valued differential equations. Order conditions appear in many contexts [35, 36] and complex coefficients may offer potential pathways to circumvent them. With the advent of scientific machine learning tools, there has been renewed activity in discovering numerically optimal specialized integrators in various contexts [37, 38, 39, 40, 31] where it might be beneficial explore complex solutions.

2.9. Expanded regions of stability with complex time-steps

In the previous sections, we designed paths in the complex plane to achieve higher order. In this section, we leverage complex paths in an alternative way and expand the time steppers's region of absolute stability. First, we discuss the linear stability regions of the complex integrators developed in Sec. 2.3. Next, we expand these regions by designing the complex steps to optimize a desired stability polynomial. Expanded stability regions have been studied in a similar manner in the context of Runge-Kutta methods where the order of accuracy is sacrificed to obtain a larger stability region [41, 42, 43, 44, 45, 46, 47]. The authors of these works derived new Runge-Kutta integrators (different integration method) with stability function/polynomial corresponding to the desired stability regions. Here, we maintain the integration method and

traverse optimized paths in the complex plane to obtain the desired stability regions (different integration domain). We illustrate this approach using the forward Euler method. We then demonstrate design of complex paths with larger regions for specific problems such as complex valued differential equations in quantum systems, where stability currently limits computational efficiency [48]. As a simple example, we show $2\times$ improvement in stability for a 2-step or 2-stage first order integrator for the linear and nonlinear Schrodinger equations.

Consider the linear equation $\dot{y} = \lambda y$ studied in Sec. 2.3.1. To analyze linear stability for most integrators, including those discussed here, we convert the linear equation into the difference equation $y(t + \Delta t) = \Phi(\lambda \Delta t)y(t)$ where $\Phi(z)$ is the stability function of the integrator [49]. The region $z \in \mathbb{C}$ where $|\Phi(z)| \leq 1$ is the region of absolute stability. For a complex n -step n th order method, the stability function is given by the first $n + 1$ terms in the Taylor series of $e^{\Delta t}$. For the 3-step 3rd order method, the stability function is

$$(2.65) \quad \Phi(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{3!}.$$

The regions of absolute stability for the 1-step 1st order, 2-step 2nd order and 3-step 3rd order methods (Fig 2.12) are the same as those for the n -stage Runge-Kutta because they have the same stability functions [49]. Note that Δt in all these methods refers to the step size separating the beginning and end points of the complex path and hence a real time step. Since the 1-step, 2-step and 3-step methods require a different number of steps and hence a different number of function evaluations to take a net step of size Δt , a standard metric for comparison among methods is the effective step size $\Delta t_{eff} = \Delta t/n$ where n is the number of steps in the complex integrator case or the number of stages in the case of Runge Kutta methods [45].

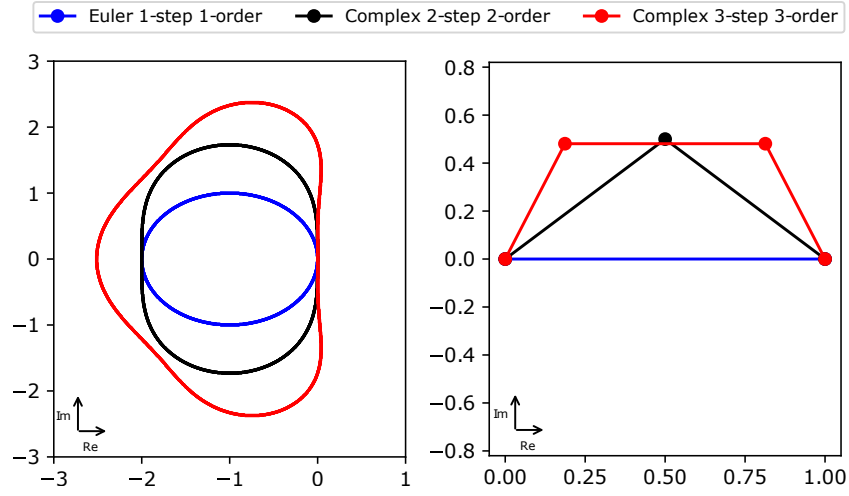


Figure 2.12. The 1-step first order, 2-step 2nd order and 3-step 3rd order complex Euler methods share the same stability regions as the 1,2 and 3 stage Runge-Kutta methods (left). Corresponding complex time-stepping paths are given on right.

We would like to expand the regions in Fig 2.12 by optimizing the stability function. If we relax our requirement for a 3-step integrator to be 3rd order accurate, and require only 2nd order accuracy, its stability function would be given by

$$(2.66) \quad \Phi(z) = 1 + z + \frac{z^2}{2} + kz^3.$$

The coefficient k in the above expression is the same as the coefficient in front of the $\lambda^3 \Delta t^3$ in equation (2.5): $w_1 w_2 w_3$. We can choose the complex steps w_1, w_2 and w_3 such that they satisfy the first and second order conditions ($w_1 + w_2 + w_3 = 1, w_1 w_2 + w_2 w_3 + w_1 w_3 = 1/2$) and set $w_1 w_2 w_3$ equal to our desired value of k . The optimal k value is the one that would allow us to take the largest stable time step. For problems with purely real and imaginary eigenvalues,

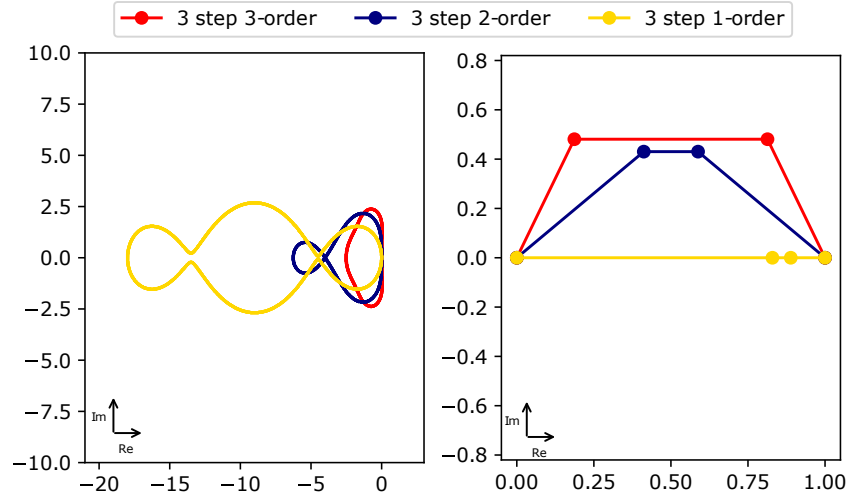


Figure 2.13. By sacrificing accuracy, one can expand the stability region (left) using complex time steps, as demonstrated by these optimized 3-step complex Euler paths (right) with 3rd, 2nd, and 1st order accuracy.

the optimal k values and the associated optimal stability functions/polynomials have been long-studied [41, 44, 43]. In the past decade, Ketcheson and co-authors [46, 47, 45] have explored optimal stability functions in the context of $\dot{y} = Ly$ where L is a matrix with arbitrary spectra. In [45], Ketcheson and Ahmadi describe an optimization algorithm to obtain the optimal stability polynomial that allows the maximal time step depending on the spectra of L . This algorithm (RKOpt [50]) is publicly available through Nodapy [24].

By choosing the value k that results in the largest permissible time step along the negative real eigenvalue line, we achieve a 3-step 2nd order method (blue) with expanded stability along the negative real axis compared to the 3-step 3rd order method (red Fig 2.13). Requiring only 1st order accuracy for the 3-step method can nearly triple the size of the optimal stability region along the negative real axis (gold) and surprisingly requires only real time-steps.

To our knowledge, our work is the first to use complex time steps to create stability polynomials, enabling us to expand the stability region. Optimal stability polynomials with complex

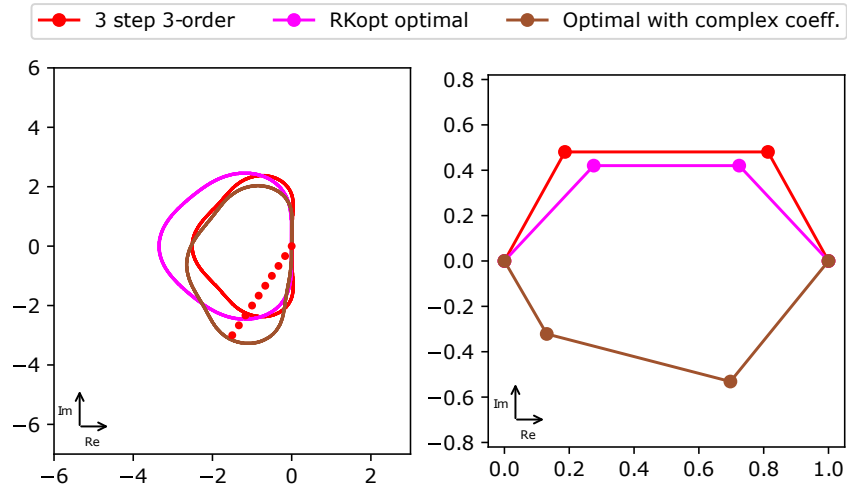


Figure 2.14. Complex coefficients in optimal stability polynomials often result in unsymmetrical stability regions (left). So, while the 3-step 2nd order complex integrator allows larger time steps along the eigenvalues (red dots, left), the 3-stage 2nd order real integrator (RKopt) is still optimal if one has to take timesteps along both the eigenvalues and its conjugate (as is the case for a real system).

coefficients may have larger stability regions than those with purely real coefficients. Through numerical experimentation, we have found that complex-valued coefficients (in the stability polynomial) often result in stability regions that are not symmetric about the real axis (Fig 2.14). Asymmetric stability regions will likely be sub-optimal for real-valued problems, since the eigenvalues of real-valued problems appear as complex conjugate pair symmetric about the real axis and asymmetric regions cannot capture both pairs. This leads us to hypothesize that the optimal stability polynomials for real-valued problems likely have purely real coefficients.

However, for complex-valued problems which have asymmetric eigenvalue spectra, complex coefficients produce larger stability regions in the relevant direction. For complex-valued problems in fields like quantum mechanics, complex time steps require no additional cost and

can often reduce computational effort by expanding stability regions to allow for larger time-steps. For example, the complex-valued Kohn-Sham equations which describe ultra-fast electron dynamics are known to require extremely small time steps for stability when using standard explicit Runge-Kutta methods [51]. In the field of complex-valued quantum systems, there is an ongoing and active search for efficient time integration techniques with large stable time steps [52, 53, 48]. Kononov *et al.* [48] review a number of numerical methods for the simulation of electron dynamics using the complex-valued Kohn-Sham equations and, given the expense of implicit methods, they express a specific need for explicit methods with expanded stability regions.

To demonstrate the expanded stability benefits of using complex coefficients, we consider a classical complex-valued problem: the linear Schrodinger equation, which has eigenvalues along the negative imaginary axis. We first show that for systems where all the eigenvalues lie either on the positive or the negative imaginary axis, the 2-stage first-order integrator that allows for the maximal stable time step is complex-valued. We then demonstrate the reduction in computational effort with the nonlinear Schrodinger equation as a numerical example.

Theorem 2. *For systems with all the eigenvalues either on the positive or the negative imaginary axis, the optimal stability polynomial for a 2-stage first-order integrator is complex-valued.*

PROOF. Let us consider a system with all the eigenvalues on the positive imaginary axis. The stability polynomial for a general 2-stage/2-step first-order integrator is given by

$$(2.67) \quad \Phi(z) = 1 + z + kz^2.$$

where the k value needs to be optimized so that the region enclosed by $|\Phi(z)| \leq 1$ is maximal on the negative imaginary axis, enabling the maximal stable time step.

For the stability polynomial with purely real coefficients, the stability condition translates to

$$(2.68) \quad k^2 y^4 + (1 - 2k)y^2 \leq 0$$

where $y = |\lambda| \Delta t$. For the integrator to be stable, we require $y^2 \in [0, \frac{2k-1}{k^2}]$. The maximal value of y is 1, which is achieved for $k = 1$. Hence, for real-valued integrators, the optimal stability polynomial for this system is $1 + z + z^2$ and the maximum stable step is $\frac{1}{|\lambda_{max}|}$.

For the stability polynomial with complex coefficients, the stability condition translates to

$$(2.69) \quad (k_r^2 + k_i^2)y^4 - 2k_i y^3 + (1 - 2k_r)y^2 \leq 0$$

where k_r and k_i are real and imaginary components of k and $y = |\lambda| \Delta t$. We can numerically show that the optimal value of k is $(\frac{1}{2} + \frac{1}{2}i)$ and $y \in [0, 2]$ satisfies the stability condition for the optimal value of k . Hence, for general integrators, the optimal stability polynomial for this system is $1 + z + (\frac{1}{2} + \frac{1}{2}i)z^2$ and the maximum stable step is $\frac{2}{|\lambda_{max}|}$ which is twice as large as its real equivalent.

Similarly, if all the eigenvalues lie on the negative imaginary axis like in the case of the linear Schrodinger equation, the optimal stability polynomial is $1 + z + (\frac{1}{2} - \frac{1}{2}i)z^2$ and the maximum stable step is $\frac{2}{|\lambda_{max}|}$. The stability region using complex coefficients covers double the length on the negative imaginary axis as the one obtained using purely real coefficients, enabling stable time steps twice as large (Fig 2.15). □

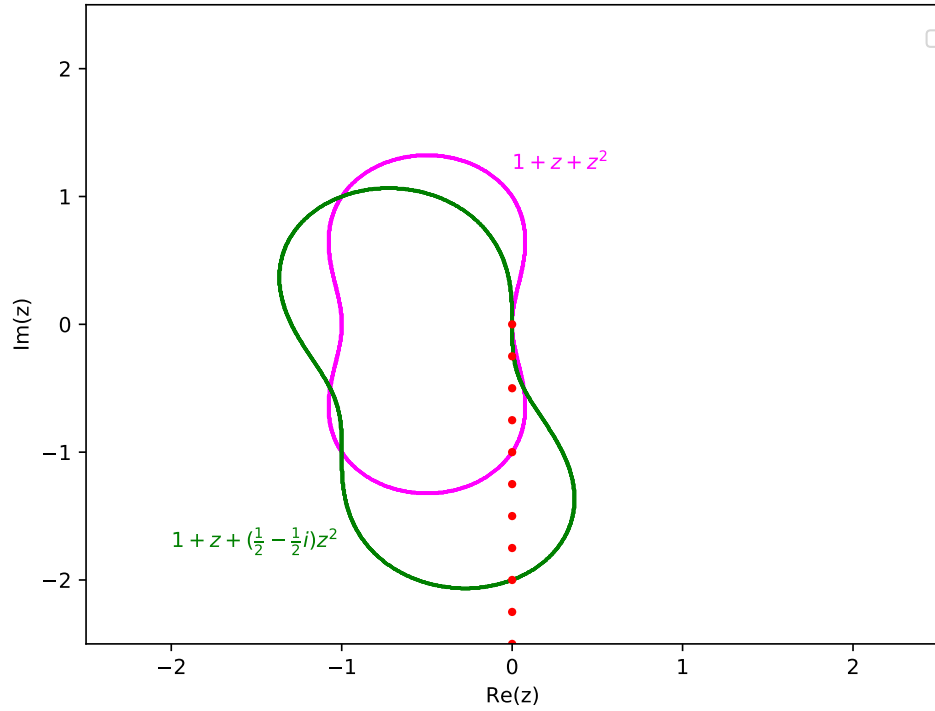


Figure 2.15. Constructing optimal stability polynomials using complex coefficients enables us to get larger stability regions than those using purely real coefficients. The optimal 2-stage/step 1st order stability polynomials for the linear Schrodinger equation using purely real and complex coefficients are shown here in magenta and green respectively. The possible timesteps for the linear Schrodinger equation along its eigenvalues are shown in red.

2.9.1. Numerical example: The nonlinear Schrodinger equation

In order to demonstrate the practical efficiency of using complex time integrators with expanded stability regions, we consider the nonlinear Schrodinger equation.

$$(2.70) \quad iu_t + \frac{1}{2}u_{xx} + |u|^2u = 0, \quad x \in [-2\pi, 4\pi], \quad t \in [0, 6]$$

The initial condition ($u_0 = u(x, 0)$) is chosen to admit the following soliton solution.

$$(2.71) \quad u(x, t) = \sqrt{2} \operatorname{sech}(\sqrt{2}(x - t)) e^{i(x + 0.5t)}$$

The spatial differentiation is done using the Fast Fourier Transform (FFT) with 100 modes. Periodic boundary conditions are imposed on the domain.

We simulate the nonlinear Schrodinger equation with the optimally stable real and complex integrators. Figure 2.16 shows the error and computational time associated with various time steps. The optimal complex integrator allows for double the largest stable step size with nearly similar error when compared to the optimal real integrator.

We show through this example that complex steps/coefficients in a time integrator can expand stability regions beyond real equivalents and allow for larger time steps. Even larger regions may be possible using methods with more stages/sub-steps, but the cost trade-off between increased evaluations and increased step size would need to be evaluated on a problem specific basis.

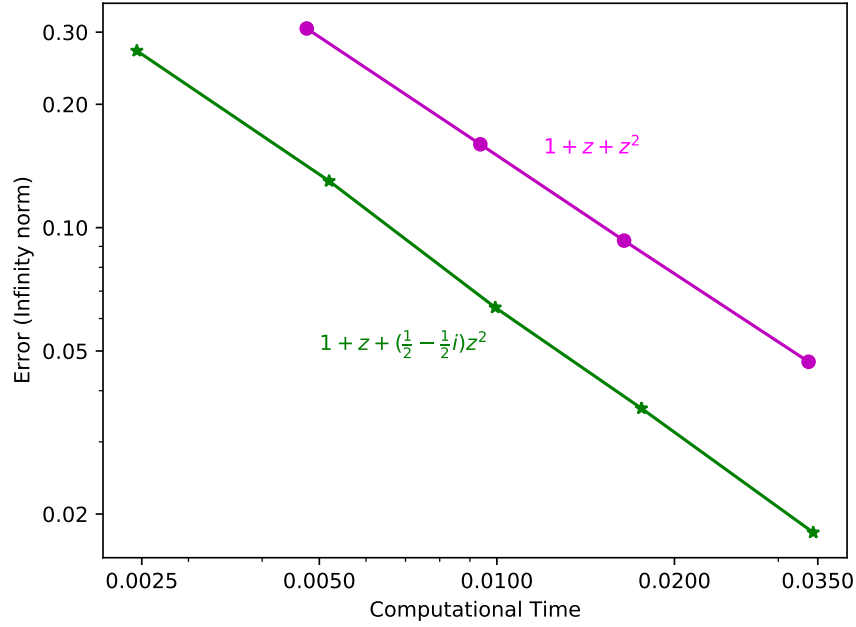


Figure 2.16. The nonlinear Schrodinger equation is simulated using the optimally stable 2-stage first-order real and complex integrators. The error and computational time (averaged across 1000 runs) is displayed. The complex integrator allows for twice as large of a time step i.e half as much computational effort for slightly better error.

CHAPTER 3

Mathematical modeling of the reaction environment in an electrocatalytic reaction environment.

3.1. Introduction

Electrochemical reactions can be a clean way of storing and utilizing energy and look like one of our best hopes for processes like carbon capture (extracting the carbon dioxide from the atmosphere) and desalination (separating salt from seawater for clean drinking water) [54, 55]. The key to these reactions being feasible is the presence of an efficient catalyst. A good catalyst in the right environmental conditions makes these reactions efficient by lowering the formation energy of the right intermediates, nudging the reaction along the desired pathways [56, 57]. The research done in this chapter and the next focuses on using mathematical methods and scientific computing tools to simulate the microscopic region where the key catalytic reactions occur with the help of our experimental collaborators, Dr. Linsey Seitz and Dr. Brianna Ruggiero from the Chemical Engineering department at Northwestern.

Catalysis is quite an enigmatic process because the key reactions happen in a nano-scale domain where observation and measurement by dynamic probing over time are difficult. There is a medley of scenarios that are hypothesized to occur in that domain. Different reaction intermediates that could be formed, various reaction pathways that could be chosen ([57]), and even the binding site/location/geometry could affect how efficient a catalyst is. This determines the rates of the reactions and the pathways taken. Furthermore, the governing laws at such a

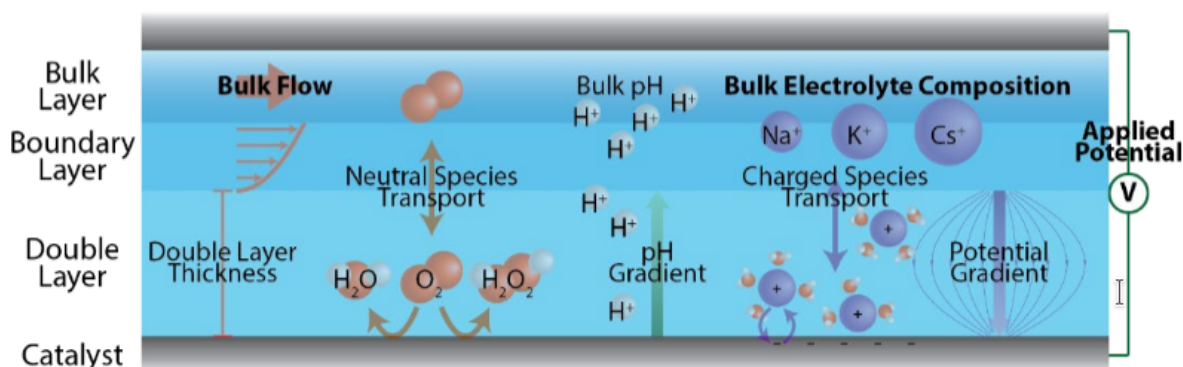


Figure 3.1. Mathematical modeling and effective numerical simulation would allow us to have a clearer understanding of the type of environment that could lead to more efficient reactions. Figure reproduced with permission from Dr. Linsey Seitz and Dr. Brianna Ruggiero. A version of this figure was previously published in [1, 2]. Reprinted (adapted) with permission from [2]. ©2023, American Chemical Society.

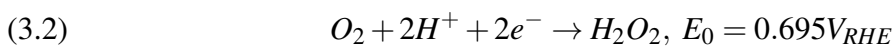
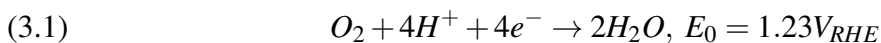
small scale are still quite unknown because many common physical assumptions break down. A big challenge is the formation of an electric double layer within the nano-scale interface for which the theory is not yet complete. Bohra and co-authors remark in [58] that "the characteristic length and time scales of the formation of the electric double layer are several orders of magnitude smaller than the length and time scales relevant for the homogeneous reaction and diffusion processes".

In order to combat the multiple spatial and time scales within the electrocatalytic reactor, both our numerical methods and our assumptions about the underlying physics and the representative equations need to be simultaneously improved upon. In this chapter, we highlight the derivation of our mathematical model, the modified Poisson-Boltzmann equation with flux which helps us resolve the time scale separation. In the next chapter, we discuss our associated numerical method which is designed to yield accurate simulations over disparate spatial scales.

3.2. What are we modeling?

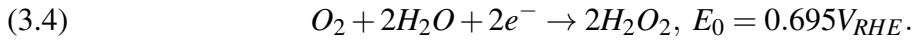
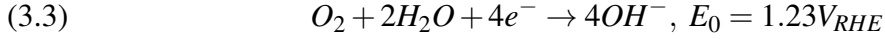
The key reaction of interest in this thesis is the Oxygen Reduction Reaction (ORR) where Oxygen is reduced either to water (H_2O) or hydrogen peroxide (H_2O_2). The mathematical work done in this thesis benefits from the ORR voltammetry data and the experimental findings of our collaborators Dr. Brianna Ruggierro and Dr. Linsey Seitz in Chemical and Biological Engineering Department at Northwestern. There is a substantial global demand for H_2O_2 , a valuable environment oxidizing agent, due to its versatile uses in the textile industry, chemical synthesis, metallurgy, wastewater treatment, bleaching amongst other applications [59]. The standard industrial method for large scale H_2O_2 production is the anthroquinone oxidation process [60] which requires large-scale infrastructure and produces a large amount of waste [59]. An efficient electrochemical pathway for producing H_2O_2 would serve as a clean, sustainable and decentralized alternative that also alleviate the logistical difficulties and hazards in storing and transporting H_2O_2 [61, 62, 63, 64]. The aim of our mathematical work and associated experimental collaboration is to understand the ideal reaction environment for a catalyst to facilitate H_2O_2 production.

The ORR can proceed via the 4 electron to produce H_2O or the 2 electron pathway to produce H_2O_2 . A review of this reaction system can be found in our associated experimental work [1]. In acidic solutions, when the potential difference is less than E_0 (the standard thermodynamic equilibrium potential), the following reactions are feasible.

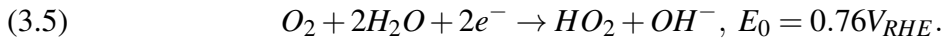


V_{RHE} represents the potential relative to the reversible hydrogen electrode (RHE). V_{RHE} is related to the standard hydrogen electrode (V_{SHE}) via $V_{RHE} = V_{SHE} + 0.0592 \times \text{pH}$ at 300 K.

In alkaline solutions, the feasible reactions are



Furthermore, for $\text{pH} > 11.63$, the H_2O_2 reaction becomes



Our goal is to effectively model and understand the electrocatalytic reaction that takes place within the rotating ring disk electrode (RRDE) setup shown in Figure 3.2 . In the figure, we can see a disk surface and a ring surface which are both electrodes. The catalyst is deposited on the disk surface of the RRDE and only the disk electrode is involved in the Oxygen Reduction Reaction. The ring electrode is kept at a potential of 1.2 V vs RHE to quantify the amount of H_2O_2 produced in the ORR reaction at the disk electrode. The ring electrode does so by oxidising the H_2O_2 produced in the ORR and measuring the current at the ring. Note that all of the H_2O_2 produced in the ORR at the disk does not arrive at the ring. To account for this, the collection efficiency of the ring is measured and determined to be around 0.224.

One can then measure the selectivity of H_2O_2 at the disk, which is percentage of the total current measured at the disk that is due the H_2O_2 pathway in the ORR (as opposed to the H_2O pathway). If we use i_{ring} and i_{disk} to denote the currents at the ring and the disk, the formula for selectivity is obtained from [1] and given by

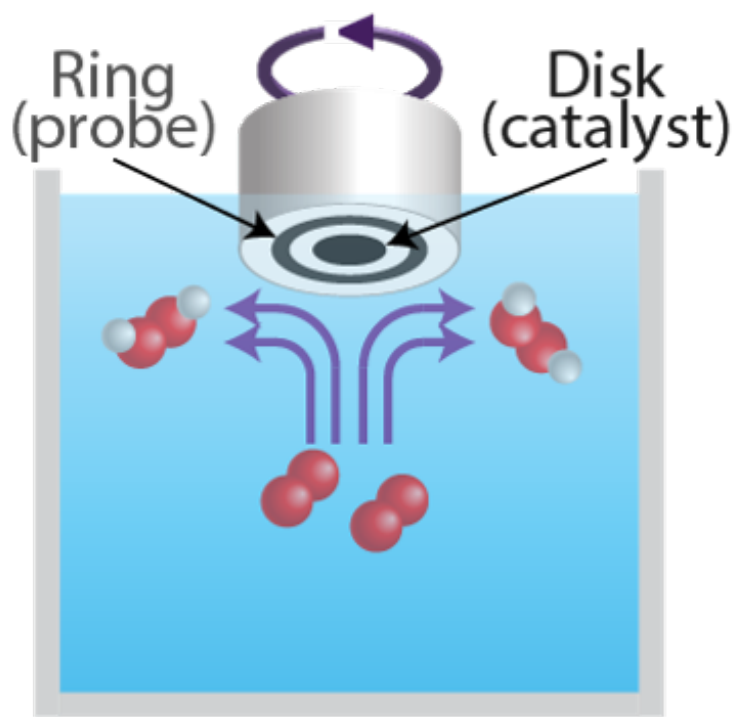


Figure 3.2. The Rotating Ring Disk Electrode (RRDE). Figure reproduced with permission from Dr. Linsey Seitz and Dr. Brianna Ruggiero. A version of this figure was previously published in [1, 2]. Reprinted (adapted) with permission from [2]. ©2023, American Chemical Society.

$$(3.6) \quad H_2O_2(\%) = 200 \times \frac{\frac{i_{ring}}{0.224}}{|i_{disk}| + \frac{i_{ring}}{0.224}}.$$

The RRDE thus allows us to run the ORR and measure the current involved in the H_2O_2 pathway. The rotating electrode creates a laminar flow of the species in the electrolyte to the catalyst surface. By varying the rotation rate of the electrode, the experimentalist can control the length of the diffusion layer (length from the catalyst surface to the region where the species are at their bulk concentrations) and the limiting current density. A pH sensing probe can also

be incorporated into the RRDE which can be used to quantify the changes in the local proton concentration [2]. Further details about our RRDE setup can be found in [1, 2] .

Our goal is to effectively model this experimental setting and simulate it in order to understand the optimal reaction environment for facilitating H_2O_2 production in the ORR.

3.3. A journey through key mathematical models in electrochemistry

This section explores the historical development of mathematical equations employed to model electrochemical processes. The mathematical and computational limitations of different models are emphasized, leading us to the model we have developed in Section 3.4.

3.3.1. Gouy-Chapman theory

3.3.1.1. The Poisson-Boltzmann (PB) equation. Modeling electrochemical processes requires a model for the electric double layer that forms near the electrode surface. Helmholtz[65] first introduced the concept of the electric double layer. Gouy [66] and Chapman [67] modified the Helmholtz model to describe the equilibrium concentrations of ionic species by their Boltzmann distributions. The Gouy-Chapman theory naturally leads to the Poisson-Boltzmann equation given below.

$$(3.7) \quad \nabla \cdot (\epsilon \nabla \Psi) = -F \sum z_k c_k$$

$$(3.8) \quad c_k = c_k^0 e^{-z_k F \Psi / RT}$$

where c_k is the concentration of species k across the electrolyte domain, c_k^0 is the reference concentration, z is the charge number, u is the mobility, ϵ is the permittivity of the medium, Ψ

is the potential across the electrolyte, F is Faraday's constant, R is the universal gas constant and T is the absolute temperature.

The Poisson-Boltzmann is considered the standard classical model for electric double layers, particularly in dilute solutions. It is often solved with Dirichlet boundary conditions to give the potential profile across the electrolyte. However, since the Poisson-Boltzmann equation models point charges with no volume, as the electrode potential increases, the concentration of counterions near the electrode can reach unphysically large values. Additionally, the counterions can get arbitrarily close to the electrode. Stern [68] modified the Gouy-Chapman model to include a Helmholtz layer of counterions at the electrode surface, now known as the Stern layer. The Inner Helmholtz Plane (IHP) and the Outer Helmholtz Plane (OHP) divide the Stern layer into two constitutive layers (Figure 3.3). Bard, Faulkner and White [69] describe the IHP as the “the locus of the electrical centers of specifically adsorbed ion” and the OHP as the “locus of the centers of the nearest solvated ion”. The Stern modification essentially prevents ions from getting arbitrarily close to the electrode. The region outside the Stern layer where the ions are less tightly packed, is called the diffuse layer. Most of the potential drop is assumed to occur in the diffuse layer (around 10 nm in length). The diffuse layer is not to be confused with the diffusion layer which is the domain in between the Stern layer and the bulk region which has a length on the order of μms . The length of the diffuse layer depends predominantly on the permittivity of the electrolyte and the concentrations of the various species. The length of the diffusion layer depends on the experimental setting. For example, in the RRDE setup, increasing the rotation rate drives the bulk concentration closer to the electrode, decreasing the length of the diffusion layer. The Gouy-Chapman-Stern model can be described by the following equations.

$$(3.9) \quad \nabla \cdot (\epsilon \nabla \Psi) = 0 \quad \text{inside the Stern layer}$$

$$(3.10) \quad \nabla \cdot (\epsilon \nabla \Psi) = -F \sum z_k c_k^0 e^{-z_k F \Psi / RT} \quad \text{inside the Diffuse layer}$$

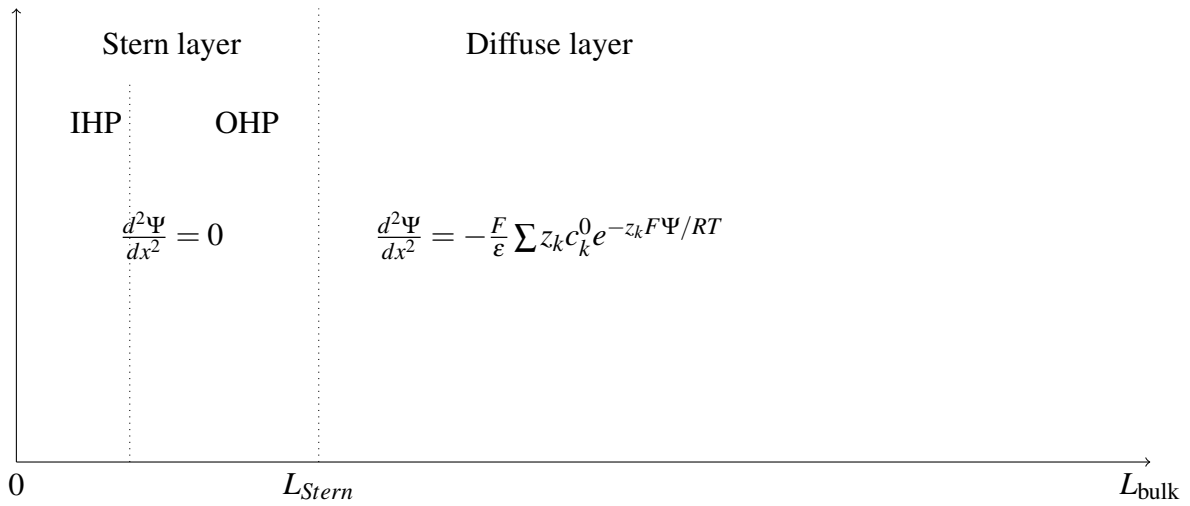


Figure 3.3. Gouy-Chapman Theory: The Poisson-Boltzmann equation.

3.3.1.2. Poisson-Nernst-Planck (PNP) equations. The Poisson-Boltzmann equation describes equilibrium concentrations of the potential and the concentration of various species across the electrolyte. For a time-dependent description of potential and concentration, the Gouy - Chapman theory can be combined with conservation laws to form the Poisson-Nernst-Planck equations [70] shown below.

$$(3.11) \quad \frac{\partial c_k}{\partial t} + \nabla \cdot \left(-D_k \nabla c_k - \frac{z_k D_k F}{RT} c_k \nabla \Psi \right) = 0$$

$$(3.12) \quad \nabla \cdot (\epsilon \nabla \Psi) = -F \sum z_k c_k$$

where D_k is the diffusivity of species k .

For species that react at the electrode, we have the following flux condition at the electrode boundary.

$$(3.13) \quad -D_k \frac{\partial c_k}{\partial x} - z_k u_k F c_k \frac{\partial \Psi}{\partial x} = -\frac{i}{n_k D_k F A}$$

where i is the current, A the area of electrode surface and n_k the number of ions of species k corresponding to the number of electrons involved in the reaction.

For non-reacting species, we have the following no-flow boundary condition.

$$(3.14) \quad -D_k \frac{\partial c_k}{\partial x} - z_k u_k F c_k \frac{\partial \Psi}{\partial x} = 0$$

Dirichlet or mixed boundary conditions are often used for the Poisson equation (Equation 3.12).

Like the Poisson-Boltzmann equation, the Poisson Nernst-Planck equations assume the atoms to be point charges with no volume. This leads to unphysical concentrations of counterions near the electrode surface at large applied voltages. These equations are unable to realistically capture the physics near the catalyst surface. Since the equation assumes the atoms are point masses, some ions can accumulate to unphysical concentrations in the boundary layer. Consider the simple case of a solution of NaCl simulated using the Poisson-Boltzmann equation (Equation 3.8) with no reactions at the boundaries. The bulk concentration of both Na^+

and Cl^- ions is $1\text{mol}/L$ and the length of the domain is 2nm . For this simple system, we see in Figure 3.4 that the concentration of Cl^- ions reach impossible values near the boundaries.

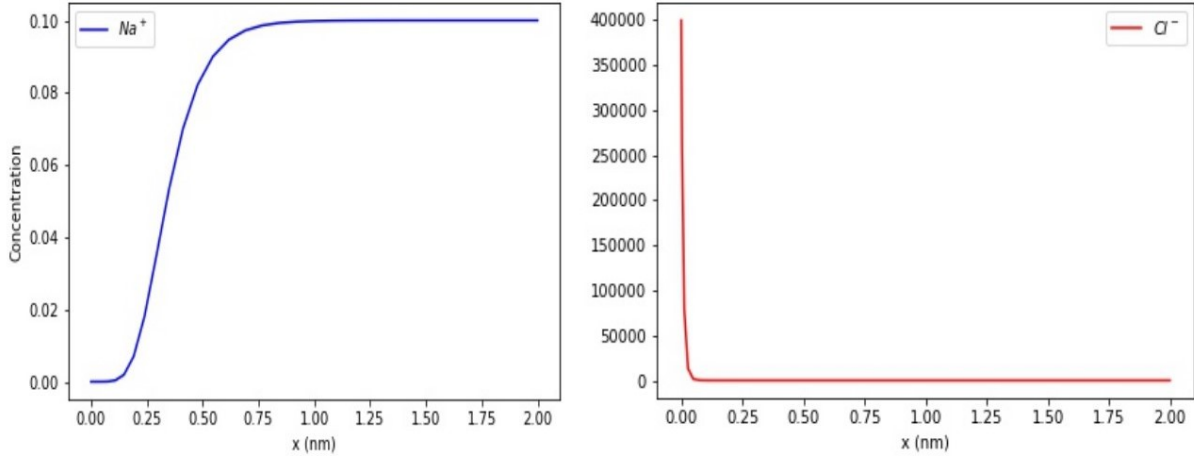


Figure 3.4. Equilibrium concentrations with two oppositely charged ions. We see the Cl^- ions reach unphysical values of concentration. This is because the mathematical model (the Poisson-Boltzmann equation) assumes atoms are points with zero volume.

The large buildup of counterions makes it numerically difficult to simulate both the Poisson-Boltzmann and Poisson-Nernst-Planck equations using classical methods like finite difference methods or finite volume methods especially when the length scale of the domain is large (on the order of micro-meters).

The Poisson-Nernst-Planck equations, in addition, are numerically unstable when the domain length is of the order of micro-meters, which means that an explicit time-stepping method only permits an extremely small and unfeasible time step. More expensive implicit methods are typically used in the simulation. In [58], the Python package Fenics [71] is used to solve the Poisson-Nernst-Planck equations using finite element methods and the Backward Euler method for time integration. The PNP equations have to be stabilized using a Petrov-Galerkin method in order to get a tractable stable time step. A robust and efficient numerical method that can

adapt to diverse input conditions is crucial for the effective operation of a virtual laboratory dedicated to understanding catalyst performance. Hence, one of the aims of our work is to build a simplistic numerical framework to obtain the steady-state potential and concentration profiles across the reactor domain.

3.3.2. Steric effects: Modified Poisson-Boltzmann equation and Modified Poisson-Nernst-Planck equations.

There have been many attempts to incorporate steric effects (volume effects) into the PB and PNP equations which we review in this section. One can find broad reviews of modified models for describing the electric double layer in [72, 73, 74]. Note that variants of the PB equation are used usually only for systems with no flow (blocking boundary conditions) at the boundaries [75, 76]. For systems with flow at the boundary, the variants of the PNP equations are used and the flow is incorporated through the boundary conditions for the time-dependent part of the PNP equations [58, 77].

Bikerman proposed one of the earliest Modified Poisson-Boltzmann models [78]. Since then, there have been many different approaches to deriving Modified Poisson-Boltzmann equations. In [79], the authors propose a Langmuir-type correction to account for the finite sizes of the ions. The authors later extend their model in [80, 81, 82] to include a distance of closest approach to the ions and consider non-constant permittivity to model ions as insulating spheres. [74] contains a delightful overview of the various reformulations of the Bikerman model which occurred independently from all around the world.

Multiple papers [74, 83, 84] discuss modifications to these models because the Langmuir-type correction/activity coefficient in the Bikerman model has too weak of an impact at low concentrations when using hydrated radii and results in a wrong limiting value [85, 86]. In [83], a modified Poisson-Boltzmann equation using the "Boublik–Mansoori–Carnahan–Starling–Leland" equation-of-state [87, 88] is introduced which allows it to describe ions of unequal sizes. Some other approaches to incorporate steric effects have included using a variational integral term [89] in the PNP equations, the use of multiple Stern layers for ions of different sizes [90] and the incorporation of a hydration potential [91].

The starting point for our work is the Modified Poisson-Boltzmann equation introduced by Borukhov and co-authors in [75] as it appears to naturally extend the Poisson-Boltzmann equation using statistical mechanics theory. The authors in [75] use the variation of free energy to derive a simplistic model incorporating steric effects into the steady-state concentration distribution of ions. The Modified Poisson-Boltzmann equation for a symmetric $z : z$ electrolyte (contains two oppositely charged species with equivalent valency and bulk concentrations) is shown in Equation 3.15.

$$(3.15) \quad \nabla \cdot (\epsilon \nabla \Psi) = -zF c_0 \frac{2 \sinh(\frac{zF\Psi}{RT})}{1 + 2\nu \sinh^2(\frac{zF\Psi}{2RT})}$$

where c_0 is the bulk concentration of both ions, $\nu = 2a^3 c_0$ is the total bulk volume fraction and a is effective size of the ion.

Resimulating the NaCl solution with no reaction at the boundaries using the Modified Poisson-Boltzmann equation (Equation 3.15), we can get more realistic concentration profiles as seen in Figure 3.5.

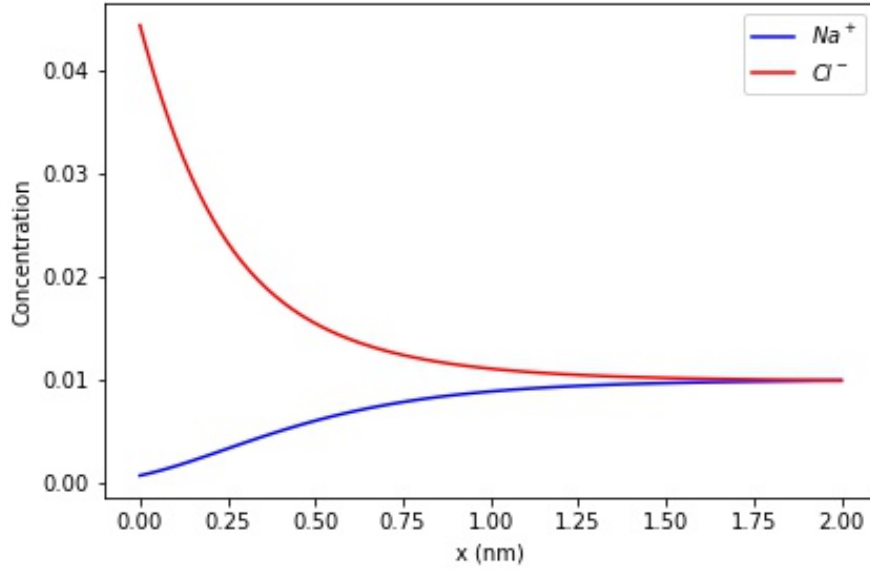


Figure 3.5. The modified equations derived allow us to produce more realistic simulations of the NaCl system without unphysical concentrations.

Kilic and co-authors later extended this Modified Poisson-Boltzmann equation in [76] to its time-dependent version, the modified Poisson Nernst-Planck equations. For the symmetric $z : z$ electrolyte, the equations describing the concentration of positive and negative species are given by Equation 3.16.

$$(3.16) \quad \frac{\partial c_{\pm}}{\partial t} + \nabla \cdot \left(-D \nabla c_{\pm} \mp \frac{zDF}{RT} c_{\pm} \nabla \Psi - N_A a^3 D \frac{c_{\pm} \nabla (c_+ + c_-)}{1 - c_+ N_A a^3 - c_- N_A a^3} \right) = 0$$

where N_A is Avogadro's constant

In [58], the authors use the generalized version of the above equations, the Generalized Modified Poisson-Nernst-Planck equations to model the dynamics of multiple asymmetric charged

species in the CO_2 reduction reaction. The Generalized Modified Poisson-Nernst-Planck equations are shown in Equation 3.17.

$$(3.17) \quad \frac{\partial c_k}{\partial t} + \nabla \cdot \left(-D_k \nabla c_k - \frac{z_k D_k F}{RT} c_k \nabla \Psi - D_k c_k \frac{N_A \nabla \sum a_k^3 c_k}{1 - N_A \sum a_k^3 c_k} \right) - R_i = 0$$

where the subscript k refers to the species k and R_k is the rate of production of species k from various reactions occurring in the electrolyte.

More recently, a size-modified set of Poisson-Nernst-Planck equations have been used to account for volume exclusion due to the solvent species [77].

3.4. The modified Poisson-Boltzmann equation with flux

The current state-of-the-art equations for modeling electrochemical phenomena are variants of the modified Poisson-Nernst-Planck equations [58, 77, 92]. Because of the various spatial and time scales present in these models, numerically simulating them is an error-prone process. In this thesis, we simultaneously modify both our mathematical model and our numerical method to combat the difficulties caused by the multiple spatial and time scales.

In this section, we choose not to work with a variant of the modified Poisson-Nernst-Planck equations. Instead, we focus our attention on the steady state description - the Poisson-Boltzmann/Modified Poisson-Boltzmann equation and derive a version of it that incorporates flux from the bulk region. This is novel because the Poisson-Boltzmann/Modified Poisson-Boltzmann equation have typically been used only in no-flux (blocking) conditions and it is standard practice to use the dynamic Poisson-Nernst-Planck equations in cases where there is flux at the boundaries. Focusing on the steady state problem has the major benefit of removing

the time scale separation completely from the model. Furthermore, even in situations where the experimental parameters are varied dynamically, the rate of formation of the electric double layer is on a much faster timescale [93, 94] than the experimental changes that a quasi-steady state assumption is reasonable. In the next chapter, we explore approaches to numerically simulate this steady state model that can deal with spatial scale separation.

3.4.1. Derivation of the Poisson-Boltzmann equation with flux

This subsection shows the derivation of the Poisson-Boltzmann equation with flux. Notably, there are no steric effects in this model which makes the derivation much simpler than that in the next subsection (which includes steric effects and follows a similar derivation). The following equation describes how the concentration of various species change according to the Poisson Nernst Planck equations.

$$(3.18) \quad \frac{\partial c_k}{\partial t} + \frac{\partial}{\partial x} \left(-D_k \frac{\partial c_k}{\partial x} - z_k u_k F c_k \frac{\partial \Psi}{\partial x} \right) = R_k$$

where c_k is the concentration of species k in space and time, R_k is the generation of species, D is the diffusion coefficient, z is the charge number, u is the mobility, Ψ is the electrolyte potential and F is Faraday's constant. For a non-charged species, z would be zero. For reacting species, this equation has flux boundary conditions at the electrode. Otherwise, there is no flow of new species into our system.

If we assume that the species generation throughout the electrolyte solution does not have a significant impact in our reaction, Equation 3.18 becomes

$$(3.19) \quad \frac{\partial c_k}{\partial t} + \frac{\partial}{\partial x} \left(-D_k \frac{\partial c_k}{\partial x} - z_k u_k F c_k \frac{\partial \Psi}{\partial x} \right) = 0$$

Another way to write the above equation is

$$(3.20) \quad \frac{\partial c_k}{\partial t} + \frac{\partial q_k}{\partial x} = 0$$

Here, q_k represents the flux of species k . The above equation means that the concentration of the species over time changes according to the spatial differences in the flux.

At steady state, we have

$$(3.21) \quad \frac{\partial c_k}{\partial t} = 0$$

And hence,

$$(3.22) \quad \frac{\partial q_k}{\partial x} = 0$$

So, at steady state, the flux of species doesn't change over space and is equal to the flux at the boundary.

$$(3.23) \quad \frac{\partial q_k}{\partial x} = 0$$

$$(3.24) \quad q_k = -\frac{i}{n_k F A}$$

Expanding q_k , we get

$$(3.25) \quad -D_k \frac{\partial c_k}{\partial x} - z_k u_k F c_k \frac{\partial \Psi}{\partial x} = -\frac{i}{n_k F A}$$

Since ionic mobility $u = \frac{DF}{RT}$, this becomes

$$(3.26) \quad D_k \frac{\partial c_k}{\partial x} + z_k \frac{DF}{RT} c_k \frac{\partial \Psi}{\partial x} = \frac{i}{n_k F A}$$

Let $\beta = \frac{F}{RT}$

$$(3.27) \quad e^{z_k \beta \Psi} \frac{\partial c_k}{\partial x} + e^{z_k \beta \Psi} z_k \beta c_k \frac{\partial \Psi}{\partial x} = e^{z_k \beta \Psi} \frac{i}{n_k D_k F A}$$

$$(3.28) \quad \frac{\partial}{\partial x} \left(e^{z_k \beta \Psi} c_k \right) = e^{z_k \beta \Psi} \frac{i}{n_k D_k F A}$$

Integrating this equation from L to x , we get

$$(3.29) \quad c_k(x) = c_k(L) e^{-z_k \beta \Psi(x)} + e^{-z_k \beta \Psi(x)} \int_L^x e^{z_k \beta \Psi(s)} \frac{i}{n_k D_k F A} ds$$

In the diffusion layer, the equations are

$$(3.30) \quad \frac{d^2\Psi}{dx^2} = -\frac{F}{\varepsilon} \sum z_k c_k, \quad \Psi(0) = \Psi_0, \quad \Psi(L) = 0$$

where

$$(3.31) \quad c_k(x) = c_k(L)e^{-z_k\beta\Psi(x)} + e^{-z_k\beta\Psi(x)} \int_L^x e^{z_k\beta\Psi(s)} \frac{i}{n_k D_k F A} ds$$

3.4.2. Modified Poisson-Boltzmann equation with flux

Since the model doesn't have the size of the various species incorporated, the counterions can accumulate at an impossible concentration near the catalyst. So, in order to incorporate the steric effects, our derivation starts from the General Modified Poisson-Nernst-Planck equations ([76]).

$$(3.32) \quad \frac{\partial c}{\partial t} + \frac{\partial}{\partial x} \left(-D \frac{\partial c}{\partial x} - \frac{DcF}{RT} \frac{\partial \Psi}{\partial x} - Dc \left(\frac{N_A \sum_{i=1}^n N_A a_k^3 \frac{\partial c_k}{\partial x}}{1 - N_A \sum_{i=1}^n N_A a_k^3 c_k} \right) \right) = 0$$

where N_A is the Avogadro number and a_k is the effective solvated diameter of the species i .

For a system without reactions or flux at the boundary, it is straightforward to convert it to its steady state version. These become the General Modified Poisson Boltzmann equations.

$$(3.33) \quad \frac{d^2\Psi}{dx^2} = -\frac{F}{\varepsilon} \sum z_k c_k, \quad \Psi(0) = \Psi_0, \quad \Psi(L) = 0$$

The right hand side in the above equation can be expressed as

$$(3.34) \quad \sum_k z_k c_k = \frac{\sum_k z_k \alpha_k e^{-z_k \beta \Psi}}{1 + \sum_j \alpha_j e^{-z_j \beta \Psi} a_j^3}$$

However, when there is flow at the boundaries, the system gets more complex and the process of deriving the right hand side is not so trivial. With flow at the boundaries, following the same process as the previous subsection, the equations become

$$(3.35) \quad \frac{d^2 \Psi}{dx^2} = -\frac{F}{\varepsilon} \sum_k z_k c_k$$

where

$$(3.36) \quad \frac{c_k}{1 - \sum N_A a_k^3 c_k} = A_k e^{-z_k \beta \Psi} + e^{-z_k \beta \Psi} \int_L^x \frac{ie^{z_k \beta \Psi}}{n_k D_k F A (1 - \sum N_A a_k^3 c_k)} dx$$

The A_k s are constant values which can be obtained using information about the concentration of the species at the boundary of the diffusion layer. When we set $x = L$ in the above equation, we get

$$(3.37) \quad \frac{c_k(L)}{1 - \sum N_A a_k^3 c_k(L)} = A_k$$

What makes this system difficult to solve is that the right hand side of Equation 3.36 features an integral over the concentrations of various species. Thus, finding a closed form expression for $\sum_k z_k c_k$ is non-trivial. However, we have managed to derive an expression for it, giving us the following expression for the right hand side of the Poisson equation in Equation 3.35.

$$(3.38) \quad \sum_k z_k c_k = \frac{\sum_k z_k A_k e^{-z_k \beta \Psi} + z_k \phi_k g_k}{1 + \sum_k N_A a_k^3 A_k e^{-z_k \beta \Psi} + \phi_k N_A a_k^3 g_k}$$

Here, the ϕ_k s represent the effective fluxes of the reacting species

$$(3.39) \quad \phi_k = \frac{i}{n_k D_k F A}$$

and the g_k s are given by

$$(3.40) \quad g_k = e^{-z_k \beta \Psi} \int_L^x \frac{e^{z_k \beta \Psi}}{(1 - \sum N_A a_k^3 c_k)} dx$$

$$(3.41) \quad = e^{-z_k \beta \Psi} \int_L^x \frac{e^{z_k \beta \Psi}}{(1 - \sum N_A a_k^3 c_k)} dx$$

Since the g_k s contain integrals, Equation 3.35 is an integro-differential equation. One might naturally consider solving it using numerical methods for integral equations. However, because of the non-trivial dependence of the g_k s on the concentrations c_k s, this approach is not feasible except in the limiting case of $a_k \rightarrow 0$.

However, by differentiating Equation 3.40 with respect to x , we can show that g_k s however satisfy this system of ordinary differential equations which do not explicitly depend on the c_k s.

$$(3.42) \quad g'_k + z_k \beta \Psi' g_k - \sum_k \phi_k N_A a_k^3 g_k = 1 + \sum_k A_k N_A a_k^3 e^{-z_k \beta \Psi}, \quad g_k(L) = 0$$

Since this system is linear in g_k s, one might think of obtaining an analytic expression for the g_k s using matrix exponentials with integrals. The g_k s in their vector form $\mathbf{g}$ can be represented as

$$(3.43) \quad \mathbf{g} = e^{-\mathbf{J}x - \beta \mathbf{Z} \Psi} \int_L^x e^{\mathbf{J}x + \beta \mathbf{Z} \Psi} (1 + \sum_k A_k N_A a_k^3 e^{-z_k \beta \Psi})$$

However, this expression is not valid when Ψ has a nonlinear dependence on x , which is usually the case in our situation. So, we instead obtain g_k by solving the system of equations 3.42 numerically by using an accurate time integrator along our spatial domain.

3.4.3. Incorporating Butler-Volmer dynamics into the Modified Poisson-Boltzmann equation with flux

So far in our mathematical descriptions, the effective flux ϕ_k of the reacting species k is given by

$$(3.44) \quad \phi_k = \frac{i}{n_k D_k F A}$$

where D_k is the diffusivity of species k , i is the current, A the area of electrode surface and n_k the number of ions of species k corresponding to the number of electrons involved in the reaction.

A typical approach when solving the General Modified Poisson Nernst-Planck equations with flux boundary conditions [58] is to specify a current i typically for a specific potential.

A more general mathematical approach to modeling the electrochemical double layer is to not explicitly incorporate specific current/current density but to model the current density ($j = \frac{i}{A}$) as a function of potential using the Butler-Volmer relation given below.

$$(3.45) \quad j = j_0 \exp\left(\frac{\log(10)|\eta|}{A_{Tafel}}\right)$$

where j_0 is the exchange current density, η is the overpotential and A_{Tafel} is the Tafel slope. The overpotential η is potential difference between the electrode potential and thermodynamic reduction potential. The Tafel slope can be estimated from experimental data over a range of electrode potential. The calculation of the Tafel slope is as much an art as it is a science. There is an active discussion in the literature on exploring various perspectives for appropriate Tafel slope calculations [95, 96, 97, 98, 99, 100].

The Butler-Volmer relation in Equation 3.45 assumes that the concentration of the species that get consumed in the reaction is the same at the electrode surface and at the bulk. This is an accurate assumption when the transport rate of all the species from the bulk to the electrode surface is faster than the reaction rate. In the Oxygen Reduction Reaction, oxygen is often considered to be limited by its mass transport from the bulk at various applied potentials and we are specifically interested in the interaction between bulk transport rates and kinetic processes near the catalyst surface. In that scenario, the extended Butler-Volmer equation given below is more representative of the current density in the system.

$$(3.46) \quad j = j_0 \frac{c_{O_2}^*}{c_{O_2}^b} \exp\left(\frac{\log(10)|\eta|}{A_{Tafel}}\right)$$

Here, $C_{O_2}^b$ is the concentration of oxygen at the bulk and $C_{O_2}^*$ is the concentration of oxygen at the Outer Helmholtz Plane (OHP).

Incorporating the fluxes $\phi_k = \frac{j}{n_k D_k}$ of the various species into the General Modified Poisson-Boltzmann Equation with flux through the extended Butler-Volmer equation adds another level of numerical and mathematical complexity to our model. It results in the Butler-Volmer consistency condition that we derive below.

The concentration of the species k must satisfy

$$(3.47) \quad \frac{c_k}{1 - \sum N_A a_k^3 c_k} = A_k e^{-z_k \Phi} + \frac{1}{n_k D_k F} j_0 \frac{c_{O_2}^*}{c_{O_2}^b} \exp\left(\frac{\log(10)|\eta|}{A_{Tafel}}\right) e^{-z_k \Phi} \int_L^x \frac{e^{z_k \beta \Psi}}{1 - \sum N_A a_k^3 c_k} dx$$

So, the concentration of oxygen at the OHP must satisfy

$$(3.48) \quad \frac{c_{O_2}^*}{1 - \sum N_A a_k^3 c_k} = A_{O_2} + \frac{j_0}{2D_{O_2}F} j_0 \frac{c_{O_2}^*}{c_{O_2}^b} \exp\left(\frac{\log(10)|\eta|}{A_{Tafel}}\right) \int_L^0 \frac{1}{1 - \sum N_A a_k^3 c_k} dx$$

assuming n_k for Oxygen in acidic solutions

This can be rewritten as

$$(3.49) \quad g_{O_2}' c_{O_2}^* = A_{O_2} + \frac{j_0}{2D_{O_2}F} \frac{c_{O_2}^*}{c_{O_2}^b} \exp\left(\frac{\log(10)|\eta|}{A_{Tafel}}\right) g_{O_2}^*$$

The concentration of oxygen at the Outer Helmholtz Plane (OHP) $c_{O_2}^*$ has to satisfy Equation 3.49 which we will refer to hereafter as the Butler-Volmer consistency condition.

3.5. The complete model

In this section, we detail our complete mathematical description of the electrochemical reactor.

We consider a simplified one-dimensional domain that goes from the catalyst surface at the electrode to the bulk region. The full length of the domain from the catalyst to the bulk is referred to by L_{bulk} . L_{bulk} is typically on the order of $20 - 50 \mu m$. In the RRDE setup, L_{bulk} can be increased or decreased through the rotation rate. This allows it to be a tunable experimental parameter. Right next to the catalyst surface, we have the Stern layer L_{Stern} which is assumed to be on the order of $0.2 - 1 nm$. In the Stern Layer, the potential is assumed to behave according to the Poisson equation (Equation 3.50).

$$(3.50) \quad \frac{d^2\Psi}{dx^2} = 0, \quad \Psi(x=0) = \Psi_{electrode}, \quad \Psi(x=L_{Stern}) = \Psi_{OHP}$$

$\Psi_{electrode}$ is the potential applied at the electrode and Ψ_{OHP} is the potential at the Outer Helmholtz Plane (OHP), the boundary of the Stern layer away from the electrode. The solution to Equation 3.50 is

$$(3.51) \quad \Psi(x) = \Psi_{electrode} + \frac{\Psi_{OHP} - \Psi_{electrode}}{L_{Stern}}x$$

Since we know the solution in the Stern layer, the Stern layer is not part of the domain that is solved numerically. From the Stern layer to the bulk domain, the potential in our mathematical model obeys the General Modified Poisson-Boltzmann equation with flux detailed in Subsection 3.4.2.

Hence, on the domain $[L_{Stern}, L_{bulk}]$, the potential behaves according to

(3.52)

$$\frac{d^2\Psi}{dx^2} = -\frac{F}{\varepsilon} \frac{\sum_k z_k A_k e^{-z_k \beta \Psi} + z_k \phi_k g_k}{1 + \sum_k N_A a_k^3 A_k e^{-z_k \beta \Psi} + \phi_k N_A a_k^3 g_k}, \quad \Psi(x = L_{Stern}) = \Psi_{OHP}, \quad \Psi(x = L_{bulk}) = 0,$$

where

$$(3.53) \quad A_k = \frac{c_k(L_{bulk})}{1 - \sum N_A a_k^3 c_k(L_{bulk})}, \quad \phi_k = \frac{j}{n_k D_k F}, \quad g_k = e^{-z_k \beta \Psi} \int_{L_{bulk}}^x \frac{e^{z_k \beta \Psi}}{(1 - \sum N_A a_k^3 c_k)} dx$$

The g_k s satisfy

$$(3.54) \quad g'_k + z_k \beta \Psi' g_k - \sum_k \phi_k N_A a_k^3 g_k = 1 + \sum_k A_k N_A a_k^3 e^{-z_k \beta \Psi}, \quad g_k(L_{bulk}) = 0$$

One can see in Equation 3.52 that the potential value at the OHP Ψ_{OHP} is needed to solve it. However, we typically only know the potential at the electrode $\Psi_{electrode}$. One way to solve the model is to change the Dirichlet boundary condition in Equation 3.52 to the Robin Boundary condition using Equation 3.51.

$$(3.55) \quad \Psi(x = L_{Stern}) - L_{Stern} \frac{d\Psi}{dx}(x = L_{Stern}) = \Psi_{electrode}$$

A more practical approach especially when simulating over a range of applied voltages is to assume a Ψ_{OHP} , run the simulations using the Dirichlet boundary condition $\Psi(x = L_{Stern}) = \Psi_{OHP}$ and then back calculate the potential at the electrode $\Psi_{electrode}$ using Equation 3.51.

In our mathematical model, there are two ways to incorporate the flow of the reacting species from the bulk to the electrode in our simulations. One approach is to specify a set current density j [58]. The other approach is to incorporate the extended Butler-Volmer relation into our model. Doing so, the current density is formulated as

$$(3.56) \quad j = j_0 \frac{c_{O_2}^*}{c_{O_2}^b} \exp\left(\frac{\log(10)|\eta|}{A_{Tafel}}\right)$$

where $c_{O_2}^b$ is the concentration of oxygen at the bulk and $c_{O_2}^*$ is the concentration of oxygen at the Outer Helmholtz Plane (OHP). We can obtain $c_{O_2}^b$ from the Butler-Volmer Consistency Condition rewritten below.

$$(3.57) \quad g_{O_2}^{*'} c_{O_2}^* = A_{O_2} + \frac{j_0}{2D_{O_2}F} \frac{c_{O_2}^*}{c_{O_2}^b} \exp\left(\frac{\log(10)|\eta|}{A_{Tafel}}\right) g_{O_2}^*$$

Our complete mathematical model is shown illustrated in Figure 3.6.

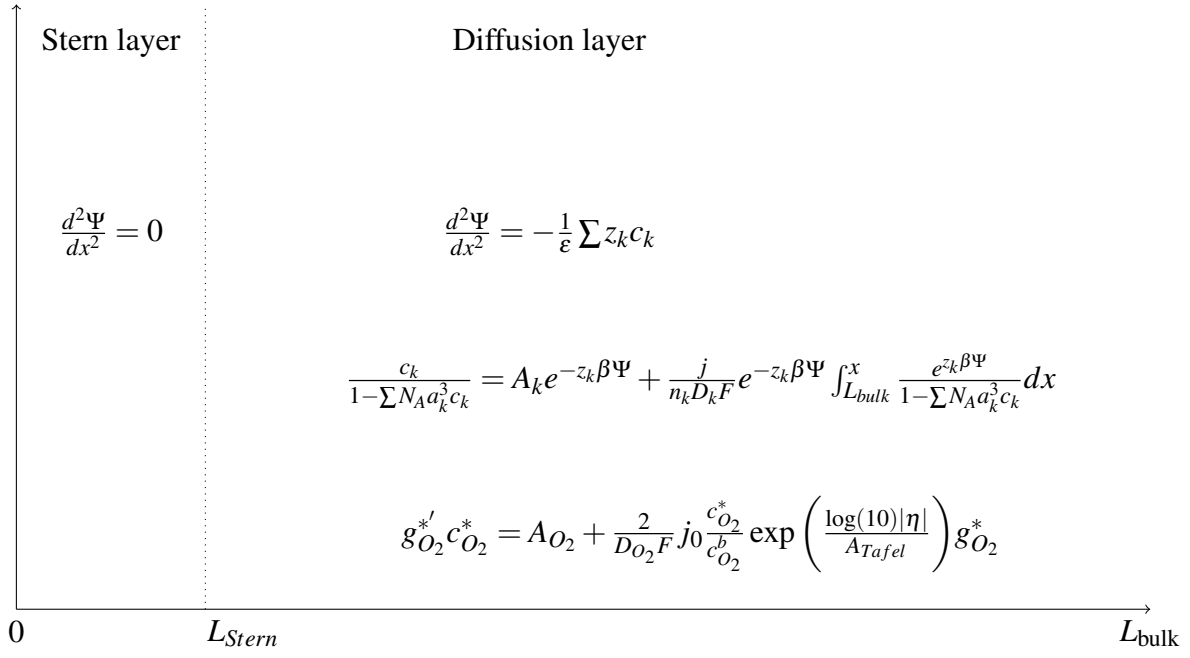


Figure 3.6. The complete mathematical model.

CHAPTER 4

A numerical method for solving the modified Poisson-Boltzmann equation with flux

In the previous chapter , we arrived at the Modified Poisson-Boltzmann equation with flux. We study the steady state problem, since most of our experiments are done at a steady state. Even in future experiments where the experimental parameters may be varied dynamically, since the rate of formation of the electric double layer is on a much faster timescale [93, 94] than the experimental changes, it would be reasonable to assume a quasi-steady state problem. By studying a steady-state problem, we eliminated time integration woes associated with the multiple time scales occurring in the electrochemical reactor. In this section, we develop a specialized numerical method for solving the Modified Poisson-Boltzmann equation with flux.

Because we are incorporating flow from a microscale domain along with the nano-scale electric double layer, there is a sharp boundary layer that is difficult to capture using numerical methods like finite differences and finite volume methods. Other works [58, 77, 92] where electrochemical reactions have been simulated using the General Modified Poisson-Nernst Planck equations [76] and its variants have had to rely on industrial finite element solvers like COMSOL [101] and Fenics [71] along with additional stabilizers. Our aim is to create a simple yet reliable approach for simulating the one-dimensional Modified Poisson-Boltzmann equation with flux. This would allow us to create a tunable virtual laboratory where the electrochemical reaction environment can be rapidly and efficiently solved.

In this chapter, we begin by rescaling the Modified Poisson-Boltzmann equation with flux and considering the Chebyshev spectral method for numerically simulating it. It turns out the Chebyshev spectral method by itself is insufficient for a well-resolved simulation. We explore two options for better resolving the multiple spatial scales from the microscale flow and the nanoscale electric double layer. We first look at a hybrid numerical model where an approximation is used to split the domain in order to separate the nanoscale and microscale phenomena. Then, we consider a mapping of our numerical domain that allows us to accurately resolve the electric double layer. Finally, we detail our numerical algorithms and showcase our simulations.

4.1. Rescaling of variables and dimensional analysis.

The General Modified Poisson Boltzmann equation depends on a lot of dimensional variables. When numerically solving systems like these, it is prudent to eliminate vestigial variables and turn the equations into a minimal non-dimensional form. The resulting minimal set of variables provide a lot of perspective on the relevant scales in the problem. We start with the dimensional version of the General Modified Poisson Boltzmann equation with flux. The dimensional versions of key variables are denoted by a tilde ($\tilde{\cdot}$) on top of them.

$$(4.1) \quad \frac{d^2 \tilde{\Psi}}{d\tilde{x}^2} = -\frac{F}{\epsilon} \sum_k z_k \tilde{c}_k$$

$$(4.2) \quad = -\frac{F}{\epsilon} \frac{\sum_k z_k \tilde{A}_k e^{-z_k \beta \tilde{\Psi}} + z_k \tilde{\phi}_k \tilde{g}_k}{1 + \sum_k N_A \tilde{a}_k^3 A_k e^{-z_k \beta \tilde{\Psi}} + N_A \tilde{a}_k^3 \tilde{\phi}_k \tilde{g}_k}$$

Multiplying both sides by $\beta = \frac{F}{RT}$, we get

$$(4.3) \quad \beta \frac{d^2 \tilde{\Psi}}{d\tilde{x}^2} = -\beta \frac{F}{\varepsilon} \frac{\sum_k z_k \tilde{A}_k e^{-z_k \beta \tilde{\Psi}} + z_k \tilde{\phi}_k \tilde{g}_k}{1 + \sum_k N_A \tilde{a}_k^3 \tilde{A}_k e^{-z_k \beta \tilde{\Psi}} + N_A \tilde{a}_k^3 \tilde{\phi}_k \tilde{g}_k}$$

Since $\beta = \frac{F}{RT}$ has units of 1 / Volt , we can express the potential as a dimensionless variable $\Psi = \beta \tilde{\Psi}$. We can also divide $\tilde{x}$ by the length scale L of our physical system to obtain another dimensionless variable $x = \frac{\tilde{x}}{L}$. The physical domain could be defined as $\tilde{x} \in [0, 2L]$. Then, the dimensionless variable x would be defined on $[0, 2]$ (This is a natural domain to be mapped to the Chebyshev domain $[-1, 1]$ where we numerically solve our model).

$$(4.4) \quad \frac{d^2 \Psi}{dx^2} = -\frac{F^2 L^2}{\varepsilon RT} \frac{\sum_k z_k \tilde{A}_k e^{-z_k \Psi} + z_k \tilde{\phi}_k \tilde{g}_k}{1 + \sum_k N_A \tilde{a}_k^3 \tilde{A}_k e^{-z_k \Psi} + N_A \tilde{a}_k^3 \tilde{\phi}_k \tilde{g}_k}$$

We can further rescale this system to reduce the number of variables and make it more dimensionless. $\tilde{A}_k$ has dimensions of concentration (mol/L). We can extract a dimensionless form $A_k = \frac{\tilde{A}_k}{c_0}$ by dividing it by c_0 , a characteristic concentration for the system (typically the bulk concentration of the species with the maximum bulk concentration). Similarly, we can divide $\tilde{a}_k$ by a characteristic ionic radius a_0 to get the dimensionless variable $a_k = \frac{\tilde{a}_k}{a_0}$.

The effective flux $\tilde{\phi}_k$ can be converted to a dimensionless form by

$$(4.5) \quad \tilde{\phi}_k = \frac{\tilde{j}}{n_k \tilde{D}_K F} = \frac{j_c}{D_0 F} \phi_k$$

where j_c is a characteristic current density for the physical system and D_0 is a characteristic diffusion coefficient for the species in the system.

The $\tilde{g}_k$ s can be converted to a dimensionless form g_k by

$$(4.6) \quad \tilde{g}_k = e^{-z_k \beta \tilde{\Psi}} \int_{2L}^{\tilde{x}} \frac{e^{z_k \beta \tilde{\Psi}}}{(1 - \sum N_A \tilde{a}_k^3 \tilde{c}_k)} d\tilde{x}$$

$$(4.7) \quad = L e^{-z_k \Psi} \int_2^x \frac{e^{z_k \Psi}}{(1 - \sum N_A a_0^3 c_0 a_k^3 c_k)} dx = L g_k$$

The system of equations for the g_k s become

$$(4.8) \quad g'_k + z_k \Psi' g_k - \sum_k \frac{N_A a_0^3 j_c L}{D_0 F} a_k^3 \phi_k g_k = 1 + \sum_k N_A a_0^3 c_0 A_k a_k^3 e^{-z_k \Psi}, \quad g_k(2) = 0$$

The initial condition for $\tilde{g}_k$ changes from $\tilde{g}_k(\tilde{x} = 2L) = 0$ to $g_k(x = 2) = 0$.

Incorporating these rescaled variables, Equation 4.4 becomes

$$(4.9) \quad \frac{d^2 \Psi}{dx^2} = -\frac{F^2 c_0 L^2}{\varepsilon R T} \frac{\sum_k z_k A_k e^{-z_k \beta \Psi}}{1 + \sum_k N_A a_0^3 c_0 a_k^3 A_k e^{-z_k \beta \Psi} + \frac{N_A a_0^3 j_c L}{D_0 F} \phi_k a_k^3 g_k}$$

$$(4.10) \quad -\frac{F j_c L^3}{\varepsilon R T D_0} \frac{\sum_k z_k \phi_k g_k}{1 + \sum_k N_A a_0^3 c_0 a_k^3 A_k e^{-z_k \beta \Psi} + \frac{N_A a_0^3 j_c L}{D_0 F} \phi_k a_k^3 g_k}$$

We can massively reduce the number of variables in the above equation.

$$(4.11) \quad \frac{d^2 \Psi}{dx^2} = -\kappa_1 \frac{\sum_k z_k A_k e^{-z_k \beta \Psi}}{1 + \sum_k \kappa_3 a_k^3 A_k e^{-z_k \beta \Psi} + \kappa_4 \phi_k a_k^3 g_k} - \kappa_2 \frac{\sum_k z_k \phi_k g_k}{1 + \sum_k \kappa_3 a_k^3 A_k e^{-z_k \beta \Psi} + \kappa_4 \phi_k a_k^3 g_k}$$

where

$$(4.12) \quad \kappa_1 = \frac{F^2 c_0 L^2}{\varepsilon R T}, \quad \kappa_2 = \frac{F j_c L^3}{\varepsilon R T D_0}, \quad \kappa_3 = a_0^3 N_A c_0, \quad \kappa_4 = \frac{N_A a_0^3 j_c L}{D_0 F}$$

The system of equations for the g_k s become

$$(4.13) \quad g'_k + z_k \Psi' g_k - \kappa_4 \sum_k a_k^3 \phi_k g_k = 1 + \kappa_3 \sum_k A_k a_k^3 e^{-z_k \Psi}, \quad g_k(2) = 0$$

The Butler-Volmer Consistency Condition (Equation 3.49) can also be rescaled to be

$$(4.14) \quad g_{O_2}^* c_{O_2}^* = c_o A_{O_2} + \frac{j_c L}{2D_0 F} \frac{1}{D_{O_2}} j_0 \frac{c_{O_2}^*}{c_{O_2}^b} \exp\left(\frac{\log(10)|\eta|}{A_{Tafel}}\right) g_{O_2}^*$$

where D_{O_2} is the rescaled diffusion coefficient for oxygen given by $\tilde{D}_{O_2}/D_0$ where $\tilde{D}_{O_2}$ is the diffusion coefficient for oxygen .

The Butler-Volmer Consistency Condition can be made more compact by using the dimensionless variable κ_5 .

$$(4.15) \quad g_{O_2}^* c_{O_2}^* = A_{O_2} + \kappa_5 \frac{j_0}{2D_{O_2}} \frac{c_{O_2}^*}{c_{O_2}^b} \exp\left(\frac{\log(10)|\eta|}{A_{Tafel}}\right) g_{O_2}^*, \quad \kappa_5 = \frac{j_c L}{c_0 D_0 F}$$

4.2. A primer on the Chebyshev spectral method

We considered a variety of numerical methods to solve the Equations 4.11-4.15. These equations represent a sharp spatial boundary layer . To a large extent, the value of κ_1 dictates the sharpness of the boundary layer. The characteristic concentration c_0 is chosen to be 1 mol/L in our simulations which is typical of the maximum bulk concentration of the various species in our system. κ_1 can be rewritten in terms of the Debye length [69, 102] as

$$(4.16) \quad \kappa_1 = \frac{F^2 c_0 L^2}{\epsilon R T} = \frac{L^2}{L_{Debye}^2} = 4.216 \times 10^{20} L^2.$$

The Debye length is representative of the length scale of the boundary layer [69] and can be calculated to be

$$(4.17) \quad L_{Debye} = \sqrt{\frac{\epsilon RT}{F^2 c_0}} = \sqrt{\frac{1}{4.216 \times 10^{20} L^2}} = 4.87 \times 10^{-11}.$$

Since our model incorporates flow from the microscale, the length of the domain is around $50\mu m$. For these values, κ_1 is 1.054×10^{12} which results in an extremely sharp boundary layer in our simulations.

Standard implementations of finite difference or finite volume methods are not capable of resolving such a sharp boundary layer. Finite element methods and spectral methods are typically better options to use in such situations. In particular, spectral methods are known to have exponential convergence for smooth solutions. This means that the discretization error scales exponentially and often reaches machine for a relatively few number of grid points [103, 104].

Spectral methods work by using global basis functions to represent numerical functions. These basis functions are typically orthogonal. The most well-known basis functions are Fourier series (used for periodic problems) and Chebyshev polynomials (typically used for non-periodic problems) [103, 104]. Furthermore, there are two main ways to implement spectral methods: the Galerkin approach and the collocation approach [103, 104]. The Galerkin approach features working with the weak formulation of the partial differential equation and using the orthogonality of the basis functions to obtain the spectral coefficients of each variable in the differential equation. The Galerkin approach is very practical to use with complicated boundary conditions which can be implemented easily using the spectral coefficients. However, it is slightly harder

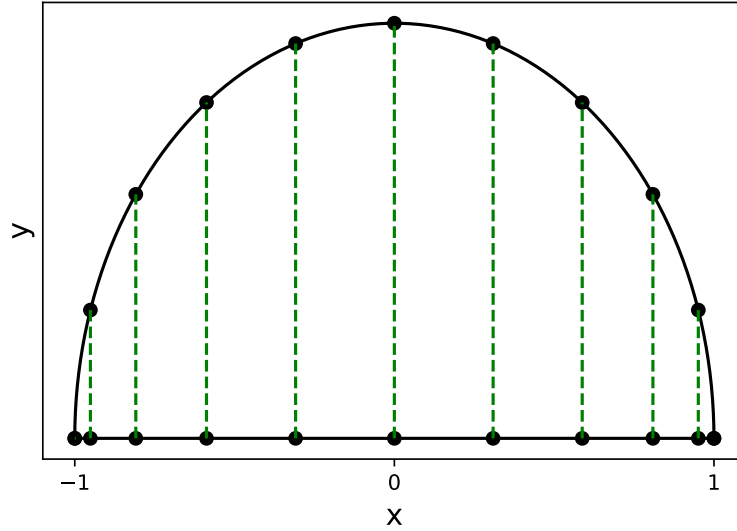


Figure 4.1. The Chebyshev collocation points.

to implement it when there are nonlinear terms in the differential equation due to additional convolutions involved.

The collocation approach features a specific domain of grid points known as collocation points. The numerical functions are represented by a combination of the basis functions defined by the numerical values at these collocation points. It is natural to think of such a function as a polynomial of certain degree interpolated at these collocation points. The derivatives of these functions can be computed from this global representation by exactly differentiating the basis functions analytically. Nonlinear terms are no different from linear terms in this representation. This representation also allows for a helpful mapping of the domain which we will discuss in Section 4.4.

For all these reasons, we work with the Chebyshev collocation method in this chapter. The Chebyshev collocation points can be generated by taking a uniform grid $z \in [-1, 1]$ and using

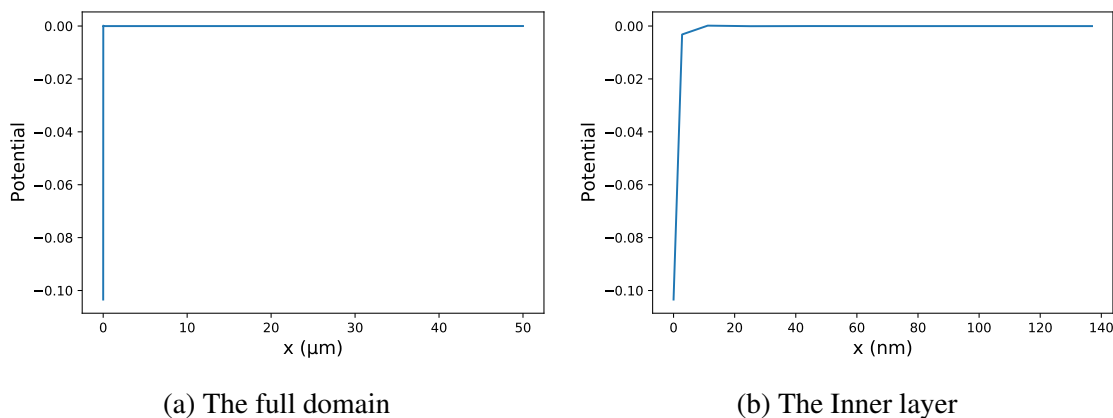


Figure 4.2. A direct implementation of the Chebyshev spectral results in an under-resolved simulation.

the cosine function to map them to $\cos(z)$. An example of collocation points are shown in Figure 4.1. While the points on the semicircle are uniformly distributed, their projections, the collocation points are distributed non-uniformly and clustered more at the boundaries. A uniform grid would, in fact, result in erroneous oscillations at the boundaries. This is known as Runge phenomenon [105, 106] and a non-uniform grid allows for a far more stable polynomial representation.

The standard Chebyshev domain is $x \in [-1, 1]$. However, any finite domain can be mapped to this standard domain using a simple linear transformation. For infinite domains, rational Chebyshev functions can be employed [107].

Figure 4.2 highlights the result of applying the Chebyshev spectral method directly to our problem. The simulated system is the same as the one described in Section 4.6 and Table 4.1. In Figure 4.2 (a), we can see an extremely sharp boundary layer in our numerical representation potential Ψ from Equation 4.11. As we zoom into the inner layer at the nanoscale level where our key reaction environment exists, we can clearly see that our numerical simulations are quite under-resolved. In the next two sections, we discuss two possible approaches to better resolve

our numerical solution at the nanoscale level: a hybrid numerical model with a split domain and a mapping of our numerical domain that allows us for better resolution of boundary layers.

4.3. A potential approach to combat the spatial scales separation: The hybrid numerical model

The reason for the under-resolved simulation is because we need to incorporate flow from the microscale domain however, the electrical double layer is on the order of nanometers. If there was no flow in the system (like in the case for modeling batteries [76]), our simulation domain would be the nanoscale domain and we would not have any scale separation.

We do know that most of the potential drop occurs in the diffuse layer. Perhaps in order to resolve the spatial scales and the behavior at the catalyst boundary, we need to consider a hybrid numerical model with our domain split into two. We can divide our entire domain up into two domains, one where the potential drops to a near-zero value and one where the potential effects are negligible.

Figure 4.3 shows the our domain for complete model (illustrated in Figure 3.6 further divided up into

- The Stern layer which is around $0.4nm$ in length
- The diffuse layer with length ($\approx L_\delta$) which is around $20nm$ in length, where most of the potential drop happens
- and the Outer Diffusion Layer where the potential is negligible and we can ignore steric effects, which is around $30 - 50\mu m$ in length.

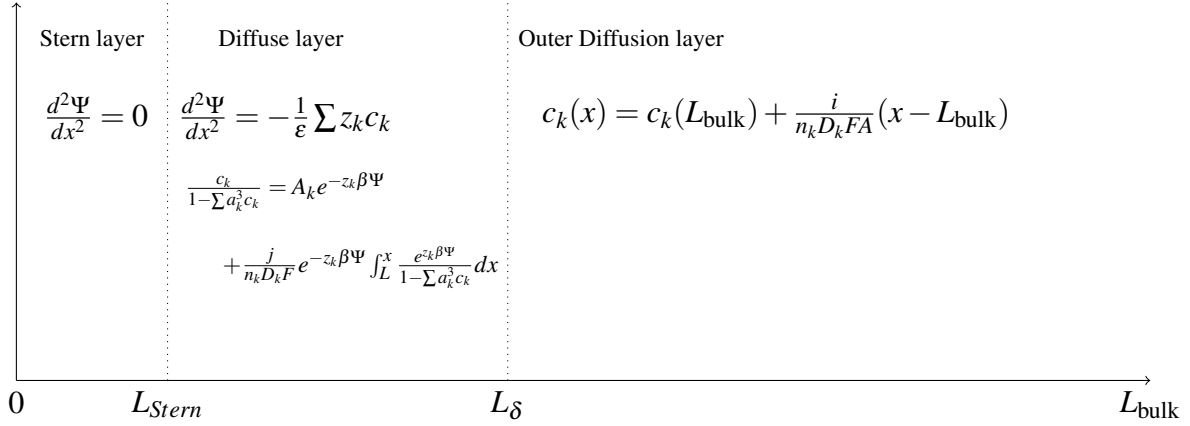


Figure 4.3. If we assume most of the potential drop occurs in the inner layer, we can split the domain into the diffuse and the outer diffusion layers and solve separate but coupled numerical models for each of them.

While this approach would solve the scale separation issue, it still makes an assumption that $\Psi(x > L_{\delta}) = 0$ which carries some error with it. In order to even quantify the error made through this assumption, a reliable “full” model is needed for reference. In the next section, we explore a mapping of our numerical domain that allows us to get a well-resolved solution to the Modified Poisson-Boltzmann equation with flux. This allows us to use one numerical model over the complete domain. This approach enables the utilization of a single numerical model across the entire domain. The hybrid approach, as described in this section, is not further explored in this thesis. However, in future research, we aim to investigate whether incorporating the hybrid approach with improved analytic descriptions could enhance the accuracy and robustness of simulations compared to the reference approach discussed in the following section.

4.4. Mappings of Chebyshev polynomials for resolving boundary layers

As previously mentioned, the Chebyshev grid has more points at the boundaries than in the middle of the grid. One might ponder that such a spatial distribution of grid points might allow

for better numerical approximations at the boundaries rather than the middle. Indeed, this turns out to be true. The increased accuracy near the boundaries does not seem to be related to the distribution of grid points but rather stems from more general spectral properties of polynomial approximations [108]. Nevertheless, there exists a rich history in the field of numerical analysis, of exploring carefully crafted mappings of grid points to achieve spatial resolution suited for specific problems [109, 110, 111, 112, 113, 114, 115]. In this section, we look at a mapping introduced in [115, 116] and discuss how it could be applied to our problem.

Consider we want to represent the function in Equation 4.18 numerically on the domain $x \in [-1, 1]$.

$$(4.18) \quad y = \tanh(\sigma(x - 1))$$

For large values of σ , the function in Equation 4.18 has a region of rapid variation around $x = 1$. This poses difficulties in representing the function and its derivatives accurately and leads to numerical issues.

We can map the domain $x \in [-1, 1]$ to a domain $s \in [-1, 1]$ using the following mapping from [115, 116].

$$(4.19) \quad s = s_0 + \frac{\tan^{-1}(\alpha_1(x - \alpha_2))}{\lambda}$$

The inverse of this mapping is given by

$$(4.20) \quad x = \alpha_2 + \frac{\tan(\lambda(s - s_0))}{\alpha_1}$$

α_1 is the key parameter of this mapping that needs to be selected or optimized for in order to represent the function in Equation 4.18 on the domain s without having a region of rapid variation. The variable α_2 corresponds to the location of the region of rapid variation. In this case, α_2 is equal to 1. The values for s_0 and λ can be calculated using formulas given below (taken from [115]).

$$(4.21) \quad s_0 = \frac{\kappa - 1}{\kappa + 1}, \quad \kappa = \tan^{-1}(\alpha_1(1 + \alpha_2)) / \tan^{-1}(\alpha_1(1 - \alpha_2))$$

$$(4.22) \quad \lambda = \tan^{-1}(\alpha_1(1 - \alpha_2)) / (1 - s_0)$$

As α_2 goes to 1,

$$(4.23) \quad s_0 \rightarrow 1$$

$$(4.24) \quad \kappa \rightarrow \infty$$

$$(4.25) \quad \lambda \rightarrow \frac{\tan^{-1}(2\alpha_1)}{2}$$

Selecting α_1 to be 200 (a value on the same order as σ), our mapping becomes

$$s = 1 + \frac{\tan^{-1}(200(x - 1))}{0.7841482}$$

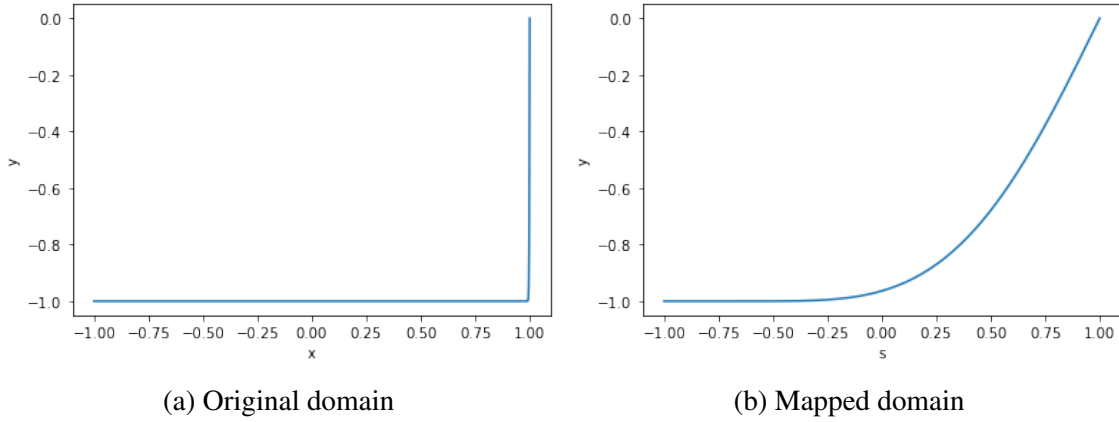


Figure 4.4. The mapping allows better resolution of the function $y = \tanh(400(x-1))$ in the mapped domain.

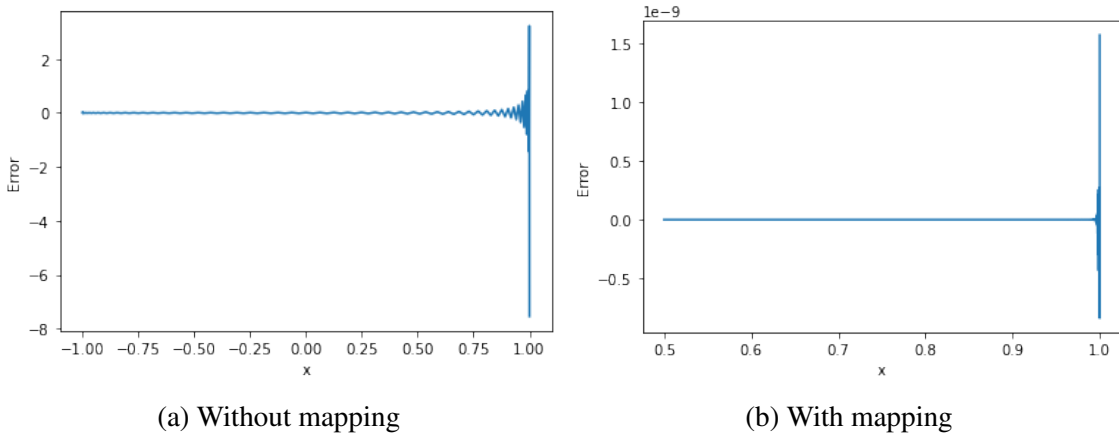


Figure 4.5. The error in the numerical computation of the derivative of the function $y = \tanh(400(x-1))$ is highlighted. The mapping allows for more accurate computation of the derivative.

Figure 4.4 highlights the function in its original domain and mapped domain. Figure 4.5 shows how the mapping results in significantly reduced error when computing derivatives. A Chebyshev grid with 100 points is used for these computations.

These results show promise for using such a mapping in order to better resolve the boundary layer in our problem. In fact, we can see in Figure 4.6 that the mapping results in a far better

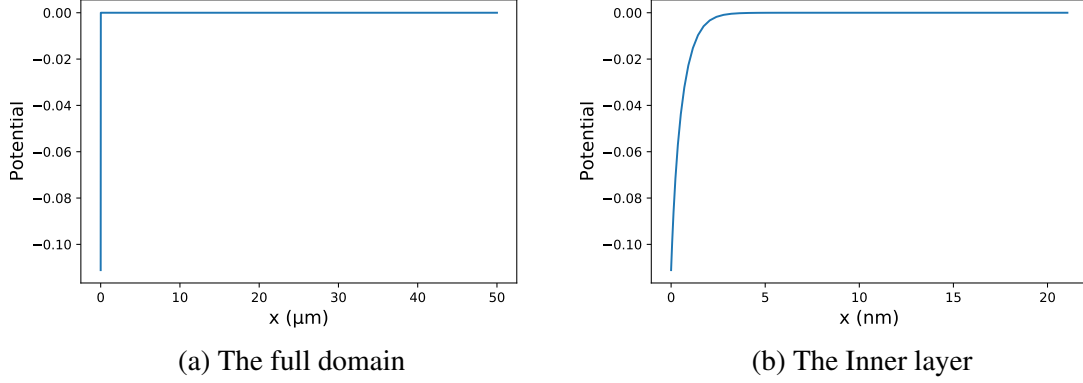


Figure 4.6. The mapping allows for better resolution at the nanoscale level for the modified Poisson-Boltzmann equation with flux.

resolved solution than that in Figure 4.2. This simulated system is described in detail in Section 4.6. In the next section, we get into the details of implementing the mapping.

4.5. Solving the General Modified Poisson-Boltzmann equation with flux

In this section, I describe how the mapping is used to solve the General Modified Poisson-Boltzmann equation with flux. The General Modified Poisson-Boltzmann equation with flux is a steady state differential equation described in Chapter 3 that relates the potential Ψ across the domain $[0, L]$ to the concentrations of various species c_k , their sizes a_k and their fluxes ϕ_k . We showed how the equation and the variables involved can be rescaled to a compact dimensionally reduced form.

$$(4.26) \quad \frac{d^2\Psi}{dx^2} = -\kappa_1 \frac{\sum_k z_k A_k e^{-z_k \beta \Psi}}{1 + \sum_k \kappa_3 a_k^3 A_k e^{-z_k \beta \Psi} + \kappa_4 \phi_k a_k^3 g_k} - \kappa_2 \frac{\sum_k z_k \phi_k g_k}{1 + \sum_k \kappa_3 a_k^3 A_k e^{-z_k \Phi} + \kappa_4 \phi_k a_k^3 g_k}$$

$$(4.27) \quad A_k = \frac{c_k(L)}{1 - \kappa_3 \sum a_k^3 c_k(L)}, \quad g_k = e^{-z_k \Psi} \int_L^x \frac{e^{z_k \Psi}}{1 - \kappa_3 \sum a_k^3 c_k} dx$$

The g_k s in the equations above are computed by solving the following the dimensionless system of equations

$$(4.28) \quad g_k' + z_k \Psi' g_k - \kappa_4 \sum_k a_k^3 \phi_k g_k = 1 + \kappa_3 \sum_k A_k a_k^3 e^{-z_k \Psi}, \quad g_k(2) = 0$$

The domain of the problem goes from 0 to 2. The standard Chebyshev grid goes from -1 to 1. The boundary of our physical rescaled domain at 0 is mapped to $x = 1$ boundary of the Chebyshev grid and the boundary at 2 is mapped to $x = -1$ boundary of the Chebyshev grid. Since the catalyst is at 0, the boundary layer appears at the $x = 1$ on the Chebyshev grid. 120-170 grid points are typically used for the Chebyshev grid.

The General Modified Poisson-Boltzmann equation with flux has a region of rapid variation in our numerical domain at $x = 1$. The reason for this region of rapid variation is that electrostatic forces and the diffusion related flow occur over two widely different scales. The mapping discussed in the previous section would allow us to "zoom" into this region of rapid variation. The domain x has a sharp region of rapid variation which is present in the actual physical domain. The domain s allows us to express our numerical functions without as sharp of a region of rapid variation. Hence, s is the domain over which the General Modified Poisson-Boltzmann equation with flux is solved. So, we first define $s \in [-1, 1]$ as a Chebyshev grid with typically 100-200 grid points. We then find our corresponding x domain by using equation 4.20. Both $x = 1$ and $s = 1$ correspond to the boundary at the Outer Helmholtz Plane (OHP). $x = -1$ and $s = -1$ correspond to the boundary at the bulk.

The General Modified Poisson-Boltzmann equation with flux is a nonlinear steady-state problem. An iterative method like Newton's method is a typical approach to solving it. We start with an initial guess for Ψ defined as some function on x that satisfies at least the boundary conditions $\Psi(x/s = 1) = \Psi_{OHP}$, $\Psi(x/s = -1) = 0$ where Ψ_0 is the rescaled potential difference between the potential at the Outer Helmholtz Plane (OHP) and the potential in the bulk region.

Once we have a guess for Ψ , we can use that to compute g_k s by solving the system of equations in equation 4.28. Equation 4.28 can be solved by a time-stepping scheme with the 'initial condition' $g_k(x = -1) = 0$. Please note that although the system of ordinary differential equations are solved using a time integrator, the domain over which they are solved is a spatial domain and does not have any time. The path/domain for the time-stepping method should include all the Cheybshev grid points in s in order to accurately obtain the g_k values at those points for solving equation 4.26. It is prudent to include more points between the Chebyshev grid points in the domain for the time integrators so g_k s are sufficiently resolved. The value of Ψ at any point within the Chebyshev grid (not just the grid points) can be calculated using the Clenshaw algorithm [117].

With Ψ and g_k known, we can compute the right-hand side of equation 4.26. We can use the Chebyshev differentiation matrices to compute $\frac{d^2\Psi}{ds^2}$. The left hand side of equation 4.26, $\frac{d^2\Psi}{dx^2}$ can then be calculated using

$$(4.29) \quad \frac{d^2\Psi}{dx^2} = \frac{d^2\Psi}{ds^2} \left(\frac{ds}{dx} \right)^2 + \frac{d\Psi}{ds} \frac{d^2s}{dx^2}$$

Using a Newton step and equation 4.26, we can find the next iteration of Ψ and continue this process until we find a Ψ function that satisfies equation 4.26 to some relative or absolute algebraic error.

4.6. Numerical results with a specified current density

A standard approach to incorporate the flux of reacting species in an electrochemical reaction is to simply set a current density value j like in [58]. Suppose we specify the current density j to be 0.1 mA/cm^2 . Then, the flow of the reacting species in the electrochemical reaction (H^+ , O_2 and H_2O_2) would be given by j/n_k where n_k refers to the number of moles of electrons involved in the reaction per mole of species k produced or consumed. For convenience, let us refer to j/n_k as the ionic density of species k .

Species	H^+	CS^+	OH^-	SO_4^{2-}	O_2	H_2O_2
Bulk concentrations (mol/L)	0.01	0.19	1e-12	0.1	0.00025	0.0
Charges	1	1	-1	-2	0	0
Volumes (nm^3)	0.175616	0.28489	0.216	0.43552	0.064	0.16638
Ionic Density (mA/cm^2)	-1	-	-	-	-0.5	0.5
Diffusivity ($10^{-9} \text{ m}^2/\text{s}$)	9.311	-	-	-	1.5	1.6

Table 4.1. Reaction environment details for simulating the Modified Poisson-Boltzmann equation with flux.

If we specify the potential difference between the Outer Helmholtz Plane and the bulk to be -0.12 V , for the reaction environment specified in Table 4.1, we can solve the Poisson-Boltzmann equation with flux to obtain the potential Ψ . Figure 4.7 shows the profile of the potential Ψ across the microscale domain.

One can see a very sharp boundary in Figure 4.7. Despite the sharp boundary layer, we are able to get a well-resolved potential profile at the nanoscale level with the help of the mapping (Equation 4.20) as seen in Figure 4.8.

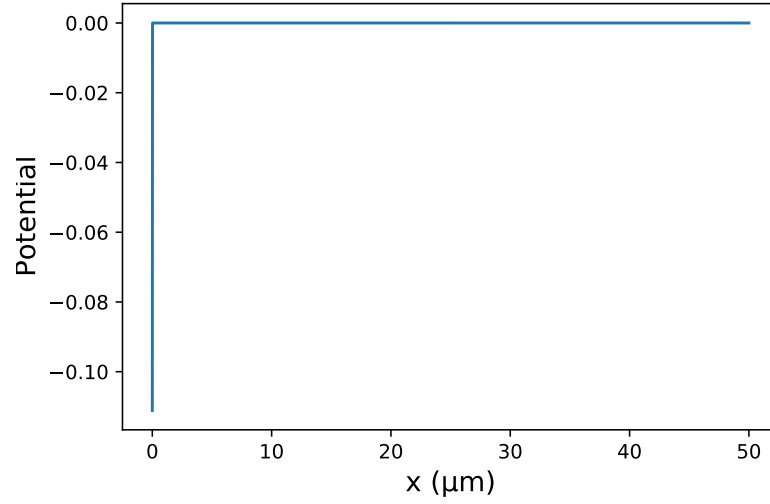


Figure 4.7. The solution to the General Modified Poisson-Boltzmann equation with flux - The potential distribution across the domain.

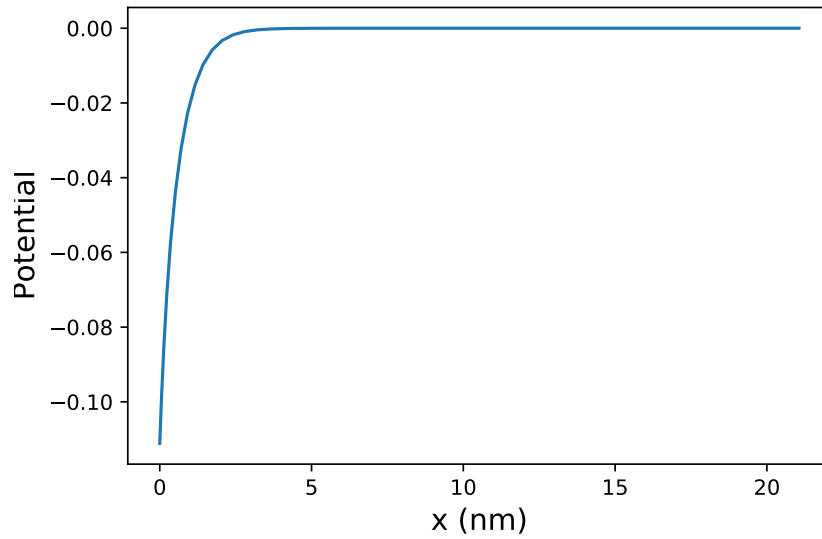


Figure 4.8. The potential distribution at the nanoscale level.

An α_1 value of 300 is used for the mapping (Equation 4.20) in the figure 4.7. Note that the α_1 value for the mapping (Equation 4.20) needs to scale with the length of the domain in order to sufficiently resolve the boundary layer .

From the potential shown in Figure 4.7 for the domain of length $L_{bulk} = 50\mu m$ and current density of $0.1mA/cm^2$, we can recover the concentration profiles of the various species over the domain using the following equation which can be derived from Equation 3.36 .

$$(4.30) \quad \tilde{c}_k = \frac{\tilde{A}_k e^{-z_k \beta \tilde{\Psi}} + \tilde{\phi}_k \tilde{g}_k}{1 + \sum N_A \tilde{a}_k^3 \tilde{A}_k e^{-z_k \beta \tilde{\Psi}} + \sum N_A \tilde{a}_k^3 \tilde{\phi}_k \tilde{g}_k}$$

The non-dimensional version of the formula above is

$$(4.31) \quad c_k = \frac{\tilde{c}_k}{c_0} = \frac{A_k e^{-z_k \Psi} + \kappa_5 \phi_k g_k}{1 + \kappa_3 \sum a_k^3 A_k e^{-z_k \Psi} + \kappa_4 \sum a_k^3 \phi_k g_k}$$

Figure 4.9-4.14 show the concentration profiles of the various species in the electrolyte at the microscale domain and at the nanoscale domain right near the electrode. All the concentration profiles plotted in this thesis have units of M (mol/L).

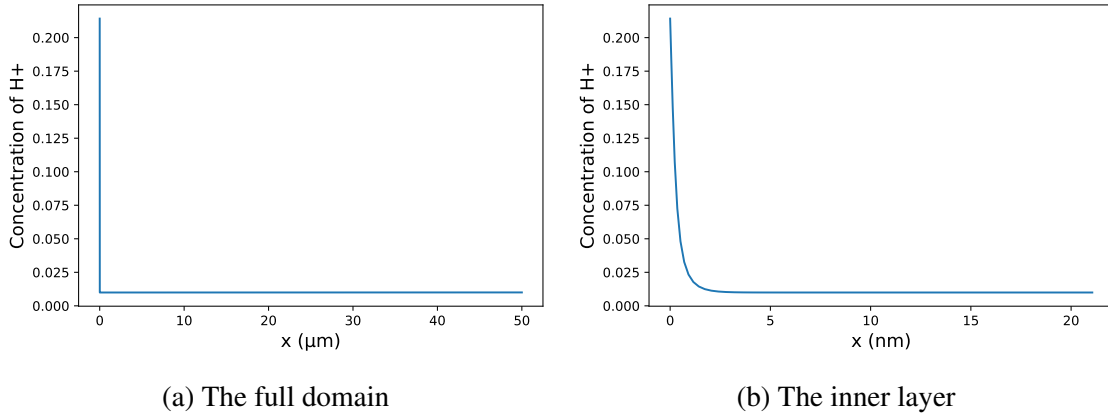


Figure 4.9. The concentration profile of H^+ across the full domain and zoomed in near the electrode.

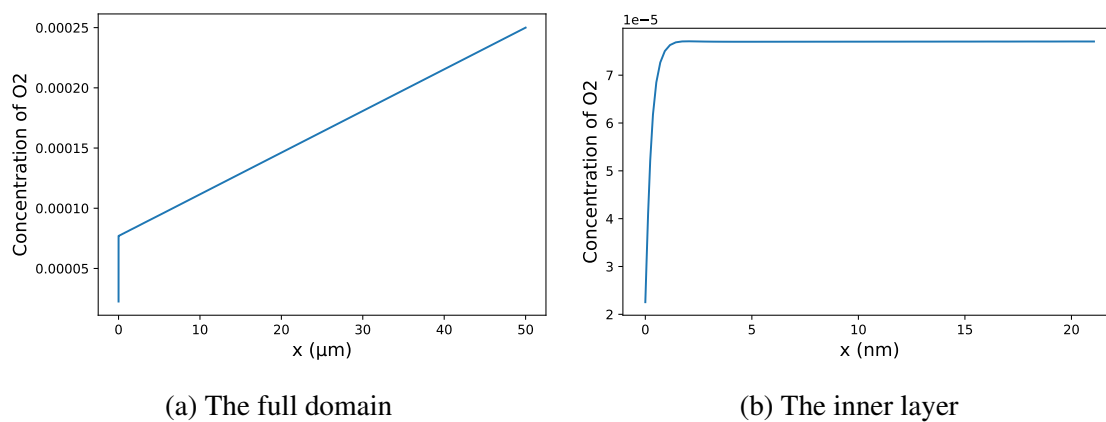


Figure 4.10. The concentration profile of O_2 across the full domain and zoomed in near the electrode.

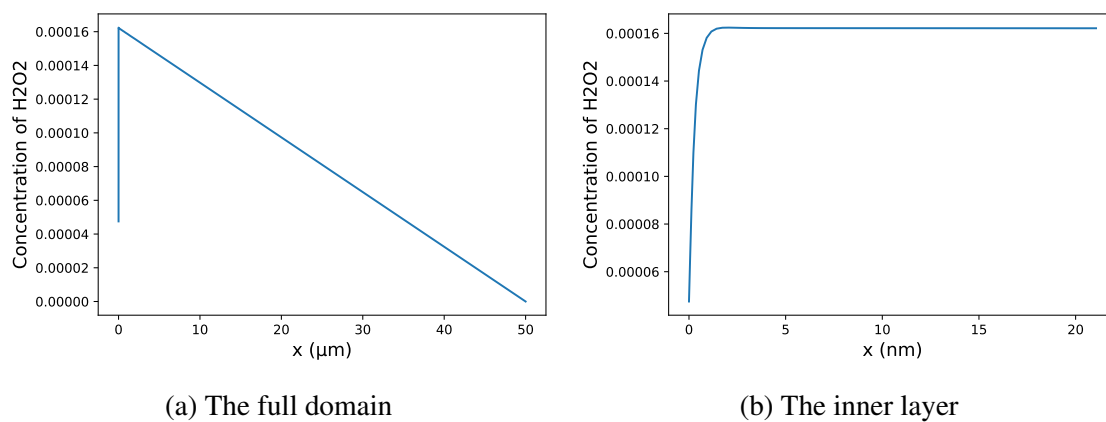


Figure 4.11. The concentration profile of H_2O_2 across the full domain and zoomed in near the electrode.

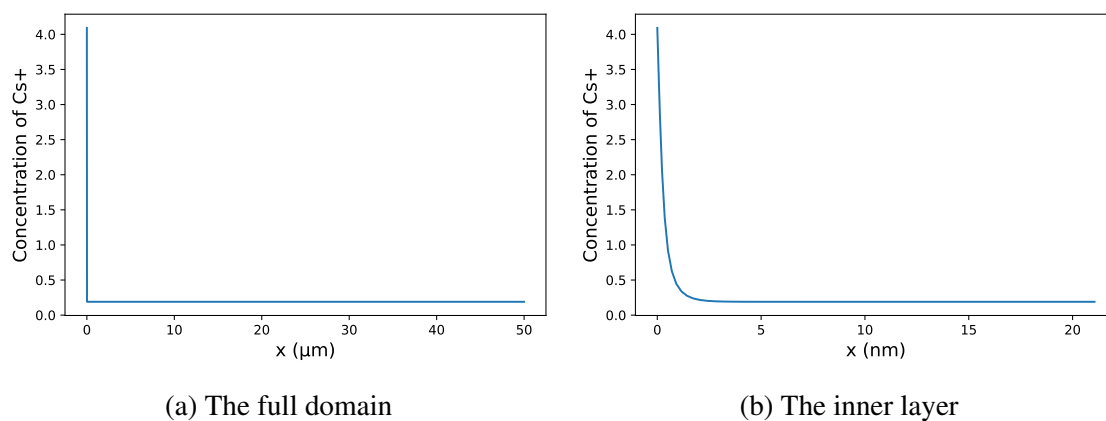


Figure 4.12. The concentration profile of Cs^+ across the full domain and zoomed in near the electrode.

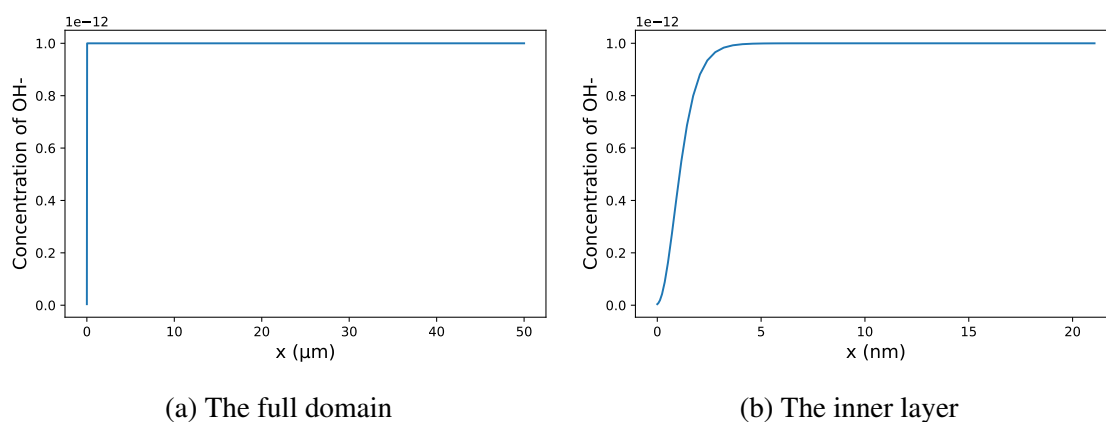


Figure 4.13. The concentration profile of OH^- across the full domain and zoomed in near the electrode.

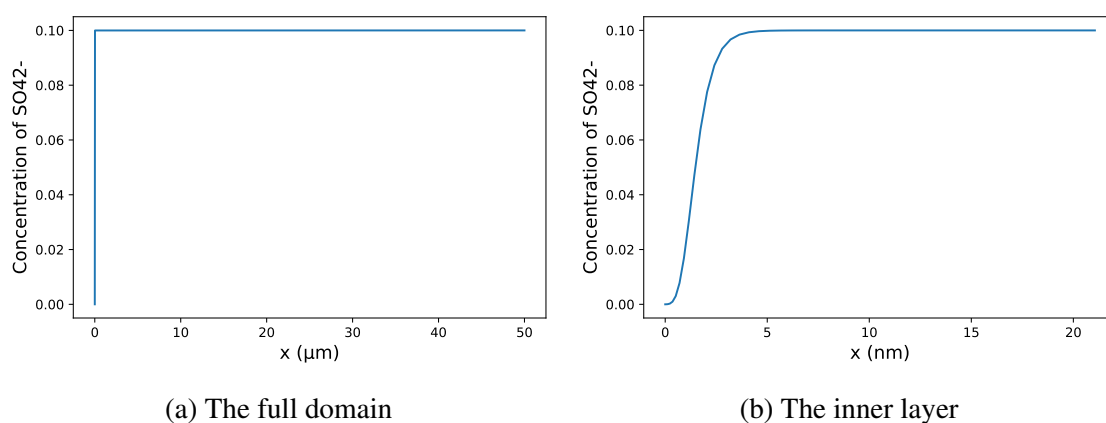


Figure 4.14. The concentration profile of SO_4^{2-} across the full domain and zoomed in near the electrode.

If we use a different cation instead of Cs^+ , the other species are impacted differently through their steric interactions. Table 4.2 shows the volumes of the different cation options.

Species	Cs^+	K^+	Na^+	Li^+
Solvated ionic diameters (nm)	0.658	0.662	0.716	0.764
Volumes (nm^3)	0.28489	0.29012	0.36706	0.44594

Table 4.2. Volumes associated with the various cations used in the ORR.

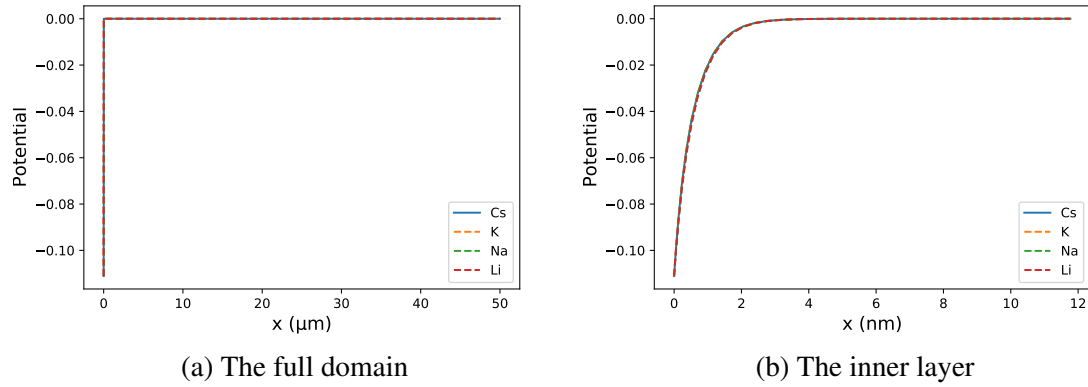


Figure 4.15. The potential profile across the full domain and zoomed in near the electrode.

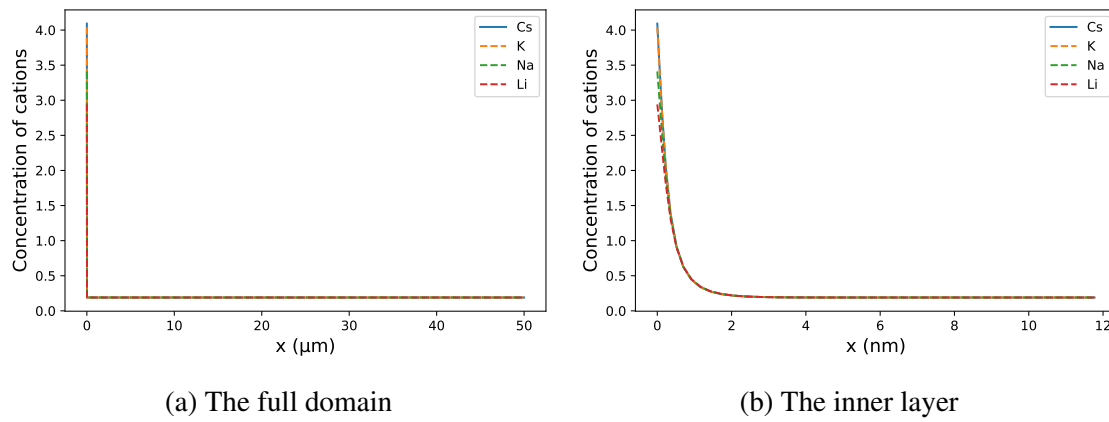


Figure 4.16. The concentration of the various cations across the full domain and zoomed in near the electrode.

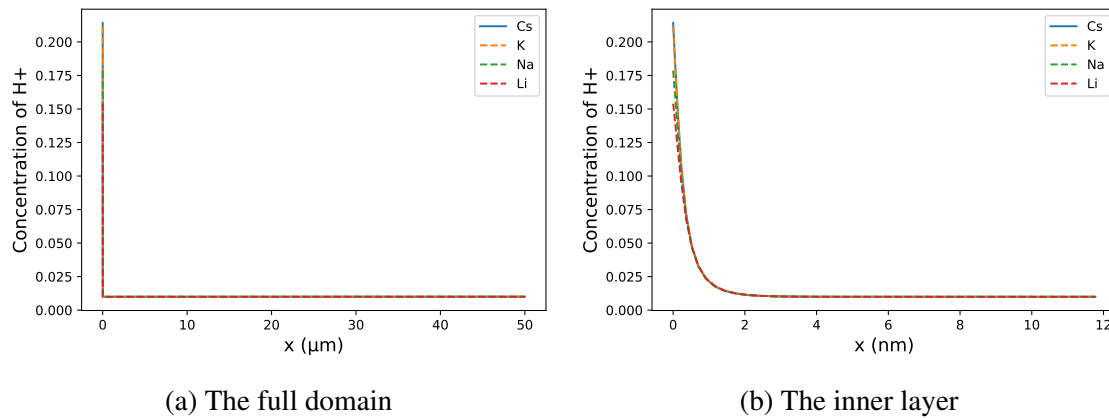


Figure 4.17. The concentration of H^+ across the full domain and zoomed in near the electrode.

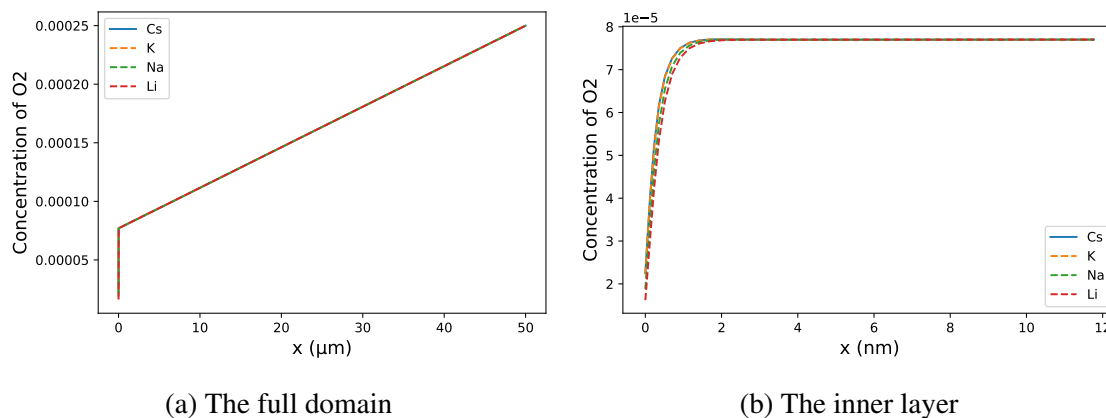


Figure 4.18. The concentration of O_2 across the full domain and zoomed in near the electrode.

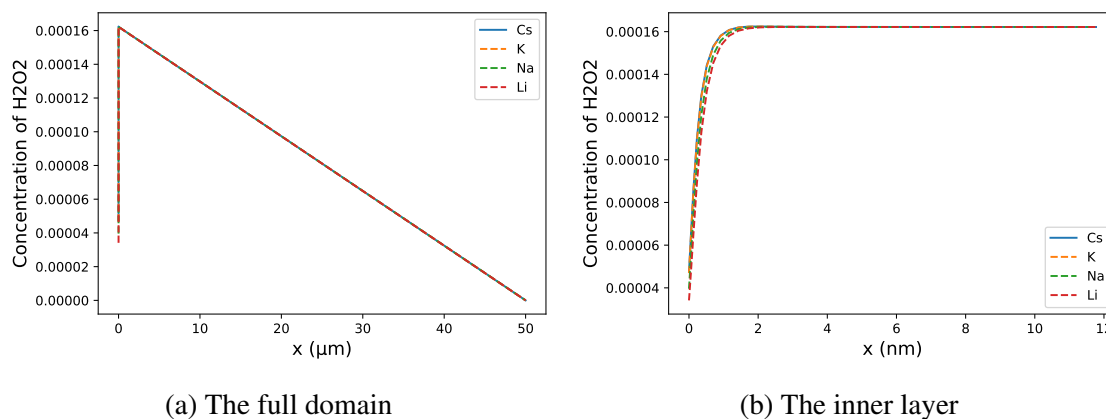


Figure 4.19. The concentration of H_2O_2 across the full domain and zoomed in near the electrode.

We can see in Figure 4.15 the potential profile is not visibly impacted by changing the cation species at this applied voltage. We can however see that the different concentrations of the cations near the electrode (Figure 4.16) do impact the local concentrations of oxygen, protons and peroxide (Figure 4.17 - 4.19).

This impact is more clearly visible in Figure 4.20 when a larger potential difference of $-0.3333V$ is specified at the the outer Helmholtz plane. The increased potential attracts the

protons and the cations to their steric limit and creates an exclusion region where O_2 species cannot enter. Since the current density is pre-specified in this simulation, the H_2O_2 production remains undiminished despite O_2 being occluded from the catalyst surface by the positive ions.

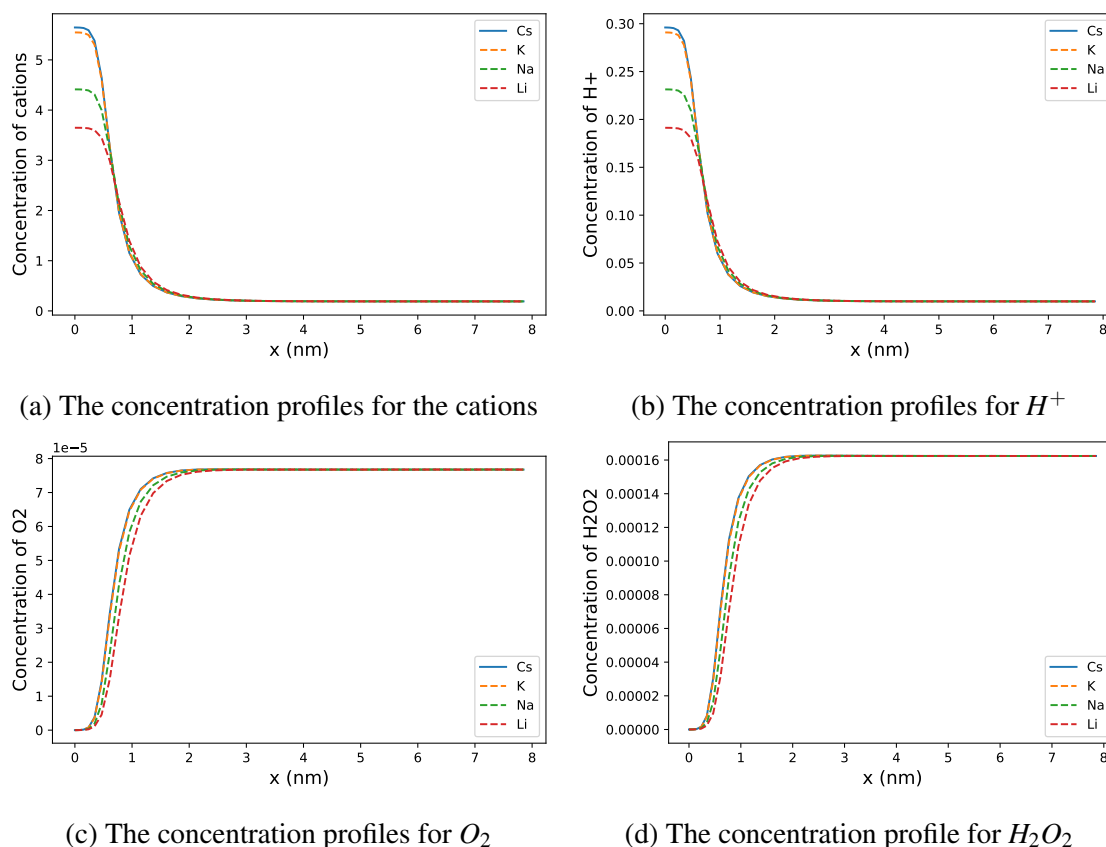


Figure 4.20. The concentration of the various species near the electrode for a larger potential difference.

If we try running the simulations with a larger current density of $0.2mA/cm^2$, we see non-physical results. While the concentration profile of H^+ looks relatively unchanged in Figure 4.21, we see that the increased current density results in increased H_2O_2 production in Figure 4.22. The oxygen concentration in Figure 4.23 is especially surprising because the concentration reaches negative values. Not only are the negative concentration values unrealistic, it also

means that there is no oxygen available at the electrode for the ORR. The H_2O_2 production increases despite the lack of oxygen needed for the reaction (Figure 4.22) This suggests that specifying the current density beforehand is too strong of a constraint for our model and it can result in over-determined and unrealistic scenarios. This highlights a clear need for the current density to be modelled a non-negative function which is dependent on the amount of oxygen available near the electrode for the ORR.

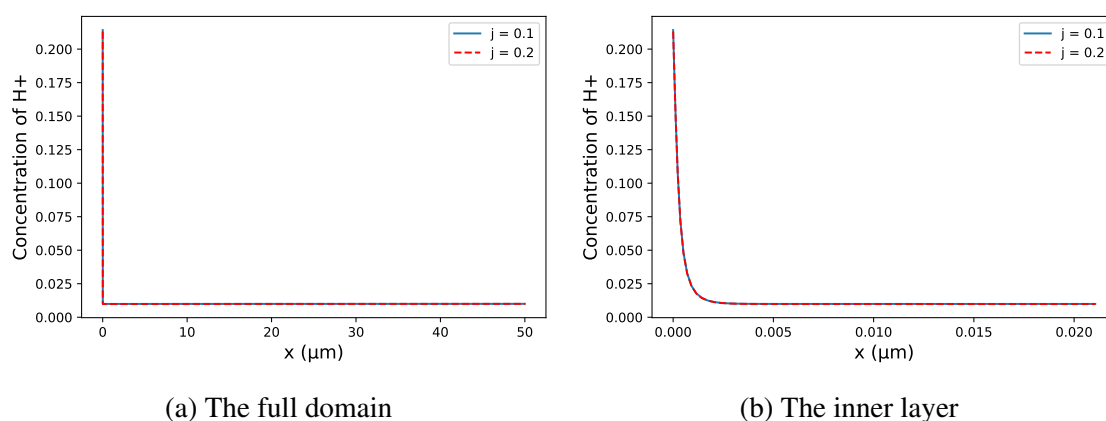


Figure 4.21. The concentration profile of H^+ across the full domain and zoomed in near the electrode.

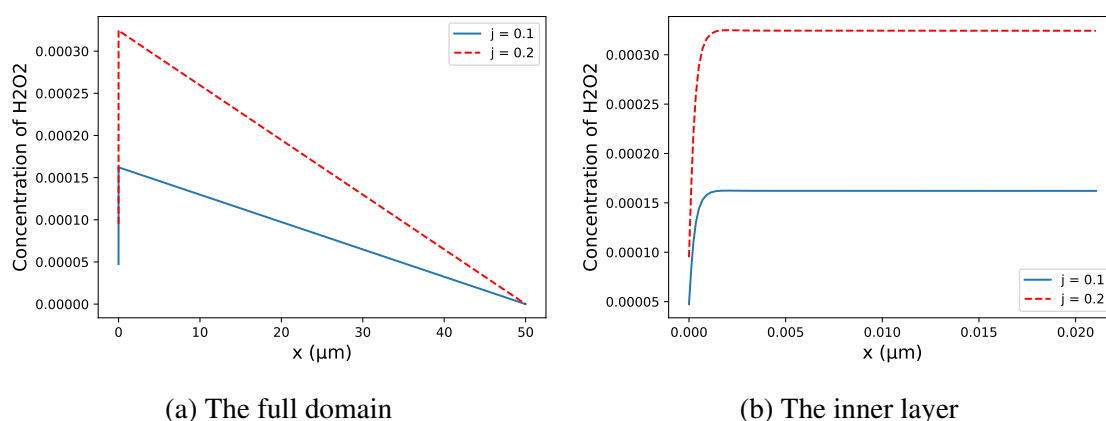


Figure 4.22. The concentration profile of H_2O_2 across the full domain and zoomed in near the electrode.

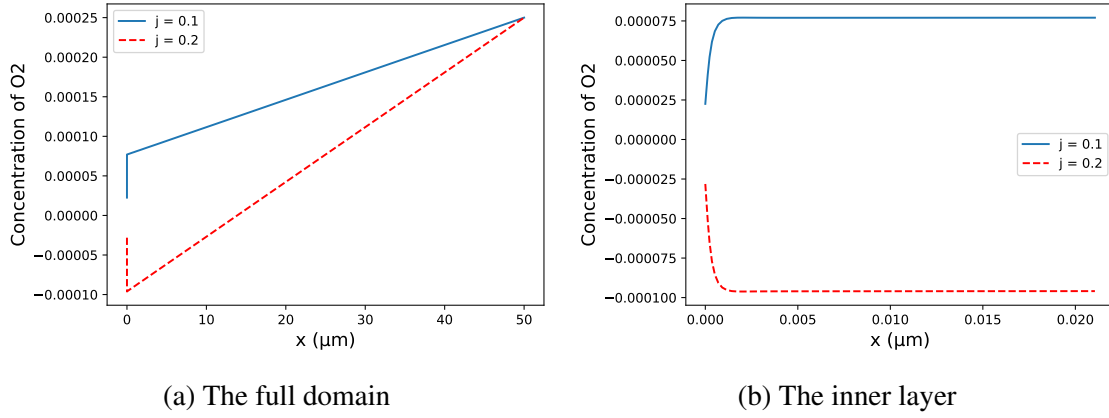


Figure 4.23. The concentration profile of O_2 across the full domain and zoomed in near the electrode.

4.7. Incorporating Butler-Volmer dynamics into the Modified Poisson-Boltzmann equation with flux

In this section, we discuss how to incorporate the fluxes ϕ_k s of the various species into our model without explicitly specifying a current density by using the extended Butler-Volmer equation. The extended Butler-Volmer equation, given by Equation 3.46, expresses the current density (and thus the effective fluxes) in the electrochemical system as a function of applied potential and the concentration of the limiting species at the electrode surface.

Incorporating the extended Butler-Volmer relation into our model results in Equation 4.32, the Butler-Volmer consistency condition which must be satisfied by the concentration of oxygen at the Outer Helmholtz Plane.

$$(4.32) \quad g_{O_2}^{*'} c_{O_2}^* = A_{O_2} + \kappa_5 \frac{1}{2D_{O_2}} j_0 \frac{c_{O_2}^*}{c_{O_2}^b} \exp\left(\frac{\log(10)|\eta|}{A_{Tafel}}\right) g_{O_2}^*$$

At first glance, one might think of extracting an closed-form expression for $c_{O_2}^*$ by rewriting the above equation as

$$(4.33) \quad c_{O_2}^* = \frac{A_{O_2}}{g_{O_2}^{*'} - \kappa_5 \frac{1}{2D_{O_2}} j_0 \exp\left(\frac{\log(10)|\eta|}{A_{Tafel}}\right) g_{O_2}^*}$$

However, $g_{O_2}^{*'} (and thus $g_{O_2}^*$) depend recursively on the other g_k s .$

$$(4.34) \quad g_{O_2}^{*'} - \kappa_4 \sum_k \phi_k N_A a_k^3 g_k = 1 + \kappa_3 \sum_k A_k N_A a_k^3 e^{-z_k \Psi}$$

The g_k s all depend on the effective fluxes ϕ_k s and thus on $c_{O_2}^*$ through the extended Butler-Volmer relation.

$$(4.35) \quad g_k^{*'} = 1 - z_k \Psi' g_k + \kappa_3 \sum_k A_k N_A a_k^3 e^{-z_k \beta \Psi} + \kappa_4 \sum_k \phi_k N_A a_k^3 g_k$$

Because of this dependence, we have to use an iterative root-finding method to calculate the $c_{O_2}^*$ value and the $g_k(c_{O_2}^*)$ values that satisfy Equation 4.32 .

The dependence of g_k s on the effective fluxes ϕ_k s (and thus on $c_{O_2}^*$) is weak because in Equation 4.35, κ_4 which appears in the coefficient of the term which contains ϕ_k is typically on the order of 10^{-3} for our problem. So, Equation 4.33 serves a very good initial guess for the iterative rooting finding process. Once $c_{O_2}^*$ is estimated, the effective fluxes can be calculated through the current density given by Equation 3.46.

4.8. Numerical results with fluxes defined through the Butler-Volmer relation

When using a pre-specified current density in Section 4.6, we saw that steric occlusion of reacting species at the OHP had no impact on H_2O_2 production and also lead to negative concentration profiles. We know in our problem that the rate of the reaction and the current density depends on the O_2 available near the electrode and must be limited by it. Thus, incorporating flux through the extended Butler-Volmer relation, which couples the current density to the concentrations and vice versa, is more representative of the underlying physics.

In our model, the potential difference is specified through the voltage applied at the electrode $\Psi_{electrode}$ and the open circuit potential (the potential in the bulk region) which is 0.785 V from experimental measurements. The current density is given by the extended Butler-Volmer dynamics (Equation 3.46), where j_0 and A_{Tafel} can be estimated from experimental data over a range of applied electrode potential.

For the reaction environment specified in Table 4.3, we simulate the potential and concentration profiles for a range of applied electrode potentials. We can compare the current density from our simulation with experimentally obtained current densities for the same applied potentials. Please note that all the current density values have the units mA/cm^2 and the potential values are in volts.

Species	H^+	CS^+	OH^-	SO_4^{2-}	O_2	H_2O_2
Bulk concentrations (mol/L)	0.01	0.19	1e-12	0.1	0.00025	0.0
Charges	1	1	-1	-2	0	0
Volumes (nm^3)	0.175616	0.28489	0.216	0.43552	0.064	0.16638
Diffusivity ($10^{-9} m^2/s$)	9.311	-	-	-	1.5	1.6

Table 4.3. Reaction environment details for simulating the Modified Poisson-Boltzmann equation with flux.

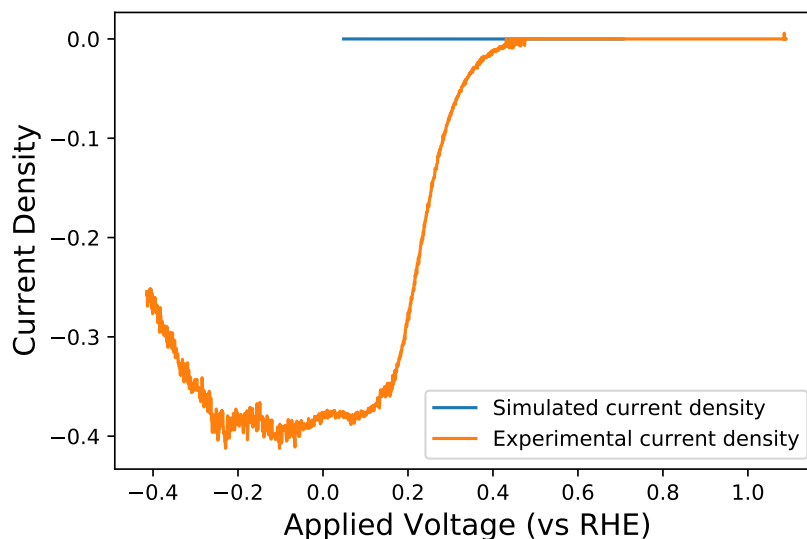


Figure 4.24. Experimental and simulated current density vs voltage (jV) curves.

Figure 4.24 compares the experimental current density vs the experimentally applied voltage at the electrode, the jV curve to that obtained through our simulations. It appears as though there is no current produced in our simulations (and thus no H_2O_2 production) as opposed to considerable experimental H_2O_2 production.

Looking closer at our simulated jV curve in Figure 4.25, we see that there is some amount of current density and reaction happening in our simulations but it is negligible in comparison to the experimental current density. The simulated current density appears to decay to zero for high overpotentials (as the applied electrode goes further to the left of the thermodynamic reduction potential of 0.695V). The experimental current density appears to reach a limiting value at high overpotentials in Figure 4.24.

If we look at the concentration of oxygen present at the OHP in our simulations, we can see in Figure 4.26 that it decays to zero at high overpotentials. This means that there is less oxygen

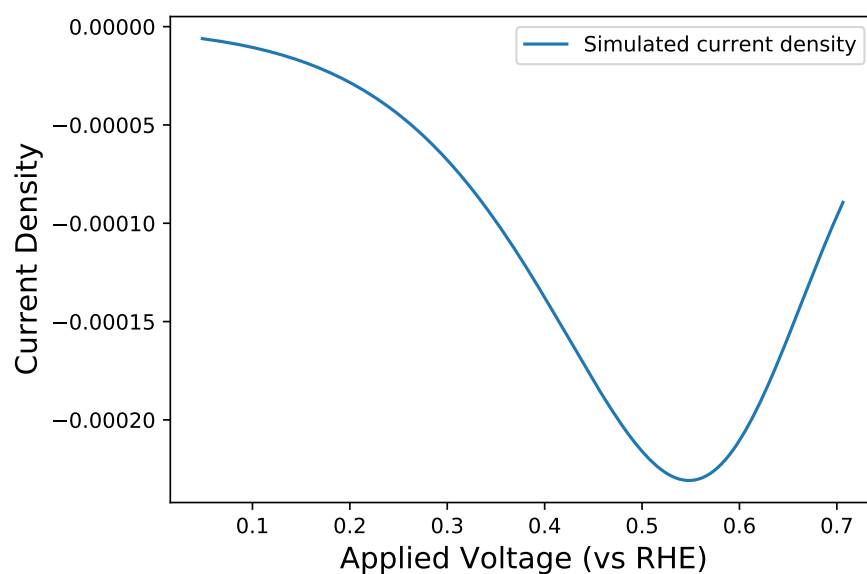


Figure 4.25. Simulated jV curve.

available for the reaction at high overpotentials. This is why the current density decays to zero at high overpotentials. in Figure 4.25.

In order to understand why concentration of oxygen present at the OHP decays to zero at high overpotentials, we look at two simulations in Figures 4.27 and 4.28 , one at high overpotential with applied voltage at 0.3 V vs RHE and one at low overpotential with applied voltage at 0.66 V vs RHE.

In Figure 4.27, we see that the concentration of oxygen drops to zero at high overpotential, while for lower overpotential, the drop is not so drastic. Furthermore, at high overpotential, there seems to be an occlusion layer which beyond which oxygen cannot get closer to the OHP. Looking at Figure 4.28, we see that the proton concentration reaches its steric limit at the high overpotential and forms a small layer next to the OHP where its concentration is at its maximum possible value. The same behaviour is seen for the other cation (Cs^+) in the system (not shown).

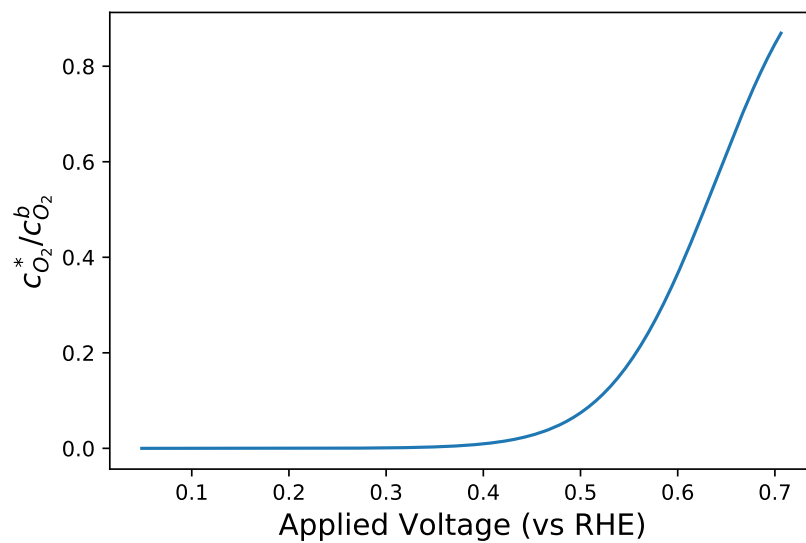


Figure 4.26. The ratio of O_2 concentration at the OHP vs the bulk O_2 concentration for a range of applied electrode potentials.

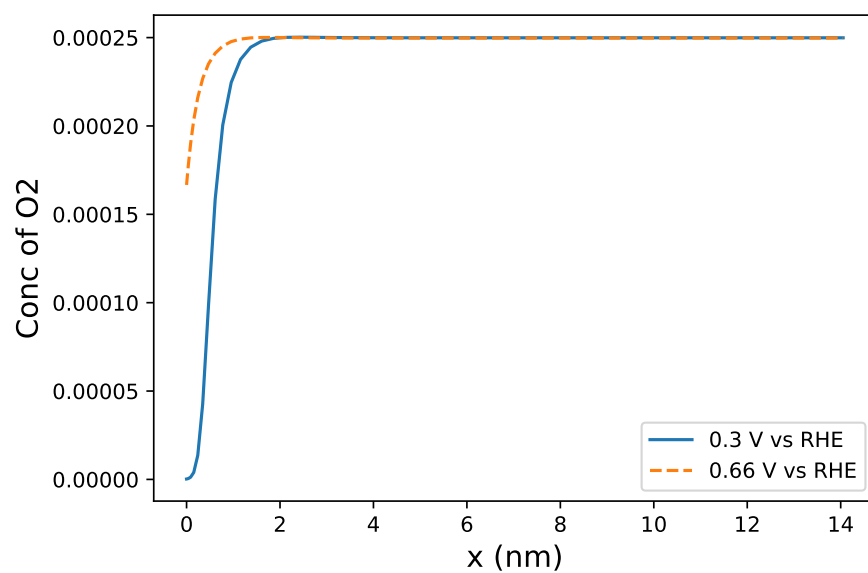


Figure 4.27. Concentration profiles of O_2 at two different overpotentials.

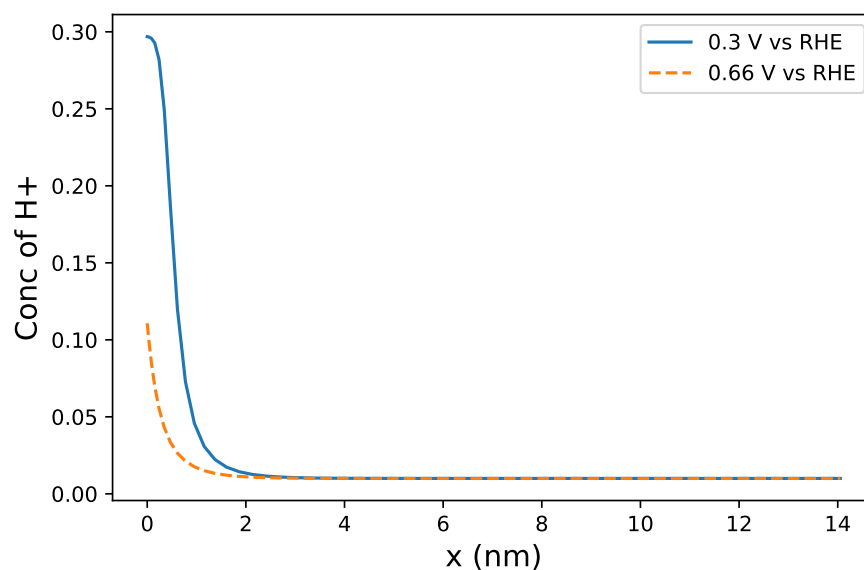


Figure 4.28. Concentration profiles of H^+ at two different overpotentials.

It seems as though the steric effects of the H^+ and Cs^+ ions dominate our model more than they should, resulting in a current density which is quite different from the experimentally seen current density. Perhaps our mathematical model needs to be modified in order to realistically capture the steric effects. An obviously direct way to reduce the overdominance of steric effects is to simply reduce the sizes of the various species involved. There has been a long and unresolved debate about how to incorporate size effects [79, 84, 85] and what the ‘correct’ sizes of the various species even means for a realistic electrochemical model. In one of the most recent and comprehensive reviews, Bazant and co-authors show in [74] discuss and compare the bare ionic diameter, the solvated diameter and the ‘hard-sphere’ diameter of an ion to the experimental data of [118] in an attempt to understand what an appropriate effective ion size is. This work was back in 2009 showing this is a recent and ongoing debate in this very young field.

Despite the debate on realistic atomic radii, we do not have rely solely on changing the volumes in our model to match experimental trends. Even in our simplistic model, we have a lot of parameters that can be varied slightly to make a big difference in the simulated current density. Consider for example, the impact of changing the length of the Stern layer. It has become almost standard practice [58, 77] when simulating models like these to assume that the Stern layer is of length 0.4nm . In reality, the width of the Stern layer is far from set in stone. Classic textbooks like [69] describe the OHP to be "locus of the centers of the nearest solvated ions". Other works describe the use of multiple Stern layers corresponding to cations of different sizes [90]. Since the solvated diameters of the various cations in our system (given in Table 4.2) are bigger than 0.4nm , this lends some argument towards increasing the size of the Stern layer in our model.

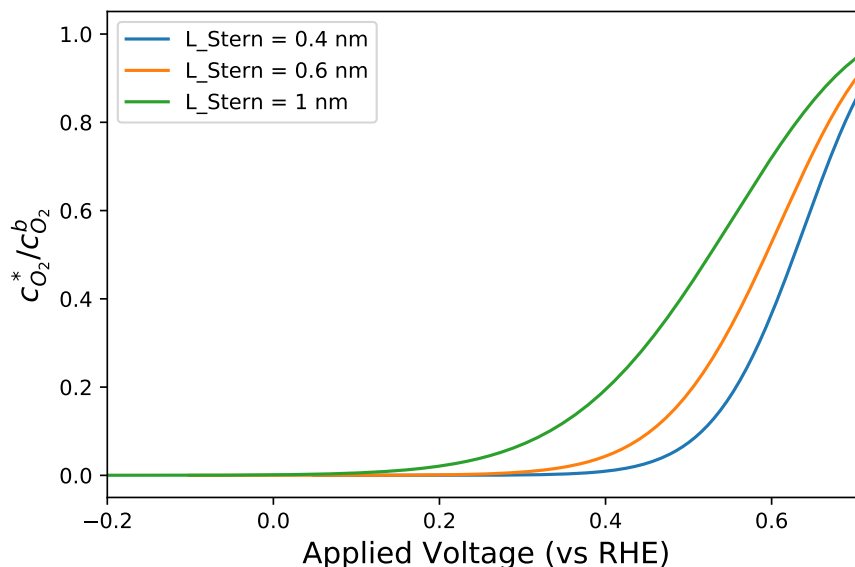


Figure 4.29. The ratio of O_2 concentration at the OHP vs the bulk O_2 concentration for different lengths of the Stern layer.

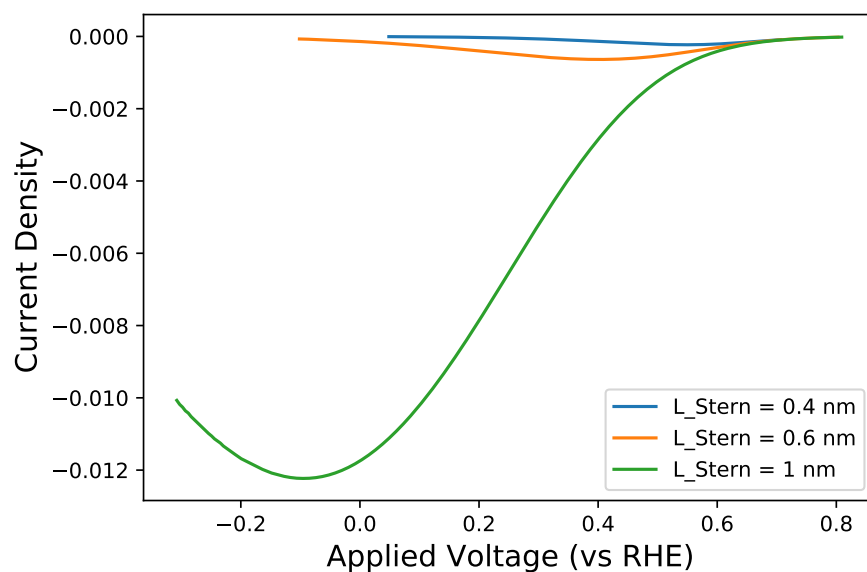


Figure 4.30. Simulated jV curves for different Stern layer lengths.

In Figure 4.29, we can see that changing the Stern layer from 0.4nm to 0.6nm and 1nm has a visible improvement on the concentration of oxygen available at the OHP for reaction. We can see in Figure 4.30 that this, in turn, has a big impact on the current density returned through our simulations. However, even with a Stern layer of length 1nm , the simulated current density is still quite far from the experimental current density. The simulated current density also seems to decay at high overpotentials instead of reaching a limiting plateau.

Despite changing the Stern layer, the steric effects still appear to be too dominant in our model. In Figures 4.31 and 4.32, we explore what happens when the volumes of the species in our model are reduced by $\frac{1}{4}$, $\frac{1}{10}$ and $\frac{1}{100}$ to become $\frac{a_k^3}{4}$, $\frac{a_k^3}{10}$ and $\frac{a_k^3}{100}$. We see in Figure 4.31 that this results in higher concentrations of O_2 being able to reach the OHP at high overpotentials. Furthermore, for the cases of $\frac{a_k^3}{10}$ and $\frac{a_k^3}{100}$, the steric occlusion effects seem to only become dominant to the left of the applied voltage of 0V . This results in current densities that are in the

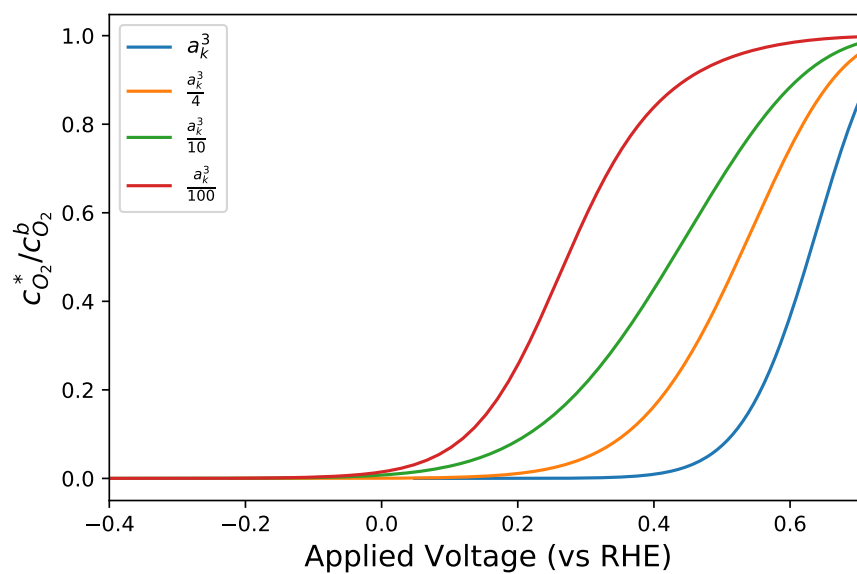


Figure 4.31. The ratio of O_2 concentration at the OHP vs the bulk O_2 concentration for species of different volumes

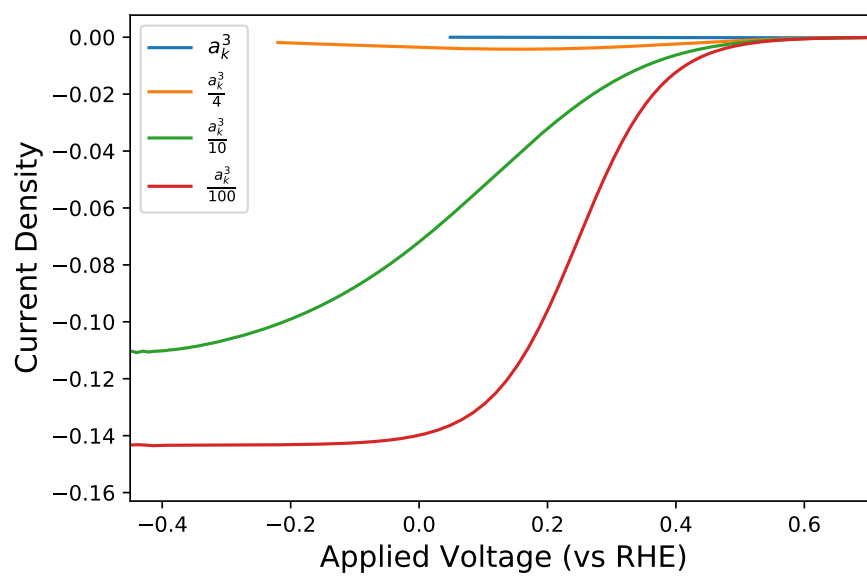


Figure 4.32. Simulated jV curves for species of different volumes.

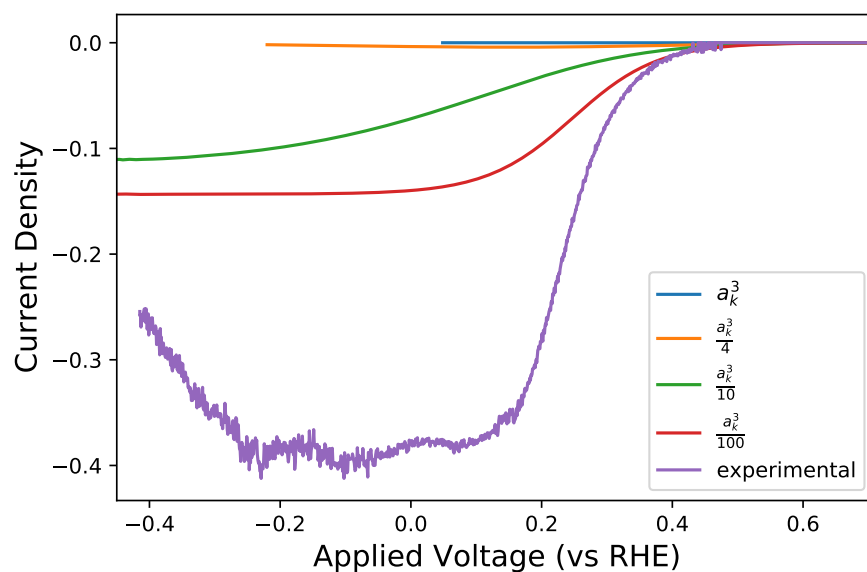


Figure 4.33. Simulated jV curves for species of different volumes compared with the experimental jV curve.

same order as the experimental data as seen in Figure 4.32. We also see that for the case of $\frac{a_k^3}{100}$, a limiting plateau is reached. However, this plateau of -0.14 mA/cm^2 is still far from the experimental limiting current density of -0.386 mA/cm^2 as seen in Figure 4.32.

Reducing the volumes of the various species in our model to a hundredth of their original values is perhaps too drastic of a step to make in a mathematical model. A model like ours is impractical if it is far from realistic. It is worth pondering if a combination of smaller changes to different effects could result in a realistic model. For Figure 4.34, we simultaneously reduced the volumes of the various species to a tenth of their original values and increased the length of their Stern layer to 0.6 nm . This combination of changes results in the current density reaching the same limiting value as the red curve in Figure 4.33 without needing to reduce the volumes of the species to a hundredth of their original value.

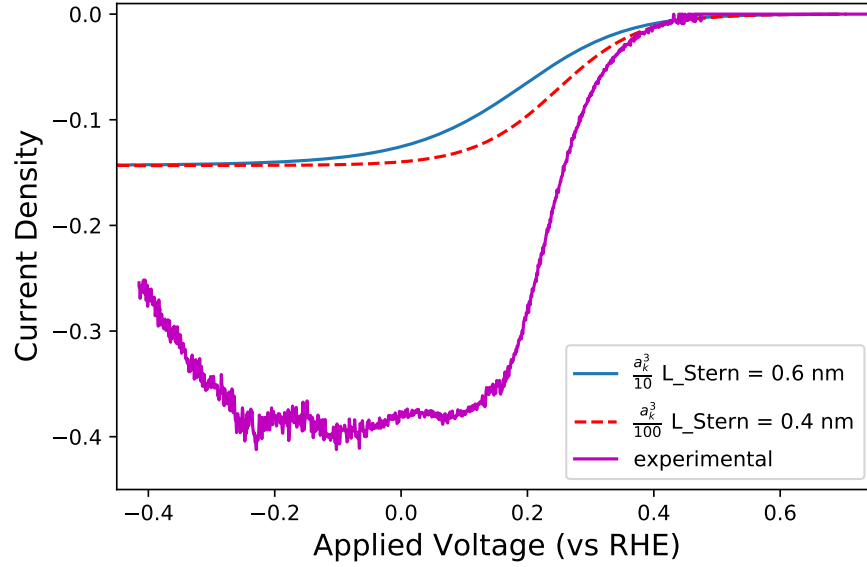


Figure 4.34. Simulated jV curves for species of different volumes compared with the experimental jV curve.

The limiting current density from in our simulations appears to be around -0.14 mA/cm^2 while the experimental current density is around -0.386 mA/cm^2 . We can do a simple back-of-envelope calculation to see why this is the case. Let us try to estimate the maximal current density for which the oxygen from the bulk is completely depleted by the time it reaches the diffuse layer. In the region between the diffuse layer and the bulk, we can assume that the potential is approximately zero (like we did in Section 4.3). The concentration profile of oxygen can then be approximated by

$$(4.36) \quad c_{O_2}(x) = c_{O_2}(L_{\text{bulk}}) + \frac{j}{n_k D_k F} (L_{\text{bulk}} - x)$$

To approximately calculate the concentration of O_2 at the boundary of the diffuse layer, we can set x to 0 in the above equation

$$(4.37) \quad c_{O_2}(0) = 0.25 + \frac{jL_{bulk}}{2 \times 1.5 \times 10^{-9} \times 96485}$$

All values above are in SI units. If we set L_{bulk} to be $50\mu m$ and j to $-0.144728mA/cm^2$, the concentration of oxygen goes from $0.25 mol/m^3$ at the bulk to $0 mol/m^3$ at the diffusion layer. This means that for this bulk concentration and length of the diffusion layer, the maximal current density is approximately $-0.144728mA/cm^2$. For this current density, all of the oxygen in the bulk is consumed by the reaction by the maximum rate allowed by diffusion. Hence, in order to further match our experimental results, we would have to change more assumptions of our model, particularly our assumptions about the length of the diffusion layer and the bulk concentration of oxygen. If we set the length of the diffusion layer in our model to be $10\mu m$, we can see in Figure 4.35 that the simulated limiting current density gets a lot closer to the experimentally observed limiting current density.

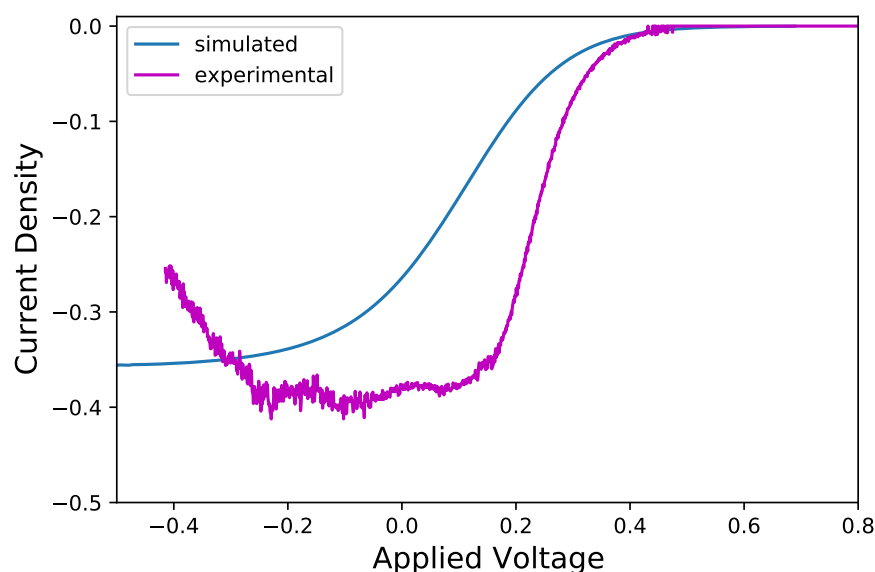


Figure 4.35. Simulated jV curve when the diffusion layer is set to have a length of $10\mu\text{m}$ compared with the experimental jV curve.

We still see in Figure 4.35 that despite being close to the experimental limiting current density, the rate of change of the simulated current density with respect of applied potential is qualitatively different from that seen experimentally. The rate of change of the simulated current density in our model depends heavily on the Tafel slope (and the exchange current density) in Equation 3.46. Although these are parameters that can be calculated from the experimental data, there is an active debate on what a representative Tafel slope is and how to calculate it [95, 96, 97, 98, 99, 100]. Furthermore, it is debated whether it is the Tafel slope in low overpotential regions or high overpotential regions that provides a better description [100]. We have been using a low overpotential Tafel slope 142 mV/dec in our simulations so far. Switching to a reduced Tafel slope of 88 mV/dec which is more representative of high overpotential regions, we arrive at the simulated current density seen in Figure 4.36.

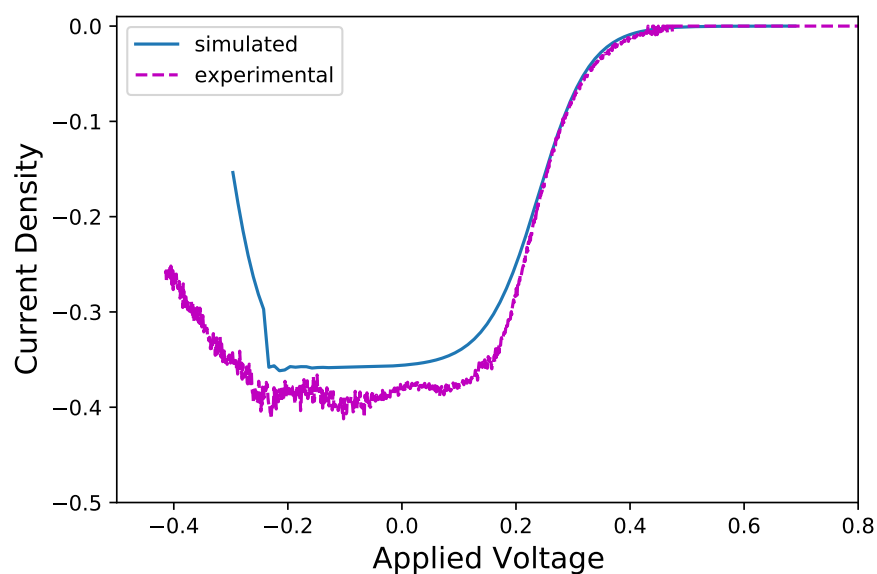


Figure 4.36. Simulated jV curve when the Tafel slope is increased to be 26.2 compared with the experimental jV curve.

In this thesis, we developed a mathematical model, an associated numerical method and qualitatively changed our modeling assumptions until we could capture the behaviour seen experimentally. One thing is for certain, having such an easily tunable virtual laboratory has allowed us to raise many interesting questions about the assumptions that are typically made when modeling electrochemical reactions.

CHAPTER 5

Discussion and Outlook

5.1. On complex coefficients for time integration.

In this thesis, we explored the benefits and trade offs of leaving the real line to perform numerical time integration in the complex plane. Our goal was to explain the process of developing a complex time stepper and its potential benefits in an accessible manner. We demonstrate how to systematically derive paths in the complex plane that increase the order of accuracy of any Runge-Kutta type integrator for general nonlinear differential equations, break the order barrier for real-valued systems, and achieve expanded stability regions. In principle, this process can be followed to derive complex paths with other desired properties.

In Section 2.3, we show that it is possible to get higher order accuracy by taking complex time steps with a single stage method like the Euler method. An unmentioned advantage of this is that we only need to store one stage at every step. For example, third order Runge Kutta methods usually require 3 intermediate stages ($3N$ registers) to stored at every step while the 3-step 3rd order Euler method only needs one stage of size $2N$ (because of the additional N imaginary values). Storage may not be much of an issue for simple systems but it becomes quite relevant in various practical settings like in [119] where partial differential equations like the Navier Stokes equations are solved in 3 dimensions at high resolution. There are already low-storage algorithms for Runge-Kutta schemes that use fewer registers [120, 119, 121] but

complex time integrators can provide an additional axis to create low-storage schemes with desired properties.

It is also possible to derive higher-order paths for specific differential equations. For example, we show in Section 2.6 that the 2-step Forward Euler path with steps $(1 - \frac{1}{\sqrt{2}}i)$ and $(\frac{1}{\sqrt{2}}i)$ is 3rd order accurate for the differential equation $\dot{y} = -y^2$ with real initial conditions. There has been recent work [31] using neural networks to obtain these uniquely accurate integrators for specific problems. Complex steps and coefficients in this context may reveal even better specialized integrators.

Complex time steppers enable us to circumvent the Runge-Kutta order barrier [23]. For real-valued differential equations, complex time steps and complex Runge-Kutta coefficients allow us to increase the number of free parameters available to satisfy the order conditions and use fewer function evaluations to achieve the same order of accuracy. Since the number of order conditions grows rapidly with higher order, we believe the best benefits lie at higher order. For example, we demonstrated a Runge-Kutta-5 method with 5 function evaluations instead of the usual 6. However, an 8th order Runge-Kutta method traditionally requires at least 11 function evaluations to obtain 8th order of accuracy. Complex timesteppers could reduce the number of evaluations to 8.

For complex-valued differential equations, our attempts to find complex coefficients that precisely satisfy the order conditions for 5th-order accuracy with 5 stages were unsuccessful. However, we were able to identify coefficients that satisfied the order conditions to an error of 2.1×10^{-11} , which may be useful from a practical standpoint. Traditionally, coefficients in numerical schemes have been limited to rational numbers that precisely meet order conditions. However, this appears to be quite restrictive. I recently attended a talk titled "The Special Math

of Translating Theory to Software In Differential Equationss” by Chris Rackauckas where he highlighted how differential equations solvers [122] can make great practical use of “traditionally imperfect” numerical coefficients[39]. Considering that the set of complex solutions also encompasses real solutions, it is worthwhile to search for complex time integrators that offer reduced computational cost, particularly in the context of other order barriers [35, 36].

Lastly, we show that complex time stepping can be used to expand stability regions, allowing larger, stable time steps. Given the need for integrators with larger stability regions for simulating quantum systems, where differential equations are complex-valued and complex time-stepping requires little additional cost, this may be the most promising immediate application of our work. Expanded stability regions have been designed using Runge-Kutta methods [41, 45], but we show that complex coefficients in the optimal stability polynomial allow for even larger stability regions, particularly for complex-valued systems. The fact that you can increase the stability region by changing the integration domain rather than the integration method may also be appealing in scenarios where particular integrators with specific properties (such as nonlinear stability) for specific differential equations are desired. Although we only explored linear stability analysis, in light of our numerical results showing improvement for the nonlinear Schrodinger equation, it is possible that complex time steppers may also satisfy nonlinear stability guarantees such as Strong Stability Preserving properties [123, 124].

The work done in this thesis is intended to be an introduction to and initiation of the search for better time integrators that take advantage of paths in the complex plane beyond their current applications [4, 6, 15, 125]. We have mostly focused on creating integrators with increased

accuracy and stability, but there are many other desirable properties including energy and momentum preservation. The complex plane offers an extra dimension on which any integrator can be improved, opening up a new realm of possibilities when designing time-stepping methods.

5.2. On mathematical modeling and numerical simulation for understanding electrochemical reactions.

Typically, when attempting to investigate a scientific phenomenon with significant practical potential, the usual approaches involve experimentation and sometimes, prediction through statistical analysis. The field of electrocatalysis is presently characterized by active and fervent experimental investigations. Recently, there has also been a surge of interest in using statistical predictive models to discover optimal catalysts. For example, [126] provides an overview of attempts by companies like Meta to use deep learning to accelerate density functional theory simulations to predict optimal electrocatalysts. These approaches yield direct and immediately applicable results.

The focus of the work in this thesis is a third type of approach: we attempt to understand and describe the dominant constitutive dynamics that govern electrocatalytic reactions through the help of mathematical modeling and simulation. We tread the path laid by Helmholtz, Gouy, Chapman, and Stern in the early 20th century [65, 66, 67, 68], Bikerman and Borukhov [78, 75] in the late 20th century, and more recently by Kilic, Bazant and Bohra [76, 74, 58], amongst many others. Although slower and more indirect compared to experimentation and statistical prediction, this approach proves to be a valuable investment. The insights gained by this approach become powerful guiding principles for experimentation and often revolutionize predictive abilities by an order of magnitude.

In order to decipher the dominant constitutive laws, a tunable mathematical model and a virtual laboratory is necessary. If the model is overly complex, disentangling the key underlying processes becomes challenging. One would not want to use a relativistic model to compute the trajectory of a ball thrown by a kid. Complex models are also often too difficult and expensive to simulate. In our work, we derived a simplistic steady-state model for the concentrations of various species in an electrochemical reaction that takes into account their finite sizes and their fluxes.

Scientific computing allows us to take this mathematical model and transform it into a virtual laboratory where all sorts of experiments can be run. A reliable numerical framework is necessary for a reliable simulation. It is often surprising how conducive a well-built scientific computing environment is to mathematical insight. In my past, I used a finite volume framework called Clawpack [127] to discover and decipher mysteries associated with tsunamis. While tsunamis and electrochemical reactions are governed by radically different physical laws at vastly different scales, they share one similarity: it is often difficult to measure them experimentally. In this thesis, one of our aims was to build a reliable and robust numerical framework to simulate electrochemical reactions over a wide range of conditions.

Our model is far from finished but the good news is that our numerical framework is malleable to modification. Some of the questions we are interested in exploring include whether asymmetric electrolyte theory is necessary to more realistically capture steric interactions [83, 74] and what would be a minimal yet representative model of non-constant permittivity [80, 128, 129].

It is worth wondering what type of advancement in other fields could massively revolutionize our understanding of electrocatalysis. In industry, there is a lot of hope on deep learning

approaches that have shown promise in a variety of other fields. I am personally interested in the recent advances in improving measurements at the atomic scale. For example, one of the world's most powerful X-ray sources, the Advanced Photon Source, recently made news [130] for providing an extremely close look at the fuel spray through a nozzle to better understand cavitation in automobile engines. Experimental measurements at this scale in the context of electrochemical fuel cells and batteries would significantly improve our understanding of them.

The electrochemical study in this thesis goes down a less-traveled path compared to standard approaches. Along less-trodden paths, one needs reliable tools and well-thought out analysis. Furthermore, one needs to search bravely. It is often the case that we do not find exactly what we are looking. But in fields like applied mathematics and scientific computing, we often find very interesting things completely different from the things we were searching. Perhaps, we realize that they were the things we should have been searching for all along. Regardless of how what we may find, it is important to keep searching bravely.

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